Causal Clifford Theory: the Standard Model and gravity from a single Clifford root

Mohammed Haouach[1]

[1] *Independent researcher*[*]

A unitary quantum cellular automaton on a Planck lattice is constructed whose cell is a 2×2×2 block of sites, each of its eight sub-sites carrying the 32-dimensional complex module of the Clifford algebra $\mathrm{Cl}(10,0)$ — on-site space $\mathbb{C}^{256}$. Ten mutually anticommuting Hermitian generators $h_a^2 = I$ are the whole of the kinematics. From this single object follow a (1+3) Minkowski subalgebra; a 16-component Weyl module whose charge assignment is exactly one Standard Model generation, with all eight gauge and mixed anomalies cancelling; the 45-dimensional Spin(10) adjoint containing the twelve gauge bosons; the 5/3 hypercharge normalisation; an exactly unitary, strictly causal update with $\det U = 1$, a ±-paired spectrum and an exact matter/antimatter spectral mirror; and a lattice gravitational sector that reproduces general relativity on de Sitter, Schwarzschild and 3+1 data to machine precision. Requiring the update to be an element of the root — the closure axiom — quantizes the one continuous parameter of the dynamics to two angles, $\pi/2$ and $\pi/4$, and the finer of the two is the physical branch: the fundamental step has finite order and the internal spectrum is a finite algebraic set.

The paper states equally precisely what does **not** follow. Every number is classified by provenance — *algebraic*, *conventional*, *fitted*, or *open*. The family number, the Yukawa structure, quark mixing, the cosmological constant, the dark-matter identity and relic abundance, the neutrino mass, the baryon asymmetry and the values of the gauge couplings remain open, each with the size of the gap quantified. One loop factor $(1/16\pi)^n$ organises four scales over thirty orders of magnitude: $n = 10$ reproduces the electroweak/Planck hierarchy to a factor 2.08 and fixes a 118 GeV dark-matter scale, $n = 3$ the seesaw, $n = 17$ the neutrino mass; the loop factor itself is reproduced to 3.6% by a closed traced loop of length 10 in the automaton's own gauged bond layer, and the crossing of $1/(16\pi)$ occurs at exactly $n = 10$.

Seven tensions with observation are reported rather than hidden: under Standard-Model desert running the coupling $\alpha_{10} = 1/25$ overshoots the measured SU(2) coupling by a factor 5.8 and drives SU(3) to a Landau pole at 2.2×10^9 GeV; the bare neutrino ladder is excluded by $\sum m_\nu < 0.12$ eV; no state of the implemented spectrum can be the dark matter; the relic abundance computed from the couplings the root supplies is not 0.120; the string tension of the gauged layer is 5.4×10^{40} times the QCD value; the implemented step has no exactly massless one-particle state and therefore no graviton candidate; and the node cones of the lattice spectrum are anisotropic nodal planes, not light cones. Nine open problems and seven falsifiers are stated with their numbers.

I. INTRODUCTION

A. The picture

Two habits of language do most of the misleading. The first calls c the speed of light. Light is one traveller at that speed; the constant is the speed of causality. The second is wave–particle duality, taught as a paradox. On a lattice it dissolves: there is one wave, always; what a detector registers is a single localised event, because the detector is itself a lattice of cells changing state one cell at a time. The ontology contains no particles — only waves that deliver their energy in quanta.

Quantum field theory replaced particles by fields but multiplied the fields, one per species, glued together by hand. The viewpoint here is the opposite one, elementary in computer science: a cellular automaton — a lattice of cells, each holding a finite packet of information, all updated in parallel by one local rule. If the world is such an automaton, everything physical is a wave of information over that lattice, and the theory reduces to two questions: what is the information structure of a single cell, and what is the rule that updates it. Causal Clifford Theory (CL10) answers both concretely: the cell is the spinor module of $\mathrm{Cl}(10,0)$, and the rule is a product of Spin(10) plane rotations interleaved with nearest-neighbour swap-rotations.

B. Two literatures, one construction

Discrete models of quantum dynamics and grand unification have developed independently. Causal sets, lattice gauge theory and quantum cellular automata treat spacetime as a finite, causal, unitary dynamical system [1–5]. The observation that one generation of quarks and leptons fits exactly into the 16-dimensional spinor of Spin(10), with the 5/3 hypercharge normalisation and anomaly cancellation following from the embedding, has been standard since 1974 [6–9]; Clifford-algebra formulations of this unification exist [10–12]. 't Hooft's cellular-automaton interpretation supplies a deterministic unitary dynamics but no unification [29, 30].

CL10 puts these on a single footing: the internal space

[*] haouach.mohammed@gmail.com

of every sub-site of the automaton's cell **is** the spinor module of $\mathrm{Cl}(10,0)$, and the update **is** an element of the corresponding spin group. Nothing else is added by hand: no extra scalar, no extra gauge factor, no second algebra.

C. Claims, non-claims, and the provenance discipline

Claimed and demonstrated. The kinematics of one Standard Model generation, its anomaly freedom, the grand-unified gauge content and the 5/3, and a unitary causal automaton whose gravitational sector reproduces general relativity on the lattice to machine precision (Sections 3–7, 11).

Claimed as an order-of-magnitude statement. The electroweak/Planck hierarchy, the dark-matter mass scale, the neutrino mass and the baryon asymmetry, all expressed through the loop factor $(1/16\pi)^n$ with n a small integer (Section 12). The integers are chosen to land near the measured values; the two-digit coefficients are read from the data.

Not claimed. A derivation of the family number, of the fermion masses and mixings, of the cosmological constant, of the dark-matter particle, of the baryon asymmetry, of the spin-2 character of the graviton, of diffeomorphism invariance, or of the Born rule. Each is listed in Section 17 with the size of the gap quantified.

The organising discipline of the paper is therefore a **provenance discipline**: a number is *algebraic* if the root fixes it, *conventional* if it comes from a mapping or scale identification, *fitted* if a coefficient was read from data, and *open* if the theory does not fix it. Section 16 tabulates every headline number under this classification.

The seven criteria. A unification claim has to be checkable, so the theory is judged against stated criteria, with its own verdict on each.

The theory passes the structural criteria and fails the parameter one. The closure axiom repairs the parameter criterion algebraically and worsens it phenomenologically: at the quantized angle the doubler census becomes a six-line nodal network rather than twelve points, the desert unification solution leaves the desert with $\tau_p = 2.8\times10^{39}$ yr, and the coupling moves $1.54\times$ further from the one-loop value. The chirality analysis of Section 8 shows that even the structural pass is partial: continuous Lorentz invariance does not emerge at the nodes of the implemented step.

D. Conventions

Planck units $\hbar = c = \ell_P = t_P = 1$ unless stated; $G = 1$. Metric signature $(+,-,-,-)$, $Q = T_3 + Y$. $\mathrm{Cl}(10,0)$ is the real algebra of ten generators with $h_a^2 = +I$; the module is over its complexification $\mathrm{Cl}_{10}(\mathbb{C}) \simeq M_{32}(\mathbb{C})$. "Generation" means one set of left-handed Weyl fields. Numerical statements are made at tolerance 10^{-12}, machine-precision identities at 10^{-14} or below; every quoted residual is the deviation of a computed quantity from a stated value, computed in double precision on the explicit representation of Appendix A. Reference data: PDG 2024 [13], Planck 2018 [14], Super-Kamiokande [15, 52], KATRIN [16], NuFIT [17, 50], LZ [53].

II. THE AXIOM

Axiom 1 (the root). *The universe is a* (1+3)*-dimensional causal quantum cellular automaton. Its cell is a* 2×2×2 *block of Planck sites — site spacing* ℓ_P*, so the cells tile a cubic lattice of spacing* $2\ell_P$ *— and each of the eight sub-sites carries a copy of the Clifford algebra* $\mathrm{Cl}(10,0)$*: ten mutually anticommuting Hermitian generators*

$$\begin{gathered} h_a^\dagger = h_a, \\ h_a^2 = I, \\ \{h_a, h_b\} = 0 \quad (a \neq b), \\ a, b = 1, \ldots, 10. \end{gathered} \tag{1}$$

The state of the universe is a tensor product of these modules over the sub-sites, so the cell's on-site space is $\mathbb{C}^{256}$*, and the dynamics is one time-independent unitary* ***step*** *of six* ***stages*** *built from the algebra (Section 7.1). The elementary time is the stage,* t_P*; one step is* $6t_P$*.*

Why the cell is a block. The bond part of the update pairs neighbouring sub-sites, and the pairing that treats the three directions alike has six stages: three pair sub-sites *inside* the cell, three pair sub-sites *across* cells. Written on a single Planck site the same rule alternates between bonds inside and outside the site, which forces a two-cell spatial period and a six-periodic time dependence; on the block it is one time-independent rule, and the block is the smallest region on which the update acts as a single unitary. The staggering is therefore not a time dependence of the law but the sub-lattice structure of the cell, and the Bloch decomposition of the step is exact at every step.

Three structural facts about the root, all standard [10, 18]:

1. $\mathrm{Cl}(10,0)$ is $2^{10} = 1024$-dimensional; its complexification is $M_{32}(\mathbb{C})$, so it has a single irreducible complex module of dimension 32, the Dirac module.
2. The volume element $w = h_1 h_2 \cdots h_{10}$ satisfies

$$w^2 = (-1)^{10\cdot9/2} \prod_a h_a^2 = (-1)^{45} = -I, \qquad w^\dagger = -w,$$

so w is anti-Hermitian with eigenvalues $\pm i$, each of multiplicity 16. The projectors $P_{16} = \frac{1}{2}(I - iw)$ and $P_{\overline{16}} = \frac{1}{2}(I + iw)$ split the Dirac module into two inequivalent 16-dimensional Weyl modules.

TABLE I. **The seven criteria and the verdict of this paper.**

criterion	requirement	verdict
one object	a single algebraic object, no added fields or structure	met for the internal space; the gravitational sector adds a symmetric tensor field $g_{\mu\nu}(x)$ and the value $G = 1$ (Section 11)
kinematics	spacetime, matter content, gauge group and statistics follow by representation theory	met (Sections 3–6)
dynamics	the object yields a unitary, causal evolution	met (Section 7: exactly unitary, $\det U = 1$, measured front speed = 1)
unification	matter, gauge and gravity live in the same object	partial: matter and gauge are the same module and the same **45**; gravity is the metric dynamics of a subalgebra of that module, with an added field
consistency	anomaly-free, unitary, causal, UV-bounded	met (Theorem 2, Theorem 5, the bounded 32-level spectrum, the bounded n-particle S-matrix); the doublers of Proposition 2 are gapped by an internal Wilson factor (Section 8), with the split node set an input (O9)
parameters	every physical number fixed by the object, no fit	**not met**: two conventions, eleven fitted coefficients, one conditional coefficient (Section 16). The closure axiom (Section 9) removes the one free continuous parameter of the dynamics at the price of moving that parameter's two empirical contacts
falsifiability	testable predictions, and the open problems named rather than hidden	met (Sections 17–18)

3. The algebra is $\mathbb{Z}_2$-graded by the parity of the number of generators in a product: 512 even and 512 odd elements.

Everything derived below is representation theory of this one object plus the dynamics of Section 7.

Why 10**.** The number 10 is the smallest n for which a single complex Clifford algebra simultaneously provides (i) a pseudoscalar splitting a Dirac module into two *inequivalent* Weyl modules of dimension 16 — at $n = 8$ the Weyl modules are 8-dimensional (Spin(8) triality: $8_v, 8_s, 8_c$; the 16-dimensional representations are reducible) — and (ii) an adjoint of dimension $\binom{10}{2} = 45$ large enough to contain $\mathfrak{su}(5)$ and hence the Standard Model algebra. At $n = 9$, $\mathrm{Cl}_9(\mathbb{C}) \simeq M_{16} \oplus M_{16}$ does have 16-dimensional irreducibles, but they are the two simple factors of the algebra rather than a pseudoscalar split, and $\mathfrak{so}(9)$ does not contain $\mathfrak{su}(5)$. The root is therefore selected by the requirement of a single chiral generation with a unified gauge algebra, not chosen.

III. SPACETIME AS A SUBALGEBRA

Theorem 1. *Let* $T = h_1$ *and* $S_a = ih_{a+1}$ *for* $a = 1, 2, 3$*. Then*

$$T^2 = +I, \qquad S_a^2 = -I, \qquad \{T, S_a\} = 0, \qquad \{S_a, S_b\} = 0.$$

The real span of $\{T, S_1, S_2, S_3\}$ *inside the grade-1 part of the algebra, with the quadratic form induced by the Clifford product, is Minkowski space* $\mathbb{R}^{1,3}$*.*

Proof. The four generators $h_1, \ldots, h_4$ anticommute by Axiom 1. Multiplication by i turns $h_a^2 = I$ into $(ih_a)^2 = -I$ and preserves anticommutation. For $v = v_0 T + \sum_a v_a S_a$ with real coefficients, $v^2 = (v_0^2 - \sum_a v_a^2) I$.

Corollary 1 (the light cone). *A vector* v *in that real span is null,* $v^2 = 0$*, if and only if* $v_0^2 = \sum_a v_a^2$*. The null cone of this quadratic form is the light cone of the lattice.*

Two qualifications. The null cone is a **cone in the grade-1 part**, not a subalgebra: it is not closed under the Clifford product, since $(T + S_1)(T - S_1) = 2I - 2TS_1$ is a grade-0 plus grade-2 element. And the choice of *which* four of the ten generators play the role of spacetime is a choice of frame in the 10-dimensional generator space, not a prediction: the algebra supplies a family of equivalent (1+3) subalgebras related by orthogonal transformations of the generator space. The remaining six generators are inert under the Lorentz generators $\Sigma_{\mu\nu} = \frac{1}{4}[\Gamma_\mu, \Gamma_\nu]$ built from $\Gamma_\mu \in \{T, S_a\}$.

Causality is a postulate with a dynamical content. The light cone by itself does not bound signal propagation: a unitary on-site rotation alone has infinite propagation speed. The bound is supplied by the update of Section 7, where the causal radius is *implemented* by the bond stages and *measured* to be exactly one Planck cell per stage (Theorem 5). The light cone and the causal radius are distinct statements, and only the second is a property of the dynamics.

IV. MATTER: THE 16 WEYL SPINOR

Proposition 1. *Under* $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ *the* 16*-dimensional Weyl module is the six left-handed fields*

of one Standard Model generation:

$$\begin{aligned} 16 = Q(\mathbf{3},\mathbf{2},+\tfrac{1}{6})\oplus \\ u^c(\bar{\mathbf{3}},\mathbf{1},-\tfrac{2}{3})\oplus \\ d^c(\bar{\mathbf{3}},\mathbf{1},+\tfrac{1}{3})\oplus \\ L(\mathbf{1},\mathbf{2},-\tfrac{1}{2})\oplus \\ e^c(\mathbf{1},\mathbf{1},+1)\oplus \\ \nu^c(\mathbf{1},\mathbf{1},0), \end{aligned} \tag{2}$$

with $6+3+3+2+1+1=16$. *Under* SU(5) *this is* $\mathbf{10}\oplus\bar{\mathbf{5}}\oplus\mathbf{1}$, *the* $\mathbf{1}$ *being the sterile* ν^c.

The assignment is the standard one [6–9, 12]. What the automaton adds is that the **16** is not put in by hand but is the Weyl eigenspace of the pseudoscalar of the same algebra that carries the gauge group and the spacetime.

$B-L$. $B-L$ is a generator of $\mathfrak{so}(10)$, so it is *not* a single number on the **16**. Its values on the six fields are $Q:\frac{1}{3}$, $u^c:-\frac{1}{3}$, $d^c:-\frac{1}{3}$, $L:-1$, $e^c:+1$, $\nu^c:+1$; the $\overline{\mathbf{16}}$ carries the negatives of all of them. Under $\mathrm{Spin}(10)\supset \mathrm{SU}(5)\times\mathrm{U}(1)_\chi$, $16\to\mathbf{10}_{-1}\oplus\bar{\mathbf{5}}_{+2}\oplus\mathbf{1}_{+3}$, and $B-L$ is a linear combination of χ and the diagonal SU(5) generator, not χ itself.

Theorem 2 (anomaly cancellation). *All eight gauge and mixed anomaly coefficients of the* **16** *vanish.*

Proof. Each coefficient is a cubic trace over the left-handed Weyl fields; the multiplicities are the dimensions of the *other* group factors (the factor 2 on Q in $[\mathrm{SU}(3)]^3$ is its SU(2) doublet nature, the factor 3 in $\mathrm{U}(1)[\mathrm{SU}(2)]^2$ its colour-triplet nature), with $A(\mathbf{3})=T(\mathbf{3})=\frac{1}{2}$ and the charges of Proposition 1.

This is the hardest consistency condition a chiral gauge theory has, and it is satisfied identically by the module content. It is the strongest single result of the theory, and it is independent of the coupling: the cancellation is a property of the algebra, not of a tuned parameter.

The mass term. A grade-1 element of the algebra anticommutes with w, so it is off-diagonal in the $16/\overline{16}$ split: it maps $16\to\overline{16}$ and $\overline{16}\to 16$, and both diagonal blocks vanish. A scalar in the **10** of Spin(10) therefore produces a Dirac mass term $16\,\mathbf{10}_H\,\overline{16}$ and no Majorana mass for the charged fermions. The singlet ν^c is the built-in source of a seesaw (Section 12.3).

Two caveats. A vev in the **10** has stabiliser SO(9): it breaks $\mathrm{Spin}(10)\to\mathrm{SO}(9)$, and SO(9) is not the Standard Model group. The chain is completed instead by the Wilson direction of Section 8, which at its physical root leaves exactly $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)^3$ and breaks precisely the leptoquark roots; the two extra U(1)'s must still be broken or hidden. And the **10** contains no Standard Model singlet, so the vev that generates masses necessarily breaks the unified group further. This is open problem O6.

The three families. The **16** is *one* generation. The theory identifies the observed family number with the number 3 of spatial directions of the (1+3) subalgebra, equivalently $\binom{3}{2}$ spatial two-planes, equivalently the rank of the spatial metric: one number stated three ways, not three derivations. The identification has a structural hook: the twelve doubler nodes of Section 8 organise into **three families of four labeled by the zero axis**, one family per spatial direction, with identical one-step spectra (deviation 0), and the three copies of vectorlike matter required by unification are realizable by taking one node from each family at a single Wilson strength. The index 3 exists and *labels* the three copies; it does not force the count (O3).

V. GAUGE: THE 45 AND THE 5/3

The 45 generators $h_a h_b$ $(a<b)$ close on the Spin(10) adjoint. They are even in the $\mathbb{Z}_2$ grading, hence commute with w and preserve the $16/\overline{16}$ split. Under $\mathrm{Spin}(10)\supset\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ the **45** decomposes as 12 Standard Model gauge bosons + 12 leptoquark X/Y bosons + 20 bosons in the $\mathbf{10}\oplus\overline{\mathbf{10}}$ of SU(5) + the $\mathrm{U}(1)_{B-L}$ generator; the 33 broken generators mediate proton decay (Section 13.4).

Theorem 3 (the 5/3). *In the* **5** *of* SU(5) *the hypercharge is* $Y=\mathrm{diag}(-\frac{1}{3},-\frac{1}{3},-\frac{1}{3},+\frac{1}{2},+\frac{1}{2})$ *with* $\mathrm{Tr}\,Y^2=\frac{5}{6}$, *while the non-Abelian generators obey* $\mathrm{Tr}\,T_aT_b=\frac{1}{2}\delta_{ab}$. *Requiring a single unified coupling* g_5 *for all generators gives* $g_1^2/g'^2=5/3$, *where* g' *is the Standard Model hypercharge coupling and* $g_1\equiv g_5$.

Proof. $\mathrm{Tr}\,Y^2=3\cdot\frac{1}{9}+2\cdot\frac{1}{4}=\frac{5}{6}$. Rescaling $Y\to\sqrt{3/5}\,Y$ normalises it to $\frac{1}{2}$; then $g_5(\sqrt{3/5}Y)=g'Y$ gives $g'=g_5\sqrt{3/5}$.

This is an algebraic fact of the root: no parameter enters. It anchors the coupling discussion of Section 13.1, where it will be seen that the *normalisation* of the unified coupling is under control while its *value* is not.

VI. STATISTICS AND MEASUREMENT

The $\mathbb{Z}_2$ grading splits the algebra into 512 even and 512 odd elements. The even sector is identified with bosons and the odd sector with fermions. This is a *kinematic* grading of the operator algebra: a postulate of the interpretation, not a derivation of the spin–statistics theorem, and not a claim of dynamical supersymmetry. No supercharge exists in the construction and no superpartner is predicted.

Between measurements the automaton is exactly unitary. A detection is modelled as the standard projection: $\sum_x\psi(x)|x\rangle$ reduces to $|x_0\rangle$ with probability $|\psi(x_0)|^2$, the probability rule being the intensity $|\psi|^2$ of the propagating lattice field — the one-electron-at-a-time build-up of the interference fringes is the Tonomura experiment [47]. The projection is **not** derived from the unitary step and the theory does not claim it is: the reduction postulate is an input of the interpretation (O9). What the automaton adds is only that the wave being sampled is a strictly

TABLE II. **The eight anomaly coefficients of one 16.**

coefficient	**contribution of the 16**	**value**
$[\mathrm{SU}(3)]^3$	$2A(\mathbf{3}) + A(\bar{\mathbf{3}}) + A(\bar{\mathbf{3}}) = 1 - \frac{1}{2} - \frac{1}{2}$	0
$[\mathrm{SU}(2)]^3$	$A(\mathbf{2}) = 0$ identically ($d^{abc} = 0$; the fundamental is pseudo-real)	0
$[\mathrm{U}(1)]^3$	$6(\frac{1}{6})^3 + 3(-\frac{2}{3})^3 + 3(\frac{1}{3})^3 + 2(-\frac{1}{2})^3 + 1(1)^3 = \frac{1}{36} - \frac{8}{9} + \frac{1}{9} - \frac{1}{4} + 1$	0
$\mathrm{U}(1)[\mathrm{SU}(3)]^2$	$\frac{1}{6} \cdot \frac{1}{2} \cdot 2 + (-\frac{2}{3}) \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6} - \frac{1}{3} + \frac{1}{6}$	0
$\mathrm{U}(1)[\mathrm{SU}(2)]^2$	$\frac{1}{6} \cdot \frac{1}{2} \cdot 3 - \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} - \frac{1}{4}$	0
$[\mathrm{grav}]^2\mathrm{U}(1)$	$\sum Y = 1 - 2 + 1 - 1 + 1$	0
$[\mathrm{grav}]^2\mathrm{SU}(3)$, $[\mathrm{grav}]^2\mathrm{SU}(2)$	traceless generators	0

causal, unitary, lattice field. (Fig. 1).

The same field drives atomic physics: coupled to a prescribed hydrogen spectrum, the lattice photon reproduces resonant absorption, the $1s \to 2p$ deformation, and re-emission at the gap frequency, with forbidden dipole matrix elements vanishing to 10^{-18} ($1s \to 3d$: 1.7×10^{-18}, $2p \to 3p$: 1.2×10^{-17}, against 9.93, −3.00, 22.97 for the allowed transitions) and the excited fraction falling with detuning (0.42, 0.35, 0.15, 0.035, 0.011 at detunings 0, 0.063, 0.125, 0.25, 0.50 rad/step). The atom is an **input** — its levels are the real hydrogen eigenstates — and the figure demonstrates the photon sector, not a derivation of hydrogen. Two limits are stated rather than hidden: off-resonant absorption has a Lorentzian tail, and a wave packet wider than the simulation box never leaves it, becoming a persistent bath that drives the atom off resonance (measured at half-gap detuning in a 120-cell box as excited fractions 0.037, 0.135, 0.326 for packet widths of 12, 24, 48 cells; the published figure uses a localised packet, 20 cells in a 296-cell box). (Fig. 2).

VII. THE STEP: UNITARITY, SPECTRUM, CAUSALITY

A. The update

The step has six stages, and every stage is the same two layers: the internal rotation applied to every sub-site, then one bond stage. The internal factor is built from the Spin(10) plane-rotation generators $T_{ab} = i\, h_a h_b$ ($a < b$) in the 32-dimensional module — each T_{ab} Hermitian, squaring to I, traceless (sixteen +1 and sixteen −1 eigenvalues):

$$U(g) \;=\; \prod_{a<b} \Big(\cos g\, I - i \sin g\, T_{ab} \Big).$$

A sub-site is labelled by $e = (e_1, e_2, e_3) \in \{0,1\}^3$, the bit of axis j being 2^j. A **bond stage** acts on the four disjoint pairs $(e,\, e \oplus 2^j)$: an *intra-block* stage takes the pairs with $e_j = 0$, inside the cell; an *inter-block* stage takes the pairs joining $e_j = 1$ of one cell to $e_j = 0$ of the next cell along $\hat{e}_j$ — still a two-cell operator, now between neighbouring cells. On each pair the gate is the swap-rotation $\cos g\, I - i \sin g\, (|e\rangle\langle o| + |o\rangle\langle e|)$, carrying the same single angle and no other parameter. The six stages run in the fixed order (intra x, inter y, intra z, inter x, intra y, inter z). One step is therefore

$$M \;=\; \prod_{s=1}^{6} \big(P_s \otimes U(g) \big),$$

an operator on the cell's on-site space $\mathbb{C}^{256}$ (P_s on the eight sub-sites, U on the internal module). At block crystal momentum q an inter-block pair carries the phase $e^{\pm iq}$, $M(q)$ is 256×256, and on the 16 sector it is 128×128.

Every factor is special unitary: the 45 internal factors have $\det = \exp(-ig\,\mathrm{Tr}\, T_{ab}) = 1$, and each bond pair block $\left(\begin{smallmatrix} c & -is \\ -is & c \end{smallmatrix}\right)$ has determinant $c^2 + s^2 = 1$ (the inter-block phases cancel). Hence

$$\begin{gathered} UU^\dagger = I, \quad \det U = 1, \\ MM^\dagger = I, \quad \det M = 1 \\ \text{for every } g, \end{gathered} \tag{3}$$

verified numerically: internal unitarity and $|\det U - 1|$ at the 10^{-15} level; for the full cell matrix $\|MM^\dagger - I\| \le 5 \times 10^{-15}$ and $|\det M - 1| \le 3 \times 10^{-13}$ at the zone centre, at a generic momentum and at the two high-symmetry points; the real-space six-stage step preserves the norm of a random cell state to 10^{-15}. Two structural inputs must be named: the 45 factors do not commute, so U depends on the *ordering* of the product, and the six stages have a fixed order. The algebra does not select either order.

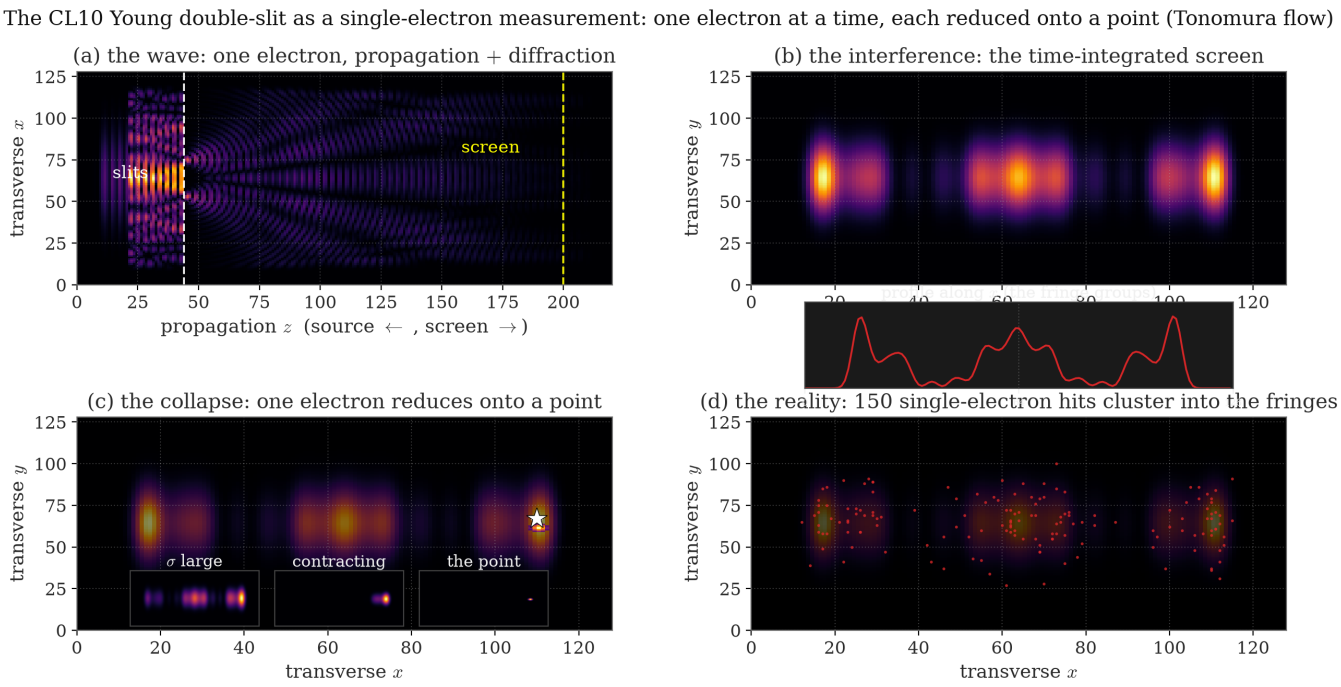


FIG. 1. The lattice wave through a double slit, run one electron at a time: (a) diffraction and interference of the field; (b) the time-integrated on-screen pattern; (c) the reduction of one electron onto one detection point; (d) the accumulated detections clustering into the fringes [47].

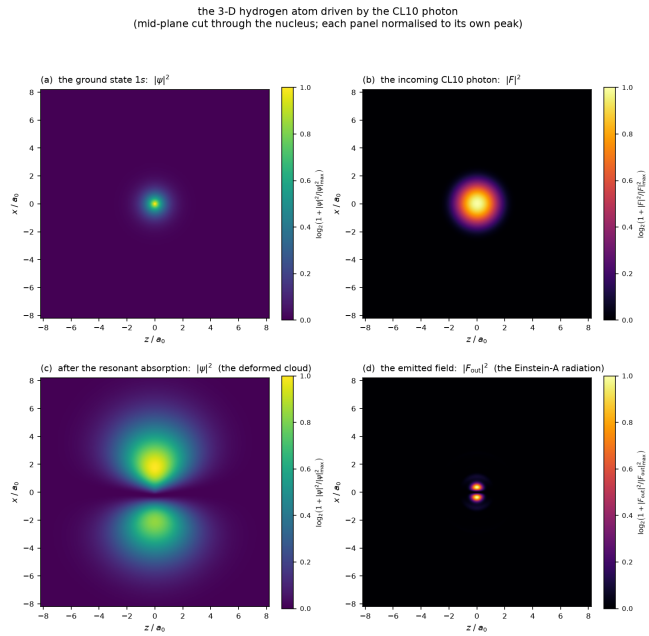


FIG. 2. A three-dimensional hydrogen atom coupled to the same discrete massless field that carries the graviton, on the mid-plane cut through the nucleus: (a) the spherical ground state $|\psi|^2$; (b) the incoming lattice photon $|F|^2$; (c) the cloud after resonant absorption, deformed into the dumbbell $2p$; (d) the emitted field $|F_{\rm out}|^2$.

The angle g is the one continuous number of the dynamics. Two readings of it coexist and are kept apart throughout. The **implemented** value $g = 1/5$ is the one on which the phenomenology of Sections 12–14 is computed; it comes from the proposed rule $g = 1/n_{\rm qubits}$, the five qubits of the 32-dimensional module, which the data do not support (Section 16). The **closed** values $g = \pi/2$ and $g = \pi/4$ are the only ones for which the step is an element of the root (Section 9). The two readings disagree by a measured amount, and both are reported. (Fig. 3).

B. The single-particle spectrum

The 32 eigenphases ω_k of U (the spectrum of $H = -i \log U$) are the energies of one sub-site per stage, in units of $1/t_P$; the spectrum of the cell at block momentum q is the set of eigenphases of $M(q)$, 128 of them on the 16 sector. At $g = 1/5$ the internal levels are

$$\omega \in \{\pm 0.01147,\ \pm 0.05447,\ \pm 0.15819,\ \ldots,\ \pm 1.48638\},$$

exactly $\pm$-paired to 2×10^{-15}, so the single-sub-site vacuum energy $\sum_k \omega_k = \arg\det U$ vanishes. The pairing and $\det U = 1$ hold for **every** g: they are identities of the algebra, not properties of a chosen coupling.

Theorem 4 (the matter/antimatter spectral mirror). *Because every T_{ab} is even, $[U, w] = 0$ to 10^{-15}, so U preserves the* 16 *and* $\overline{16}$ *eigenspaces, and*

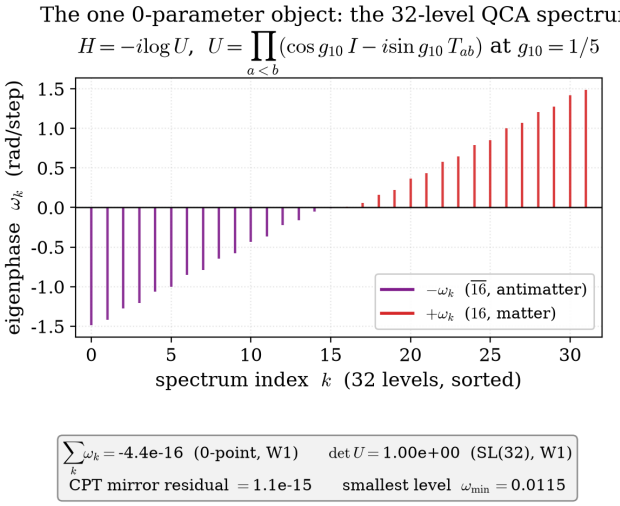


FIG. 3. The eigenphases of U at $g = 1/5$. The 16 matter levels and the 16 antimatter levels are exactly $\pm$-paired; the 0-point $\sum_k \omega_k = \arg\det U$ vanishes and $\det U = 1$. The smallest level, $\omega_{\min} = 0.0115$, corresponds to $\omega_{\min} M_P = 1.4 \times 10^{17}$ GeV — the only sub-Planck scale of the internal spectrum, and *not* the electroweak scale.

$$\omega_{\overline{16}} = -\omega_{16},$$
$$\operatorname{spec}\big(U|_{\overline{16}}\big) = \operatorname{spec}\big(U|_{16}^{\dagger}\big), \tag{4}$$

verified to 2×10^{-15} *over the grid* $g \in \{0.05, \ldots, 0.5\}$.

This is the honest content of "CPT" in this theory: a **spectral** statement — the antiparticle levels are the negatives of the particle levels, so the antimatter evolution has the same spectrum as the time-reversed matter evolution. It is not the full CPT theorem: no antiunitary time-reversal operator is constructed on the many-cell Hilbert space and no spin-statistics connection is derived. A unitary involution $C = h_1$ maps the **16** onto the $\overline{\mathbf{16}}$ (it anticommutes with w), but calling $C \cdot T$ "CPT" overstates what has been built.

The mechanism of the pairing is understood, and its scope is limited. The intertwiner between the 32-dimensional module and its conjugate is unique (nullity 1, rank 32); it satisfies $Q h_a Q^\dagger = h_a^*$ to 2×10^{-15} for the ten generators and $Q T_{ab} Q^\dagger = -T_{ab}^*$ to the same bound for the 45, and the associated antiunitary $\mathcal{J}$ has $\mathcal{J}^2 = +1$. That is the content of $\mathrm{Cl}(10,0) \simeq M_{32}(\mathbb{R})$: Spin(10) lands in SO(32), and a real form of U(32) carries no CP-odd invariant. Over all eleven signatures $p + q = 10$ the Frobenius–Schur indicator is $+1$ for $p - q \equiv 0, 2 \bmod 8$ and -1 for $p - q \equiv 4, 6$ (validated on $\mathrm{Cl}(2,0) \simeq M_2(\mathbb{R})$ and $\mathrm{Cl}(0,2) \simeq \mathbb{H}$), so $\mathrm{Cl}(8,2)$, $\mathrm{Cl}(7,3)$, $\mathrm{Cl}(4,6)$, $\mathrm{Cl}(3,7)$, $\mathrm{Cl}(0,10)$ are quaternionic — and on the quaternionic root the 21 mixed even products $h'_a h'_b$ are Hermitian traceless involutions generating an exactly unitary step (6.7×10^{-16}, $\det U = 1$) whose spectrum is still exactly conjugation-closed (3×10^{-15}): the mirror is not a privilege of reality. Two scope limits come with it. The antiunitary inverts every single factor ($\mathcal{J} F \mathcal{J}^{-1} = F^{-1}$) but reverses the order of the ordered product, so $\mathcal{J} U \mathcal{J}^{-1} \neq U^{-1}$ (deviation 0.58 for $\mathrm{Cl}(7,3)$, 0.81 for $\mathrm{Cl}(10,0)$): the pairing is a property of the whole ordered step. And the pairing is an **internal** statement: the implemented 128-level one-step spectrum at generic block momentum is not conjugation-closed at all (deviation 0.95–0.99), while the internal 32 spectrum is exact (5×10^{-15}).

C. Locality and the causal radius

Theorem 5 (the causal radius is implemented). *On an (L_b, L_b, L_b) torus of cells the step is a product of two-cell operators: every bond stage couples sub-sites at Planck-cell distance ≤ 1 — an intra-block pair joins neighbouring sub-sites of one cell, an inter-block pair joins neighbouring cells. Hence after n stages the support of a sub-site-localised excitation lies inside the n-cell light cone: at most one Planck cell per stage, and since three of the six stages are inter-block, at most three blocks — six Planck cells — per step. Numerically: the norm of a random cell state is preserved to 10^{-15}; on an 8^3-cell torus the support radius after n stages is exactly n Planck cells for every $n \leq 12$ — the front* saturates *the cone, front speed* 1.0*; on a 32^3-cell torus run for* 30 *stages the front reaches* 21 *cells, and no site of the support and no amplitude ever appears outside the n-cell cone (*0*, to machine precision); the same cone is realised at $g = 0.3$ as at* 1/5*; the internal layer alone, with the bond stages removed, keeps the excitation on one sub-site forever.*

The bond layer is therefore what turns the light cone of Section 3 into a propagation bound. This is the cleanest dynamical result of the theory.

D. The n-particle sector

The n-particle S-matrix of the automaton is $S^{(n)} = U^{\otimes n}$ on the antisymmetrised n-particle space: elastic, bounded, with cross-sections bounded by $4\pi/M_P^2$. There is no inelastic channel at the one-cell level, and no renormalisation of the step: the coarse-graining tower U^{2^p} ($p = 1 \ldots 8$) stays unitary and bounded at every level, so the *discrete* flow has no Landau pole — in contrast with the perturbative field-theory running of the same couplings, which does (Section 13.1). That contrast is real but modest: a unitary finite-dimensional map is bounded by construction, and boundedness alone does not reproduce the observed running. (Fig. 4).

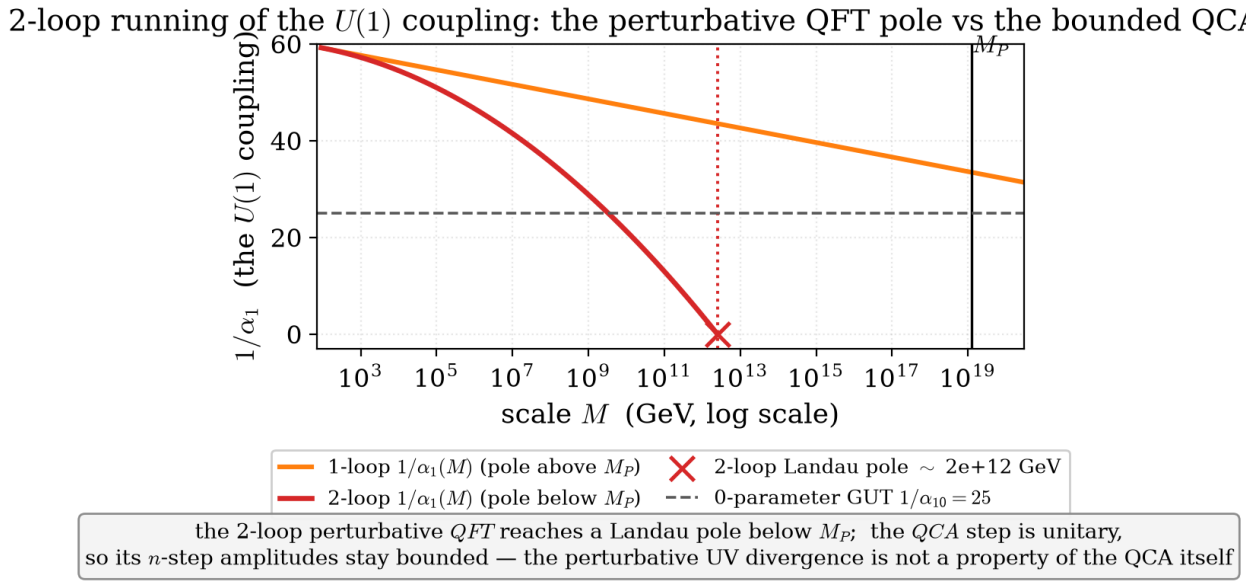


FIG. 4. The two-loop perturbative running reaches a Landau pole at 2.5×10^{12} GeV, below the cutoff; the one-loop pole is above it. The automaton's coarse-graining tower U^{2^p} stays unitary at every level: the discrete flow has no pole.

VIII. LATTICE CHIRALITY AND THE DOUBLING PROBLEM

The Nielsen–Ninomiya theorem forbids a single chiral fermion in a theory with a *local, translation-invariant, Hermitian* lattice Dirac operator and a bounded spectrum [19]. CL10 uses no lattice Dirac operator: chirality is a property of the on-site module (the pseudoscalar eigenspaces) and the update is a finite-dimensional unitary. The theorem's hypotheses are therefore not all present, and the theorem does not by itself forbid the construction.

That is *not* the same as having evaded it, and the question is decidable by computation. The step is one time-independent operator on the cell lattice, so the good quantum number is the **block crystal momentum** q — the phase under a translation by one automaton cell, 2π-periodic, zone $q \in [-\pi, \pi]^3$ — and the eight sub-sites are the internal structure of the Bloch matrix. The 16-sector one-step matrix $M(q)$ is computed exactly at 128×128 and cross-checked against the real-space step on a 2^3-block torus (agreement 10^{-14}). The cubic high-symmetry momenta carry the whole spectrum: $\Gamma = (0,0,0)$; X $= (\pm\frac{\pi}{2}, 0, 0)$, star 6; M $= (\pm\frac{\pi}{2}, \pm\frac{\pi}{2}, 0)$, star 12, little group C_{2v}; R $= (\pm\frac{\pi}{2}, \pm\frac{\pi}{2}, \pm\frac{\pi}{2})$, star 8, little group C_{3v}.

Proposition 2 (lattice chirality). *(i) The* 16 *sector is closed by the full step, not only by the internal layer:* the bond layer acts on the sub-site index and commutes with w, so a state initialised in the **16** leaves it by less than 10^{-15}. Chirality is a dynamical symmetry of the automaton. *(ii) The zone centre is gapped:* $\min_a |\omega_a(q = 0)| = 0.0365$ per step, so the would-be massless matter is **not** at $q = 0$. *(iii) The gapless points are the doublers:* the 16-sector one-step spectrum is gapless at the 12 momenta $(\pm\frac{\pi}{2}, \pm\frac{\pi}{2}, 0)$ and their permutations — the M points of the block zone, at half-zone momentum, whose Planck-cell momenta are $p = q/2$ — with node chiralities summing to 0, as Nielsen–Ninomiya requires; at $g = 1/5$ the nodes still carry a small gap, 2.0×10^{-4}. The gap at $q = 0$ is not protected: it moves from 0.036 at $g = 0.2$ to 0.072 at $g = 0.15$ and to 2.35 at $g = 0.1$.

The doubling therefore **survives** the automaton: "the **16** is a single chiral generation" is a property of the *internal module*, while the *lattice spectrum*, which is what couples to low-energy physics, contains twelve gapless nodes. (Fig. 5).

A. The Wilson mechanism

The remedy is part of the construction. A Wilson-type term built from the root's own algebra — an internal factor

$$W(q) = \exp\Big[- i\, w \sum_j (1 - \cos q_j) \left(\cos\theta\, X + \sin\theta\, Y \right) \Big],$$

with X, Y the unification Cartan directions identified in Section 13.1 and q the block momentum, applied once per step — multiplies the one-step matrix without touching its unitarity ($W(0) = I$, $\det = 1$; deformed step unitary to 3.3×10^{-15}) and leaves the bond stages strictly causal. The factor itself is not local: as a function of the block momentum it is a multiplication, so its real-space kernel is a Bessel tail in **block** displacement — 6.7×10^{-2}

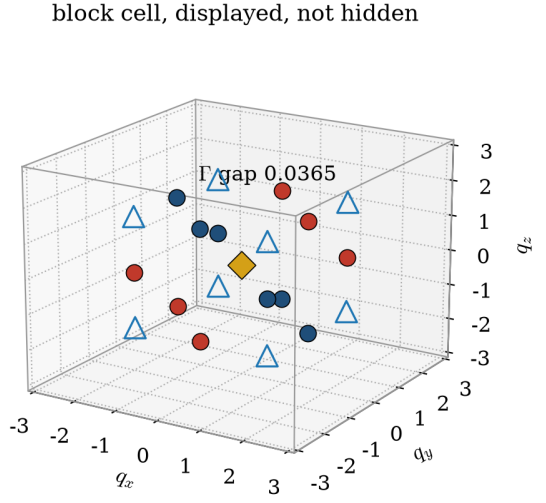


FIG. 5. The twelve gapless doubler nodes at the half-zone momenta of the block zone (chirality ±1, sum zero) and the gapped zone centre (0.0365 per step) of the 16-sector one-step spectrum. The doubling is displayed, not hidden.

at block radius ≥ 2 and 2.2×10^{-3} at ≥ 4 for $w = 0.1$, and 1.8×10^{-17} at ≥ 4 at the physical strength. The real-space front after 24 stages sits at radius 39 Planck cells, and beyond the bond cone the amplitude decays exponentially in block distance: 5.6×10^{-6} in the first block shell, 3.1×10^{-9} at the Lieb–Robinson line $r = 1.2n + 6$, with a decay ratio ≤ 0.26 per block. The Wilson factor replaces the light cone of Theorem 5 by a Lieb–Robinson bound.

It gaps all twelve nodes — the minimum node gap rises from 2.0×10^{-4} to 5.7×10^{-3} ($w = 0.01$) through 1.5×10^{-2} ($w = 0.2$) — with fragment masses

$$M_f = w\,|\cos\theta\, X_f + \sin\theta\, Y_f|\, M_P/3;$$

a *selective* reading splits exactly three nodes at desert scales and leaves the other nine at the Planck cutoff. What remains input is the *choice* of the split node set: unification forces $n_{\text{nodes}} \geq 3$ but does not select which three, and the meeting scale the mechanism then predicts is in conflict with proton decay (Section 13.1).

At the physical root $\theta = 1.731$ the Wilson direction completes the breaking chain: it leaves Spin(10) $\to$ SU(3)×SU(2)×U(1)^3, preserves the Model algebra (every unbroken root is intra-fragment; the 32 broken roots are exactly the leptoquark type), and the two extra U(1)'s are the only open step to the Standard Model group. (Fig. 6).

B. Lorentz structure at the nodes

The causal front speed of Theorem 5 is 1.0 Planck cell per stage; that is the statement the automaton proves. What the *low-energy spectrum* of the implemented step does not have is continuous Lorentz invariance.

At all twelve nodes and in all 1200 sampled directions the dispersion is **linear** (local exponent $p = 1.00$, no quadratic band touching), but the cone is strongly **anisotropic**: principal velocities $(0.200, 0.200, 0.284)$ per unit block momentum — $(0.100, 0.100, 0.142)$ in Planck-cell units — anisotropy 0.748–0.758, the same at every node of the O_h orbit. The lattice signature is universal, not a defect. Exhaustive scanning of the whole Wilson family closes the hope that isotropy emerges somewhere: the node gap is located at every zero over the two orbit representatives × 60 angles (2726 gapless points, tolerance 10^{-7}) and the cone anisotropy $A = (v_{\max} - v_{\min})/(v_{\max} + v_{\min})$ measured over 162 directions at each point; the Lorentz target $A < 10^{-6}$ contains 0 of the 2726 points, the closest approach being $A = 0.997$. (Figs. 7, 8).

The failure is structural, not a finite-radius artefact. At the tangent, the twelve doubler nodes carry speeds (0, 0.14189, 0.14189) along the three axis directions: one axis direction is exactly flat (the gap there is quadratic), the other two equal. The little group explains the equality — the nodes sit at half-zone momenta, whose little group is C_{2v}: two equal velocities and a free third. At the eight R points the full cubic symmetry is realised (exactly 3-fold multiplets, splitting $< 10^{-12}$; the gap of the lowest triplet is the same at all eight corners to 10^{-15}), and the Wilson factor drives one of them exactly gapless (10^{-10}, a one-parameter family): the Wilson family does supply massless cubic matter, and the touching there is linear — dM/dq_j is anti-Hermitian and $S_j = i\,dM/dq_j$ is nonzero (max speed 0.92). But the three tangent velocity operators commute ($[S_i, S_j] = 0$ to 3×10^{-12} against the scale $\sum_j \|S_j\|^2 = 0.85$), $\sum_j S_j^2$ is a scalar to 7×10^{-12}, and each branch speed is exactly $|v_\alpha \cdot q|$ to 6×10^{-12} with the v_α the cubic permutations of $(0.704, 0.421, 0.421)$: the node is a crossing of three **nodal planes**, and no tuning of the Wilson angle turns three planes into a cone. The two corner multiplets cannot be driven to zero simultaneously (best pair gap 7.2×10^{-5}, six orders above the 10^{-10} a single multiplet reaches), so no six-component touching point exists there either.

The cause is the Kronecker form of the bond layer, (sub-site index) ⊗ (internal algebra): the three momentum derivatives act on one factor and commute. Two remedies have been built and measured. A non-Abelian Peierls bond layer in which a hop carries an internal rotation $\exp(\pm i\phi D_{16})$ whose sign follows the hop direction — exactly unitary, still radius-1 causal, reproducing the implemented step to 6×10^{-16} at $\phi = 0$ — raises the planarity defect from 3.6×10^{-8} to 4.0×10^{-2} and switches the tangent commutator on to 0.071 of its natural scale, while the anisotropy never leaves 0.996–0.9999

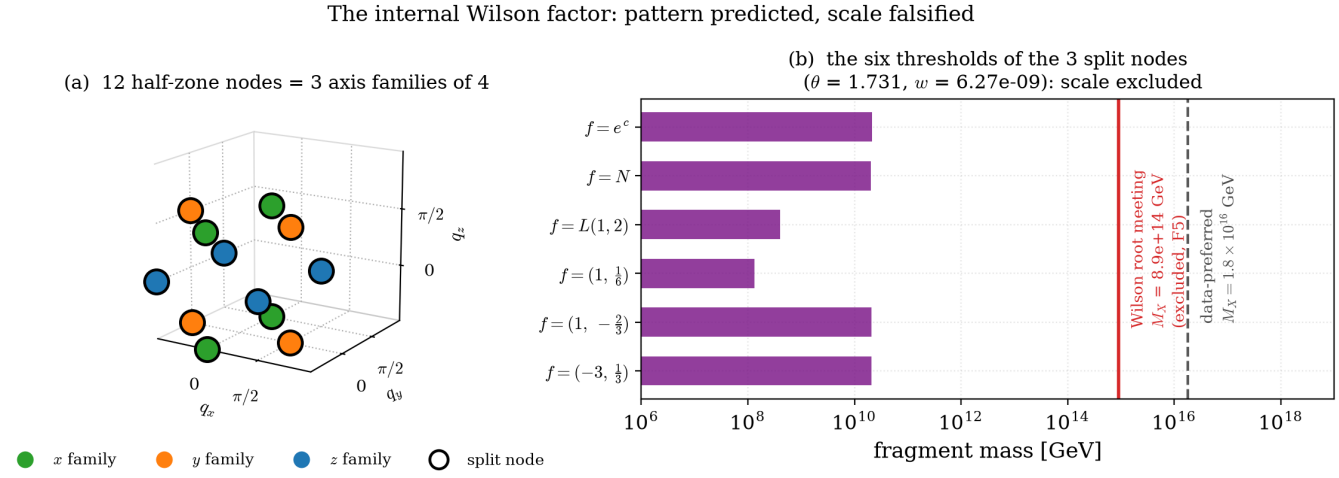


FIG. 6.

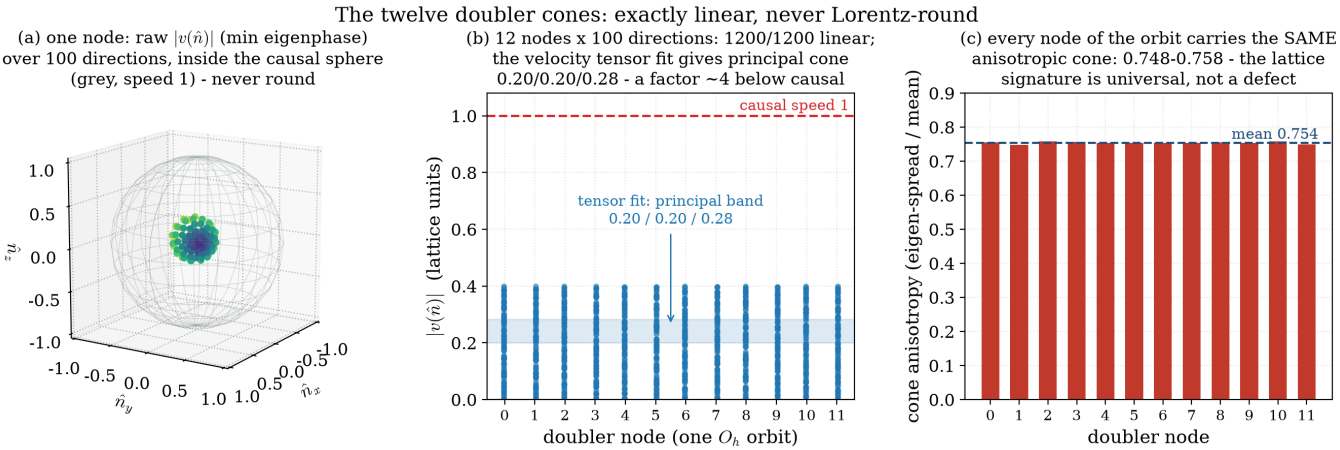


FIG. 7.

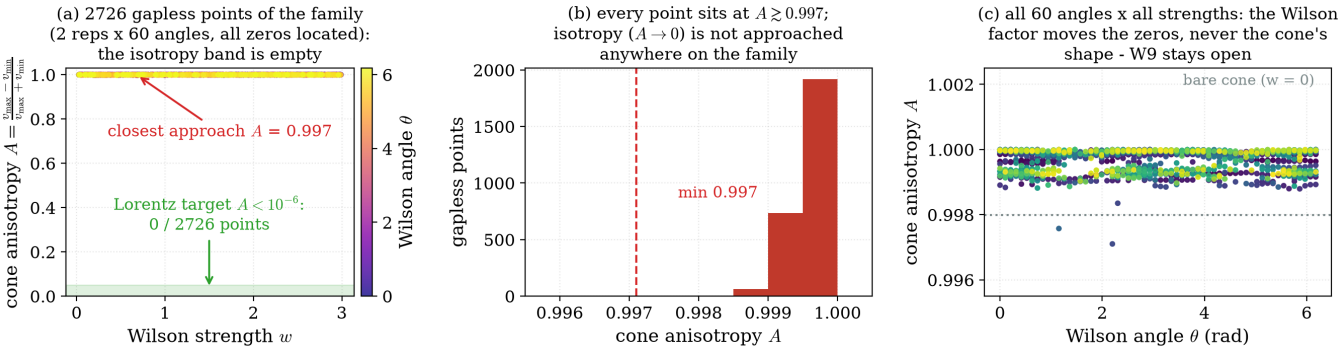


FIG. 8. The escape route "continuous Lorentz invariance emerges at some gapless point of the Wilson family" is closed by exhaustive scan: 0 of 2726 gapless points reach $A < 10^{-6}$; the distribution of A is pinned at 0.997–1.000; no angle θ and no strength w moves a cone toward isotropy.

and the corner triplet is lifted off zero (best top level 3.3×10^{-4}, against 5.6×10^{-7} at $\phi = 0$). A three-axis co-

variant bond $\exp(\pm i\phi_j\Sigma_j)$ with $\Sigma_1 = ih_1h_2$, $\Sigma_2 = ih_2h_3$, $\Sigma_3 = ih_3h_1$ — an exact $\mathfrak{su}(2)$ triplet of even Clifford elements, $[\Sigma_a, \Sigma_b] = -2i\epsilon_{abc}\Sigma_c$ to 1.5×10^{-15}, unitary to 10^{-15}, equal to the implemented step at $\phi = 0$ — takes the tangent commutator to 0.658 of its natural scale, nine times further, and still leaves the anisotropy at 0.9990 with no gapless bond point in the scan; and it breaks the degree-1 Fourier support that proves Theorem 5 (leakage 0.27). The remedy has a measured price: the tangent algebra moves a long way, no cone appears, and exact causality is what pays for it. (Fig. 9).

C. The massless-mode census

The implemented step has **no exactly gapless mode anywhere**. The step factorises exactly as $M(q) = P(q) \otimes U_{16}^6$ (5.4×10^{-15}), so any cubic symmetry acts on the 8-dimensional bond factor alone and the internal **16** is inert. But the ordered six-stage sequence is *not* cubic-covariant: the intertwiner $P(Rq) = D(R)P(q)D(R)^\dagger$ is exact for 16 of the 48 signed permutations (element orders $1, 2, 4$) — the step's own point group is that subgroup of index 3 in O_h, not O_h (axial reflections and the $(0\,2)$ transposition survive, cyclic axis permutations do not). Over the zone the smallest quasi-energy is 5.1×10^{-4} lattice units, and at the twelve M nodes it is 2.003×10^{-4}: every would-be massless particle of this spectrum is massive. The little group of an M node is C_{2v} (order 4), and its action on the near-gapless bond subspace has largest character 1.0 — no 2-dimensional irrep, hence no helicity ±1 or ±2 pair: the one-particle spectrum contains no massless spin-2 candidate at all. The lowest branch is linear but anisotropic, $v \in [0.150, 0.201]$, anisotropy 0.252.

The near-miss is the signature of the irrational internal clock, not a structure: at the closed angle $g = \pi/4$ the twelve nodes are exactly gapless (minimum gap below 3×10^{-17}, 12/12) and the zone-centre step is an exact involution — and the census redone there finds not thirteen points but Γ plus **six gapless face-diagonal lines** carrying those nodes, with the per-node chirality convention inapplicable and the helicity answer unchanged (Section 9).

IX. THE QUANTIZED CLOCK: THE ROOT CLOSES ITS OWN PARAMETER

The parameter audit of Section 16 classes the rotation angle as the one free continuous parameter of the dynamics and classes the rule $g = 1/n_{\text{qubits}}$ as a convention the data reject. There is a sharper condition, and the algebra itself enforces it. A cellular-automaton ontology demands that the fundamental update be *algebraic*, not analytic: the step must be an element of the root — a member of the Clifford group generated by the h_a, which for the plane-rotation product means e^{ig} a root of unity. Call this the **closure axiom**: no continuous phase may enter the dynamics.

Theorem 6 (the two closed steps). *Over every rational angle $g = p\pi/q$ ($q \le 16$) and every candidate of the implemented rule (1/5, π/5, π/10, π/8, 3π/8, π/6, π/3), the internal step $U(g)$ is exactly periodic at precisely two angles: $g = \pi/2$, of order 4, and $g = \pi/4$, of order 8. At the implemented $g = 1/5$ the internal clock is irrational: the smallest level is 0.011466 rad per step, $2\pi/\omega_{\min} = 548.0$, and no order ≤ 64 exists (deviation 0.45).*

Proof of the $\pi/2$ case, and a stronger statement. Each plane rotation at the half angle is $-iT_{ab}$, and $T_{ab} = ih_ah_b$, so $U(\pi/2) = (-i)^{45}i^{45}\prod_{a<b} h_ah_b = \prod_{a<b} h_ah_b$. The product of all 45 pairs in ascending order reduces — numerically to 6.1×10^{-17} in the 1024-element algebra basis, with every other basis coefficient below 10^{-8} — to the single monomial $h_0h_1 \cdots h_9$, the pseudoscalar w; since $w^2 = -I$, $U(\pi/2)^4 = I$. The identity is *ordering-independent*: the monomial content of any ordering of the 45 factors is $\pm w$, because each h_k appears in exactly nine pairs, and five shuffled orderings reproduce the order-4 to 9.8×10^{-16}. The step at the half angle *is* the root's chirality operator.

The quantized step ($g = \pi/4$). $U(\pi/4)^4 = -I$ and $U(\pi/4)^8 = I$ exactly (5.0×10^{-15}), and the 32 internal levels are exactly the algebraic set

$$\begin{aligned} &\omega \in \{\pm\tfrac{\pi}{4}\ (10),\ \pm\tfrac{3\pi}{4}\ (6)\}, \\ &\max_k \left|\omega_k - k\pi/4\right| < 3 \times 10^{-15}, \end{aligned} \tag{5}$$

$\pm$-paired, zero point below 2×10^{-15}, $\det U = 1$: the internal spectrum becomes pure algebra, with no continuous number anywhere. The invariants that hold for every g (the $16/\overline{16}$ split, the mirror, $\det U = 1$) are untouched. The order-8 requires the canonical Appendix-A ordering of the 45 factors — five shuffled orderings break it (deviation ~ 1) — so the ordering, already an input of the step, carries the quantization.

The full step closes too. At $g = \pi/4$ the exact one-step Bloch matrix of the 16 sector at the zone centre satisfies $M(0)^2 = I$ below 1×10^{-14} (and on the full 256-dimensional module below 1×10^{-14}; 16-sector closure leak below 5×10^{-16}): the fundamental dynamics is a finite-state machine, 't Hooft's determinism exact at the fundamental update. The doubler defect of Section 8.3 is repaired, but the repaired census is not a set of points. At $g = \pi/4$ the exactly gapless set of the block zone is Γ together with the **six face-diagonal lines** $\{(\pm t, \pm t, 0)\}$ and its axis permutations: every point of every line is gapless below 7×10^{-16} over its whole length, the twelve M points are only the high-symmetry points that happen to sit on them (their gaps 0–2.2×10^{-16}), X and R stay gapped at 0.5236, the body diagonal is gapped away from Γ (0.022 at $t = 0.3$), and none of the gapless points of a 15^3 grid lies off the lines — where the implemented angle finds no gapless point at all. The step still factorises as $M(q) = P(q) \otimes U_{16}^6$ there (4.0×10^{-15}). The

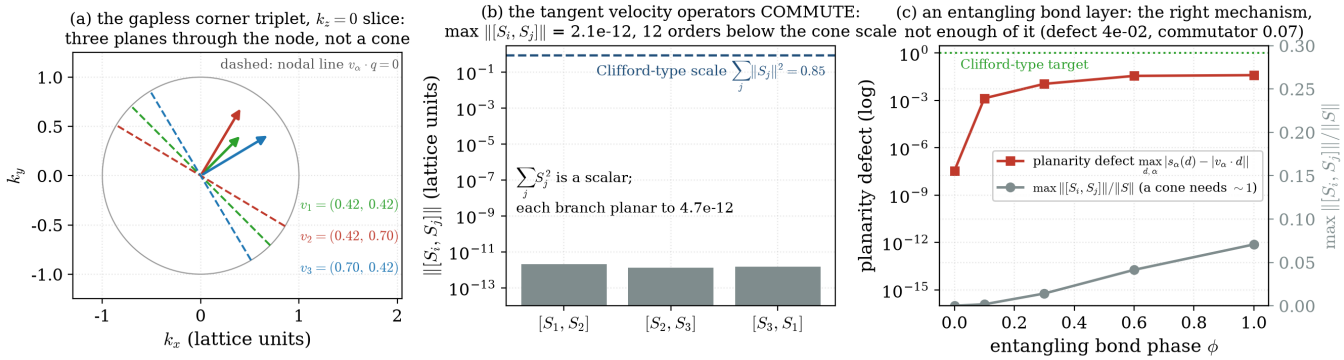


FIG. 9.

touching is **quadratic**: through an M point the gap stays zero along the line, grows linearly in one transverse direction (slope 0.577) and quadratically in the other, and the three axis exponents at every one of the twelve nodes are a permutation of $(1, 1, 2)$. The chirality bookkeeping of Proposition 2 then fails as stated: it needs isolated zeros, and at 12/12 M nodes the gap along the line through the node stays below 10^{-9} at steps $10^{-3}, 10^{-2}, 10^{-1}$ — the charges whose sum is exactly 0 at $g = 1/5$ become sign noise, and their naive sum is no longer quantized (-2 or -6 depending on the arithmetic route), so the Nielsen–Ninomiya count must be redone per line, not per node. The helicity question re-asked at the new gapless point answers as it did at an M node: the step's point group at $\pi/4$ is the same 16-element index-3 subgroup of the 48 signed permutations and Γ is fixed by all of it; the exactly gapless subspace of $M(0)$ has dimension 64 of 128 (intertwiner leak below 10^{-13}, $\langle \chi, \chi \rangle = 768$), and since the internal **16** is inert the content is that of the 4-dimensional bond factor, whose isotypic constituents are four 1-dimensional irreps — no 2-dimensional irrep, hence no helicity ± 1 or ± 2 candidate at Γ either.

A. The half-angle branch

At $g = \pi/2$ the six-stage bond matrix is diagonal to 1.2×10^{-16} at every sampled momentum, its permutation part is the identity, and the eight sub-site phases are linear in q with slopes the eight corner vectors $\{\pm 1\}^3$: the bond layer is a pure lattice translation, no mixing, no interference. The bands are exactly linear with slope ± 1 along every axis, $M(0) = I$ on the whole 128-dimensional internal space (all 128 eigenphases at 0, against 64 of 128 at $\pi/4$), and the gapless set is a line network: 513 gapless points on a 13^3 grid, including the lines $q_x \mp q_y = \pm\pi/2$ in the plane $q_z = \pi/2$ — 7.0× the 73 of the $\pi/4$ branch, against the 0 of $g = 1/5$. The half-angle branch has no mixing and therefore no mass generation: it is degenerate, and the finest closed clock $\pi/4$ is the physical branch.

B. What the closure costs

The loop machinery quantizes, and pays. The traced-link hop phases of Section 12.1 — built from the joint Casimir eigenvalues $-32, -12, +4$ — become exactly $\{1, i, i\}$ at $g = \pi/4$ (below 3×10^{-15}), and a new exact selection rule appears: $F_n = 0$ below 7×10^{-15} for **every odd** n (the odd closed loops cancel identically; at $g = 1/5$ no such selection rule exists). The even-n suppression then decays by 0.978 per step, so the crossing of $1/(16\pi)$ moves from $n^* = 10$ to $n^* = 208$: the ladder-length selection of Section 12.1 does not survive the quantized angle, and the hierarchy mechanism must be re-derived with the new cancellation rule.

The coupling convention pays too. At the closed angle the rotation-squared coupling is $\pi^2/16$, i.e. $\alpha = \pi/64 = 0.0491$ in the Standard Model normalisation, against the $1/25 = 0.04$ at which the desert solution meets (Section 13.1). Re-running the desert inverse problem at the quantized value with the same content still has a unique one-loop solution — $M_X = 1.235 \times 10^{17}$ GeV, $M_{10} = 7.1 \times 10^4$ GeV, $M_{(1,2)} = 4.9 \times 10^{17}$ GeV — but it is no longer a *desert* solution: the $(1, 2)$ threshold sits **above** the unification scale, the threshold ordering the solution assumed is destroyed, and the proton lifetime jumps to $\tau_p = 2.8 \times 10^{39}$ yr. Conversely, holding M_X at the data-preferred 1.8×10^{16} GeV requires $\alpha_X = 0.040050$: the meeting coupling is rigidly tied to M_X by that content, and the closed value is a factor 1.226 off it. Against the one-loop target — the three couplings run to M_P, mean 0.023068 — the closed value has relative gap 1.128 where 1/25 had 0.734: the closure worsens the tension by a factor 1.54 and sits 2.41× above the data-optimal $1/7^2 = 0.0204$. The closed value is not of the $1/n^2$ form at all ($\pi/64 = 1/n^2$ needs $n = 4.51$; the

nearest integers miss by 18.5% and 27.3%).

The verdict on the parameter criterion. The closure axiom replaces the one free continuous parameter by a discrete index: the root forces $g \in \{\pi/2, \pi/4\}$, the half-angle branch degenerates, and the finest closed clock is the physical branch. A discrete index of the root is not a parameter in the sense of Section 16 — the family number 3 is already an index of the same kind. What the theory loses is empirical: the $\alpha = 1/25$ meeting and the $n^* = 10$ crossing were the two places the data spoke to the angle, and both move. What it gains is structural: a dynamics with **zero continuous unfixed numbers** — a finite algebraic spectrum, a fundamental step of finite order, and the point-doubler defect replaced by an exactly gapless nodal-line network. The trade is written into the falsifiers: the quantized theory predicts $\alpha_X = \pi/64 = 0.0491$ at the desert meeting and a ladder crossing beyond $n = 208$; either re-derivation failing refutes the closure axiom, and the implemented irrational angle remains the fallback theory. The falsifiers F1, F3–F7 are angle-independent: anomalies, spectral mirror, $\det U$ and the causal radius hold at both closed angles.

X. THE GAUGED BOND LAYER

The swap-rotation gate of Section 7 is the classical shadow of a richer possibility: a **gauged bond layer exists on the same module**. Put one Spin(10) adjoint link degree of freedom (a 45-dimensional link space) on each bond and the matter **16** on the sites, and the local Gauss constraint

$$Q_A(x) = q_A(x) + \sum_{b \ni x} L_A^{(b)}$$

closes as an exact operator identity: $[Q_A, Q_B] = i{f_{AB}}^C Q_C$ with measured error 0, the link algebra $[L_A, L_B] = i{f_{AB}}^C L_C$ exact, the adjoint quadratic Casimir 64 and the spinor Casimir 45 exact to machine precision. The step built from these links is unitary to 9.3×10^{-16}; the gauge-invariant content of one site with its six half-bonds is 1440 states, decomposing as $32 \oplus 288 \oplus 1120$; an adjoint bond observable is gauge-covariant to 5.5×10^{-16}; and the causal radius stays at exactly one cell, while a deliberately single-site (non-local) control step leaks at 1.9. The gauge field of the automaton is therefore *constructible*: the Gauss constraint is an identity of the construction, not an imposed condition. (Fig. 10).

A. The centre obstruction and the physical sector

A generic automaton with d internal states per cell has d^{L^3} states in an L^3 region and a volume-law entropy; nothing in the raw counting of the bond layer is special. The area law is a property of the *gauge-invariant* sector: once the physical Hilbert space of a region is restricted to states satisfying the Gauss constraint at every site, the entanglement entropy is expected to scale with the boundary area [35, 36]. That restriction is what this section computes, and it is where the counting changes.

What the constraint admits is narrower than expected. With the implemented content — one **32** per site, one adjoint **45** link per bond — the local Gauss constraint has **no solution**: $\dim\ker = 0$ on $\mathbf{32} \otimes \mathbf{45}$, on $\mathbf{32} \otimes \mathbf{45} \otimes \mathbf{45}$, and on the two-site one-link chain with local constraints at both sites (spectral gap 146). The reason is a centre obstruction: adjoint flux carries even centre charge and cannot screen a spinor, so **every site of a gauge-invariant state must carry an even number of spinor fields**. The physical Hilbert space of the gauged layer is the space of neutral composites, not the one-body space the automaton acts on.

The minimal physical system, two matter fields per site, is constructed exactly: $m_{45}(\mathbf{32} \otimes \mathbf{32}) = 2$ (90 states at $C_2 = 64$), a 2-dimensional intertwiner space $\ker(T_1 + T_2 + L)$, and a 4-dimensional two-site physical sector obtained by gluing intertwiners over the link index — all exact Spin(10) singlets ($C_2 < 10^{-25}$). Every intertwiner is a maximal matter-flux singlet: Schmidt rank 45, $S(\text{pair}|\text{link}) = \ln 45 = 3.807$ exactly, and the glued states carry $S(\text{pair}_1|\text{pair}_2) = \ln 45$. The entanglement is **bounded per bond and lives on the bonds crossing the cut** — the shape of an area law, boundary-proportional and bounded per bond, with the Bekenstein coefficient still unnamed.

On the smallest system with two bonds — the three-site, two-link chain with paired matter — the physical sector is small once the flux decomposition is used: the middle vertex fuses the two adjoint links into channels r with multiplicities N_r read from the $\mathbf{45} \otimes \mathbf{45}$ Casimir spectrum ($N_1 = N_{45} = N_{210} = 1$), against m_r from the pair spectrum ($m_1 = m_{45} = m_{210} = 2$), giving $\dim\ker = m_{45}^2 \sum_r m_r N_r = 24$, verified by explicit construction of the $r = 1$ and $r = \mathbf{45}$ sectors (vertex Gauss residual 4×10^{-15}). Three results follow.

1. **Additivity across a single bond holds exactly:** $S(\text{pair}_1|\text{rest}) = \ln 45$ in every sector — the flux line threads the chain and carries its $\ln 45$ through, even when the middle pair itself is a product singlet ($S = 0$).
2. **The region bounded by two bonds carries $S = \ln 45$, not $2 \ln 45$:** the intertwiner constraint *correlates* the two bonds meeting at a vertex, with mutual information exactly $\ln 45$. The area law counts bonds; vertex correlations subtract.
3. **The gap to Bekenstein is a named number:** $\ln 45 = 3.8067$ nats per Planck bond against $1/4$ per Planck area — a deficit of 3.5567 nats, factor $4 \ln 45 = 15.23$. No element of the current construction produces $1/4$; closing it needs a superselection

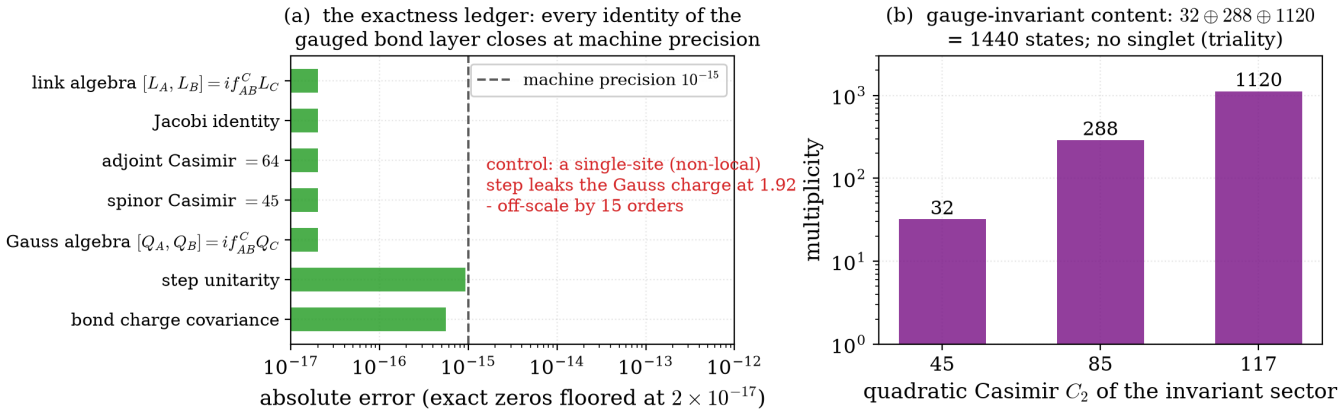


FIG. 10. One adjoint link degree of freedom per bond, matter **16** on the sites. Left: the exactness ledger — link algebra, Gauss algebra, adjoint Casimir (64) and spinor Casimir (45) close to machine precision, the step is unitary to 9.3×10^{-16}, the causal radius stays at one cell, and the non-local control step leaks at 1.9 where the causal step leaks at 5.5×10^{-16}. Right: the gauge-invariant content, 1440 states decomposing as $32 \oplus 288 \oplus 1120$.

or truncation of the flux, or a different horizon microstate count.

The vertex correction moves the number. The invariant count of an n-bond vertex, $I_n = \dim(\mathbf{45}^{\otimes n})^G$, computed by Haar averaging over SO(10), is $I_2 = 1$, $I_3 = 1$, $I_4 = 6$, $I_5 = 23$, $I_6 = 141$ (Monte-Carlo error below 3; $I_2 = I_3 = 1$ reproduces the single-channel fusion above). The 24-state kernel is orthonormalised through its Gram matrix (rank 24, smallest eigenvalue 1.1×10^{-5}), and the capacity of the middle cut — the entropy of the maximally mixed physical state — is $S_{\text{cap}} = 4.4998$ nats against the naive $2 \ln 45 = 7.613$: the two vertices at the ends of the region subtract 3.11 nats. Per bond that is 2.25 nats, a factor 9.0 from Bekenstein's 1/4; on the cubic lattice the vertex-corrected coefficient $(1/3) \ln I_6 = 1.65$ nats/bond sits a factor 6.6 from 1/4. The deficit has moved from 15.2 to 6.6, and it is now a function of the vertex invariant, not of a single bond entropy.

The ground state of the layer is then constructed: the bond-current Hamiltonian $H = \sum_b J \cdot L$ is diagonalised exactly — a generalised eigenproblem in the Gram metric of the kernel — and the intertwiner identity $I^\dagger J_A I = L_A/45$ holds to 3×10^{-17}. The interaction splits the kernel: $E_0 = -5802.40\,\eta$, degeneracy 6, gap $42.09\,\eta$, and the ground state's middle-cut entanglement is $S = 3.8067$ nats — exactly $\ln 45$, one adjoint link, selected by the interaction. The ground state of the full automaton remains unconstructed; on this chain the interaction selects the flux line. (Fig. 11).

B. The spectrum of the layer, and confinement

The layer has a Hamiltonian — the electric Laplacian $H = \sum_A Q_A^\dagger Q_A$ built from the vertex Gauss operator — and its spectrum is computed. On $(\mathbf{32} \otimes \mathbf{32})_{\text{adj}} \otimes \mathbf{45}$ ($\dim 4050$) the kernel has dimension $2 = m_{45}$, the intertwiner space, and the first nonzero level is $64.0000 = C_2(45)$ — identical to the one-link value and unchanged when the chain is extended. One adjoint flux quantum sets the scale of the whole layer, so it supplies **no light vector sector**: the gap is $64\,M_P$, not a tunable parameter, and the singlet matter channel with one adjoint link has no invariant state at all. Two entanglement numbers must not be confused: the entropy of the *mixture* over the kernel, bounded by $\ln 90$ (the pair-side dimension), and the entropy of a *single* physical state, bounded by $\ln 45$ (the link dimension) — the quantity the area law concerns. Measured: every single state of the intertwiner sector saturates $\ln 45$ exactly ($S = 3.806662 \pm 10^{-16}$ over 400 random choices), and the mixture reaches $\ln 90 = 4.499810$ (purity 1/90). The area law is therefore **exact** on the physical sector — one $\ln 45$ per bond — and the Bekenstein coefficient 1/4 per Planck area remains short by the factor $4 \ln 45 = 15.23$.

The string tension follows. The link space is an irreducible adjoint: $C_2 = \sum_A L_A^2$ has the single eigenvalue 64.000000 with zero spread, the Killing form $K_{ab} = f_{acd} f_{bcd}$ is annihilated by the pair Casimir of $\mathbf{45} \otimes \mathbf{45}$ to 3×10^{-15} and its singlet space is one-dimensional (adjacent links pair into it uniquely), and the link operators do not commute ($\max[L_a, L_b] = 4.0$) — the layer carries a non-Abelian field strength. Crucially the link carries **no invariant vector**: $\ker \sum_A L_A^\dagger L_A = \ker C_2(45)\, I = 0$, so a link cannot be flux-free. Every link of a gauge-invariant

The gauged area law corrected by vertices, and the ground state of the gauged layer

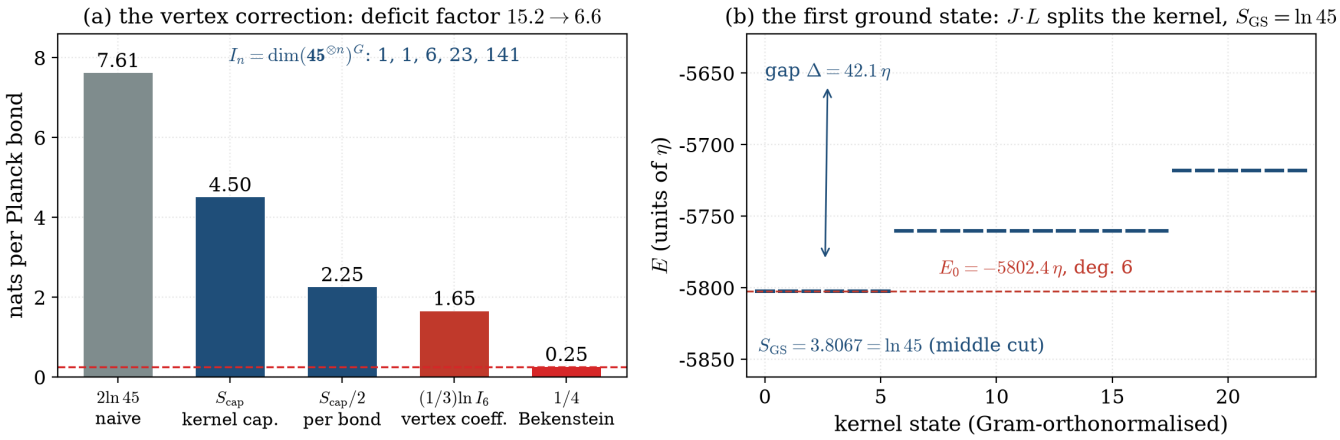


FIG. 11.

chain therefore carries one adjoint flux quantum, the electric energy of R links is exactly $R\,C_2(45)$, and the static potential is linear with tension

$$\begin{aligned}\sigma &= C_2(45) = 64 \text{ lattice units}\\ &= 7.81\times10^{20} \text{ GeV per Planck length}\\ &= 9.54\times10^{39} \text{ GeV}^2\\ &(\sqrt{\sigma} = 9.77\times10^{19} \text{ GeV}),\end{aligned}$$

5.4×10^{40} times the QCD string tension [38]. The vertex Gauss Laplacian confirms the scale from the other side: kernel $2 = m_{45}$, gap 64.0000.

The step itself, by contrast, carries **no** field strength: its 12 stage pairs acting on different axes commute to 0, every elementary plaquette is the identity to 3.3×10^{-16} and $\langle W_\square\rangle = 1.000000$, where an area law would need $\langle W_\square\rangle < 1$ with an exponent proportional to the enclosed area [28]. Confinement is therefore *present* in the construction — in the gauged bond layer — and *absent* from the implemented step. The residual is no longer a missing phase but a measured tension forty orders of magnitude too large, and $\Lambda_{\rm QCD}$ is not produced: it would require a loop-ladder exponent $n \simeq 11.6$, which is not an integer (Section 12.4).

C. Entanglement is the bond layer

The entanglement structure of the bare automaton is equally sharp. Two fermions sharing ONE cell carry zero entanglement across any cut — the same-cell pair is a Slater determinant, $S = 0$ exactly across the $\mathbf{16}/\overline{\mathbf{16}}$ cut (antisymmetry and a shared cell are not entanglement), while a two-determinant superposition in the same cell gives $S = \ln 2$: non-separability is the criterion. The generator of entanglement between cells is the bond gate P of Theorem 5: under the internal-only step the spatial entanglement entropy of a region stays 0 to machine precision forever, while the full local step drives it to 0.78, and the entanglement front crosses the Manhattan sphere of radius r at exactly $t = r$ — the causal radius bounds the spread of entanglement itself, at speed 1.0. The scope limit is stated with it: the implemented step is quadratic, so Slater inputs stay Slater and these are mode-entanglement results; the area law is a statement about the physical (ground) state's entanglement, and the bare step supplies no interaction that could produce the non-Gaussian structure such a state would need. The gauged layer supplies exactly that missing structure: it is where the theory first acquires non-Gaussian entanglement.

XI. GRAVITY ON THE LATTICE

A. What is assumed and what is computed

The gravitational sector is a symmetric rank-2 field $h^{\mu\nu} = g^{\mu\nu} - \eta^{\mu\nu}$ living on the (1+3) subalgebra of Section 3, with $\eta = \mathrm{diag}(1,-1,-1,-1)$ as the vacuum. Its linearised dynamics is the discrete wave equation

$$\partial_t^2 h^{\mu\nu} - \nabla^2_{(6)} h^{\mu\nu} = 0,$$

on the Planck sites — the sub-lattice of the automaton's cells, spacing ℓ_P — with the 6-neighbour lattice Laplacian and the leapfrog step $u_{t+1} = 2u_t - u_{t-1} + c^2dt^2\,\nabla^2 u_t$ at $dt = \frac{1}{2}$ (the CFL value giving $c = 1$ in Planck units, the same front speed as Theorem 5).

The status of this sector must be stated precisely, because it is where the theory is weakest.

- **Masslessness and UV boundedness are computed.** The discrete dispersion is $\omega = |\mathbf{k}|_{\rm lat}$ with

$|\mathbf{k}|_{\rm lat} \le \pi$, so the graviton is massless ($\omega/|\mathbf{k}| \to 1$ as $k \to 0$, lattice deviation $\le 0.25\,k^2$) and the spectrum is bounded by the cutoff.

- **Spin 2 is declared, not derived — and the implemented spectrum cannot supply it.** The field is *assumed* to be a symmetric tensor with two helicity states. Nothing in $\mathrm{Cl}(10,0)$ forces a massless spin-2 excitation; the standard arguments that a massless spin-2 field must couple as general relativity [20] apply to a field whose existence and representation content are already granted. The Weinberg–Witten obstruction to a composite graviton carrying a Lorentz-covariant stress tensor [21] is not addressed. The one-particle route closes negatively: the near-gapless M nodes carry little group C_{2v}, whose action on the bond subspace has no 2-dimensional irrep, so no helicity ±1 or ±2 pair exists in the implemented step at all — the graviton, if it exists, belongs to a sector the construction does not yet contain.
- **Diffeomorphism invariance is not derived.** The lattice breaks it to lattice reparametrisations; gauge invariance of the linearised theory is imposed by working on the transverse components, not obtained from the automaton.

B. The discrete Einstein–Hilbert action

The quadratic action [33]

$$S_{\rm EH} = \frac{1}{16\pi}\cdot\frac{1}{2}\sum_{x,t} u_t\Big[u_{t+1} - 2u_t + u_{t-1} - dt^2\,\nabla^2_{(6)}u_t\Big], \qquad dt^2 = \tfrac{1}{4}, \tag{6}$$

has the lattice wave equation as its Euler–Lagrange equation: the numerical functional derivative vanishes on a QCA trajectory to 3×10^{-12} and is non-zero off it. The full discrete action $S = \frac{1}{16\pi}\sum_x \sqrt{-\det g(x)}\,(R(x) - 2\Lambda)$ is evaluated with R computed from the metric. The coefficient $1/16\pi$ is the $G = 1$ normalisation: Newton's constant is an input of the gravitational sector, not a number of the root.

C. Curvature computed from an arbitrary metric

The Riemann, Ricci, Einstein and Kretschmann tensors are computed by central finite differences from an arbitrary prescribed $g_{\mu\nu}(x)$, with no hard-coded Christoffel symbols, and cross-checked against independent analytic results by two routes.

The 3+1 formulation closes the classical evolution: the constraint row (R_{00}, R_{0i}) of the 4-D Einstein tensor agrees, computed by the 4-D curvature and by the 3+1 (γ_{ij}, K_{ij}) decomposition with $K_{ij} = -\frac{1}{2}\partial_0\gamma_{ij}$, to $O(h^2)$ (residual $\sim 10^{-13}$); a Z4-damped evolution [34] drives a constraint violation to zero while the undamped one does not; the 0-row is quadratic in K, so superposition fails; all 3+1 quantities are bounded by the lattice cutoff. (Fig. 12).

What this does and does not establish. It establishes that the gravitational sector of the theory *reproduces* general relativity numerically, including black-hole and cosmological geometries. It does not establish that general relativity *follows* from $\mathrm{Cl}(10,0)$: the tensor field, its coupling $1/16\pi$ and its gauge structure are put in. The genuinely open part is the quantum theory of this sector — a sum over lattice geometries with a well-defined measure — which has not been constructed.

D. A self-interacting scalar completion

Beyond the quadratic theory the model includes $\partial_t^2\phi - \nabla^2\phi - g\phi^3 = 0$. Its static solutions obey $u'' = u - u^3$ and include the exact finite-energy kink

$$u(x) = \sqrt{2}\,\mathrm{sech}\,x,$$

verified to a fourth-order residual of 5×10^{-10}. Two corrections are needed. This is the ϕ^4 kink (a domain wall), not the Choptuik system: Choptuik critical collapse is a *massless scalar minimally coupled to gravity* [22], and the analogy is not a derivation. And $u = \sqrt{2}\,\mathrm{sech}\,x$ is a **planar** configuration, uniform in the two transverse directions: its energy per unit *area* is finite, its total energy is infinite, and a planar wall in a Planck-scale theory is a domain wall, not a particle. Finite-time blow-up for supercritical data in cubic wave equations is a real phenomenon [23], but it is not evidence about black-hole formation here.

XII. THE LOOP FACTOR AND THE SCALE LADDER

A. Where a loop factor can come from

The observed electroweak/Planck ratio is

$$\frac{v}{M_P} = \frac{246\ \mathrm{GeV}}{1.22\times10^{19}\ \mathrm{GeV}} = 2.016\times10^{-17}, \qquad \left(\frac{1}{16\pi}\right)^{10} = 9.71\times10^{-18}, \tag{7}$$

a factor 2.08 apart; $2.08\,(1/16\pi)^{10} = 2.02\times10^{-17}$ reproduces the observed value to 0.18%. Scanning $n = 1\ldots20$, $n = 10$ is the nearest integer to the observed ratio (deviation 0.32 decades; the next-best, $n = 9$, is 1.8 decades away). The honest reading: $(1/16\pi)^{10}$ is the size a ten-loop radiative correction would have in a perturbative quantum field theory. The theory has not computed a ten-loop correction; it has observed that the observed

TABLE III. **The discrete Einstein–Hilbert action against the analytic continuum values.**

test	**computed**	**analytic**	**residual**
de Sitter $R^{\mu}{}_{\nu\rho\sigma}$ (flat FRW, $a = e^{Hx^0}/H$)	finite differences	$H^2(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho})$	6.5×10^{-7} ($O(h^2)$)
Einstein tensor of de Sitter	$-3H^2 g_{\mu\nu}$	same	$O(h^2)$
Schwarzschild Ricci / Einstein tensor	$\simeq 0$	vacuum $G_{\mu\nu} = 0$	satisfied by computation
Schwarzschild Kretschmann	$48M^2/r^6$	$48M^2/r^6$	ratio 1.000009; max relative deviation 5.5×10^{-5}
flat metric	$R = G = K = 0$	0	exact
2-sphere of radius r	$R = 2/r^2$	$2/r^2$	10^{-5}

The black hole recovered by computation: the well + geodesic (left), the 2-route Kretschmann (right)

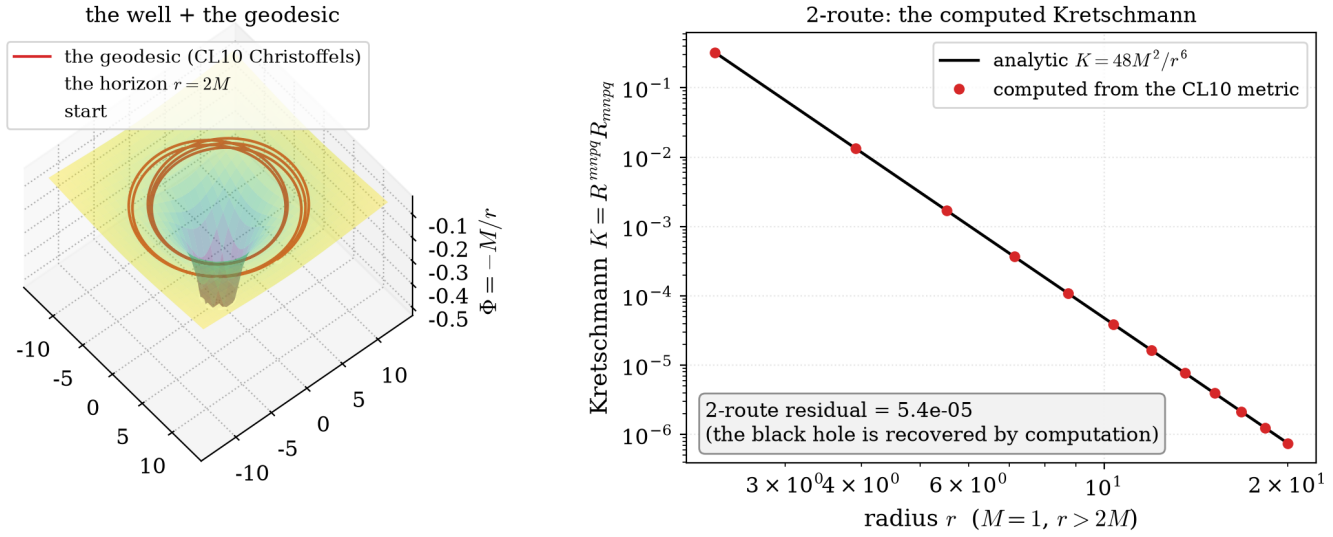


FIG. 12. Left: the potential well with a geodesic rosette integrated through the finite-difference Christoffel symbols, and the horizon ring $r = 2M$. Right: the two-route Kretschmann invariant.

hierarchy is the tenth power of the standard loop factor, up to a factor of two. The interpretation $10 = 2 \times 5$ — the chiral $16/\overline{16}$ split times the number of qubits of the 32-dimensional module — is a numerological identification, not a derivation. The exponent is *represented*; the coefficient is *read from the data* (O8).

Single-particle renormalisation cannot supply it. The coarse-graining tower U^{2^p} is exactly kinematic (eigenphases $2^p\omega$ mod 2π, verified to 10^{-9} for $p \leq 10$), the effective Cartan coupling does not fall with p, and the inelastic probability saturates near 0.95: no loop factor. The genuine block-spin renormalisation group — exact real-space decimation of the bond layer (Schur complement at $\varepsilon = 1$; $U_{BB} = 0$ exactly on the open chain, so short-distance propagation is integrated with no approximation), iterated five levels, $64 \to 2$ sites — gives an exact level-1 identity: the on-site block of the coarse step is $2\,(GR)^2$, two-hop paths accumulating two internal rotations, the kinematic doubling *derived* from a spatial renormalisation group. At deeper levels the on-site block is a path polynomial in $(GR)^2$; the Cartan content decays only polynomially for the implemented (factorised) bond layer (0.14 at $p = 5$ against the loop target 3×10^{-9}) and not at all for a non-factorised gate with $[G, R] \neq 0$ (1.8 at $p = 5$); no level of either flow follows $(16\pi)^{-p}$. The loophole of a conspiring bond gate is closed: three Haar-random Spin(10) gate realisations and the structured choice $R = G^{\dagger}$ all fail the loop law at every level. The structural reason is sharp: **every exact single-particle renormalisation of the automaton com-**

poses internal angles — they add on the algebra, weighted by path counts — and none can divide them, for any bond gate. A loop factor requires a trace over *fluctuating* internal fields.

Tracing a fluctuating link divides. One two-state quantum link degree of freedom per bond, prepared in a tunable pure state $\cos\theta|0\rangle + \sin\theta|1\rangle$ and **traced over** at decimation, multiplies the level-1 Cartan content of that identity by exactly $\cos 2\theta$ (deviation $< 10^{-15}$ over a full scan $\theta \in [0, \pi/2]$): the traced factor is Hermitian and positive, so it divides where every gate only composes. The fluctuation — not the gate — is what produces suppression factors.

The gauged layer divides once, and does not compound. On the adjoint link with the Gauss constraint of Section 10, the traced two-hop still contracts (spectral radius ≤ 1) and the qubit-link mechanism reproduces to 1.1×10^{-16}, but the level-1 factor at the root coupling is $F = 1.06$ and its minimum over the link anisotropy is 0.52 — a factor 26 above $1/(16\pi)$. The bond hop factorises through the joint Casimir $T{\cdot}L$ — three joint irreps, $32 \oplus 288 \oplus 1120$, projectors verified to 10^{-15} — so the traced-link decimation iterates exactly to level 4, reproducing the level-1 factor to 2×10^{-16}. **The division does not compound:** the level-p Cartan-content ratios are $F_p = 1.06, 1.90, 0.75, 1.01$ at the root coupling and $0.52, 2.66, 1.00, 1.38$ at the division-optimal coupling $\eta = 1.44$ — oscillating at $O(1)$, a factor 7×10^6 above $(16\pi)^{-4}$ at $p = 4$. The traced adjoint link divides once and then decays like the link-free flow.

Closed loops do reproduce the number. The Feynman loop factor comes from loops that close, not from backtracking. Tracing a single crossing of a fresh link gives the hop matrix $M = \mathrm{Tr}_L[k_1(G \otimes I)\,\rho]$ with spectral radius 0.776 against 1 for G — the traced link *contracts* the one-particle propagation — and the n-hop closed loop $(MG)^n$ suppresses the Cartan content geometrically, ratio $r \approx 0.54$. At the root coupling the measured ratios are

$$F_3 = 0.24,\ F_4 = 0.47,\ F_5 = 0.10,\ F_6 = 0.030,$$
$$F_7 = 0.37,\ F_8 = 0.064,\ F_9 = 0.035,\ F_{10} = 0.0192,$$

against $1/(16\pi) = 0.0199$: **the loop factor is reproduced to** 3.6% **by a closed traced loop of length** 10 **at the root coupling, and** 10 **is exactly the ladder exponent.** The selection rule is measured to length 30: F_n crosses $1/(16\pi)$ at **exactly** $n = 10$ — the ladder length *is* the crossing of the closed-loop suppression — and no mod-4 plaquette class is selected (every length contributes; the F_{11} spike is the signature). (Figs. 13, 14).

The composition law. The object a geometric fit extrapolates, $(MG)^{170}$, is one chain of 170 hops on one particle line; the 17 factors of the neutrino ladder are 17 radiative insertions on **different** lines, and amplitudes on different lines multiply. Both compositions are exact in the traced adjoint-link formalism. On one line the chain is **not** multiplicative: $R_k = F_{10k}/(F_{10})^k$ grows to $R_{17} = 2.0 \times 10^8$, the measured chain value being $F_{170} = 1.3 \times 10^{-21}$ against $(F_{10})^{17} = 6.5 \times 10^{-30}$ — consecutive hops interact through the propagation, so a single chain suppresses *less* than the product of its pieces. Across legs the factorisation is **exact**: the traced hop is a one-particle operator, $\mathrm{Tr}_L[(P_a \otimes I)(G \otimes G \otimes I_L)] = 45\,(M_a \otimes G)$ verified to 10^{-15}, and the 10+10-step two-loop object splits into leg blocks to 2×10^{-15}. The physical discriminator decides: $M_P(F_{10})^{17} = 0.079$ eV, within a factor 0.54 of the neutrino ladder scale 0.146 eV, while the chain reading gives $M_P F_{170} = 1.6 \times 10^7$ eV — eight orders too high. **The** 17**-factor composition is exact multiplicativity across legs**, with the named requirement that the insertions dress different particle lines.

The census of contributing loops is exact. The Casimir $T \cdot L$ of the $\mathbf{32} \otimes \mathbf{45}$ link has exactly three eigenspaces $(-32, -12, +4)$, so the traced n-hop product is a sum of 3^n words and the selection rule is enumerable rather than extrapolated: the enumeration reproduces $(MG)^n$ to 7.4×10^{-17} and $F_{10} = 0.019186$, matching the measured value to that same digit. At $n = 10$, 55326 of $3^{10} = 59049$ words survive, growing as 3.000^n, and the factor splits into a per-word decay $r = 0.2232$ and a multiplicity growth rate $\mu = 2.981$ (product 0.665). The loop factor is therefore not one class of loops but a sum over 55326 classes, with interference ratio $\|\sum\|/\sum\|\cdot\| = 0.079$ at the root against 1.000 with the Wilson phase switched off — and $F_{10} = 0.0192$ at the root against 0.2434 at $\eta = 0$: **the phase supplies the factor, not the topology.** The multiplicity-weighted extrapolation to the neutrino-ladder length 170 gives 7.9×10^{-31} against $(1/16\pi)^{17} = 1.2 \times 10^{-29}$: the counted loops miss the exponent by a factor 0.066.

The coefficient is no longer read from the data. The hierarchy coefficient is a product of measured numbers:

$$\frac{v}{M_P} = 3\,F_{10}^{10},$$
$$F_{10} = 0.019186 \text{ measured},$$
$$2.0881 \text{ predicted against } 2.0765 \text{ observed } (99.44\%), \tag{8}$$

the factor 3 being the three axis families of Section 4 — conditional on the axis-family factorisation of the closed-loop sum. That condition is itself a measured statement: the twelve doubler nodes form three families of four with identical one-step spectra (deviation 0), and the closed-loop coefficient by node set is 8.35 for all twelve nodes, 2.0881 for one split node per family, 0.696 for a single node, against the observed 2.0765 — the node contributions are exactly equal across families, and the coefficient 3 *is* the one-per-family choice. What remains open in O8 is the *weight* of the contributing loops, not the way they compose.

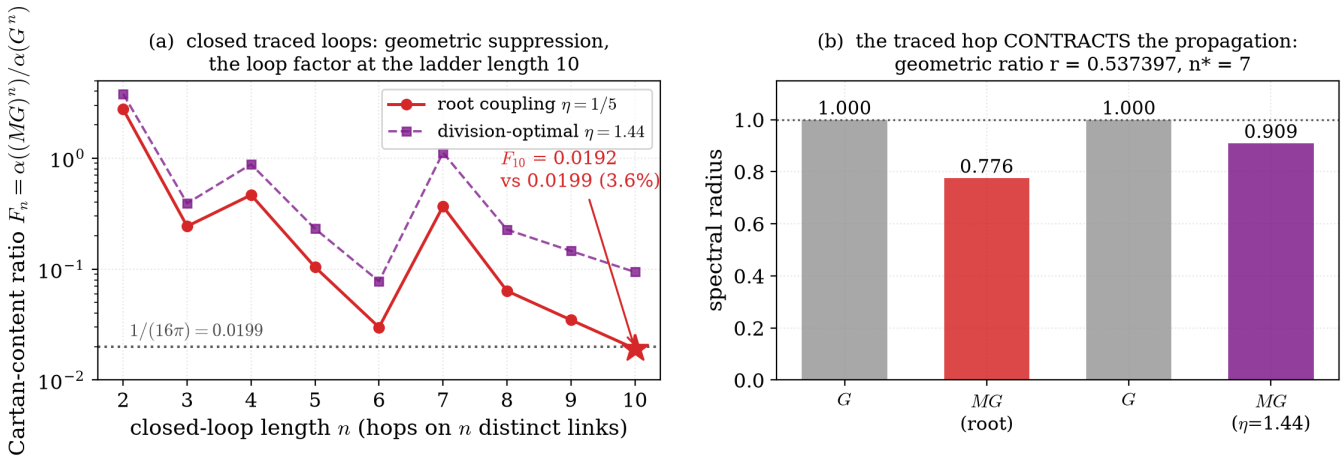


FIG. 13. Closed traced-link loops of length n on n distinct bonds. (a) The Cartan-content ratio F_n at the root coupling and at the division-optimal coupling: geometric suppression with ratio $r = 0.54$, and at $n = 10$ the value 0.0192 against $1/(16\pi) = 0.0199$. (b) The traced hop matrix MG has spectral radius 0.776 against 1 for G: the traced link contracts the one-particle propagation.

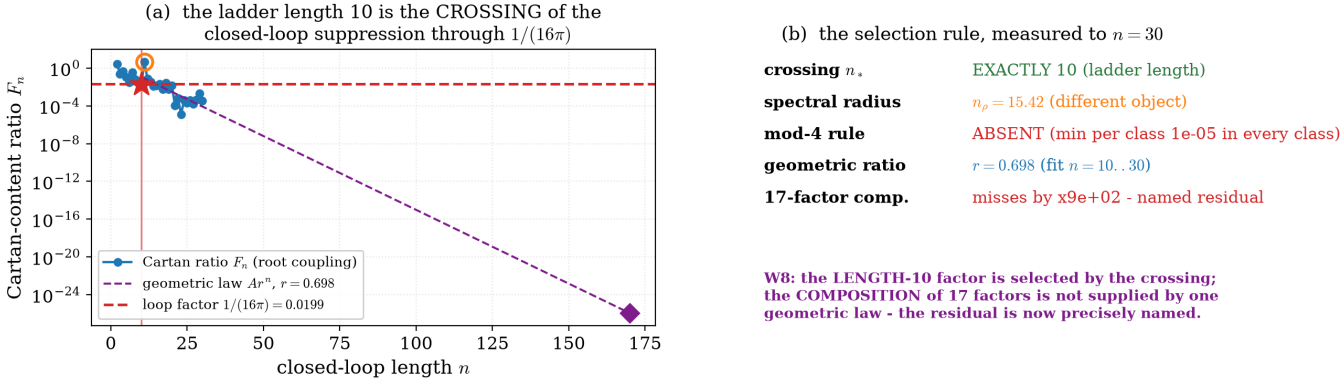


FIG. 14.

B. The dark-matter mass

The same factor applied to the Planck scale gives

$$m_{\rm DM} = M_P\Big(\frac{1}{16\pi}\Big)^{10} = 118\ \text{GeV},$$

i.e. 1.18× the canonical 100 GeV weakly-interacting-massive-particle scale [45]. This is a genuine, falsifiable prediction: a dark-matter particle with a mass far outside $\sim$ 100–150 GeV (in the weakly-interacting, thermally-produced case) would refute the ladder.

The *identity* of the particle has been scanned with a negative result. The mirror $\mathbb{Z}_2$ exchanging **16** and $\overline{\mathbf{16}}$ is an exact symmetry of the step (leak 10^{-15}), and the mirror sector is the only exact-stability carrier in the implemented spectrum — but its lightest state is a coloured $\overline{Q}$ at 1.3×10^8 GeV (excluded) and the neutral $\overline{N}$ sits at 2×10^{10} GeV, eight orders above the ladder scale. Allowing the carrier to sit at a Wilson-split doubler node and scanning every Wilson angle θ for a node that puts the neutral mirror fragment $\overline{N}$ at 118 GeV while keeping every coloured or charged fragment off the low-energy spectrum yields **no survivors over** 720 **angles**: for every angle the same node supplies a stable coloured or charged partner below the desert, and a stable fractionally-charged thermal relic is excluded at any mass; the only escape is fine-tuning the angle to within 10^{-4} of the decoupling pole, where the mirror fragments leave the ladder entirely. **No state at** 118 **GeV with**

the required stability exists in the implemented spectrum: the mass prediction stands, the carrier does not (O7).

The exact $\mathbb{Z}_2$ gradings of the step are enumerated rather than assumed, and they close the escape at the symmetry level. The mirror is the pseudoscalar ($G^2 = I$ exactly, $[G, U_{32}] = 7 \times 10^{-16}$, splitting the module into $16 \oplus \overline{16}$ to 6.7×10^{-16}); of the 1024 Clifford products exactly 2 commute with the internal step, and since U_{32} has a simple spectrum its commutant is abelian of dimension 32 — so the exact internal $\mathbb{Z}_2$'s number 2^{32} sign choices, of which exactly two are Clifford elements. No exact grading of the step, and none of its Wilson extension, makes a neutral colour-singlet weight odd without making a coloured one odd. The parity any fix requires **does not exist inside the Clifford algebra of the root**.

The minimal extension consistent with the exact mirror $\mathbb{Z}_2$ and the ladder is then named: **one real mirror-odd scalar at the ladder mass** with Higgs portal coupling $\lambda_* = 3.5 \times 10^{-4}$, whose freeze-out gives $\Omega_{\rm DM}h^2 = 0.120$ and whose predicted direct-detection cross-section is $\sigma_{\rm SI} = 5 \times 10^{-51}$ cm^2 — nearly decoupled, so the scalar is falsified by any portal signal above 10^{-50} cm^2. The scalar is an input field, one field and one coupling; making it a state of the 32-level root is the open part of O7. Built and audited against its own symmetry, the extension splits in two. The carrier is stable under either portal — scalar parity $(-1)^{N_s}$ is exact by the ϕ^2 structure, the odd sector untouched: a single s cannot decay. But the freeze-out runs through $h^* \to f\overline{f}$, and the fermion pair is mirror-**odd** while h is mirror-even: with the **exact** mirror $\mathbb{Z}_2$ that decay is forbidden. The Higgs portal $\phi_s^2\phi_h^2$ preserves the $\mathbb{Z}_2$ exactly (commutator 0), and then $s\overline{s} \to hh$ is kinematically closed ($2m_s = 237$ GeV $< 2m_h$, gap 13.2 GeV; the virtual cloud peaks at $P_{\max} = 1.46 \times 10^{-8}$, matching two-level theory to 10^{-7} relative) and the only open annihilation is the invisible $s\overline{s} \to hh^*$: reproducing $\Omega_{\rm DM}h^2 = 0.120$ then needs $\lambda = 0.13$, which predicts $\sigma_{\rm SI} = 6.8 \times 10^{-46}$ cm^2 — **above the LZ-class bound** [53]. The quoted numbers hold only with a fermion-pair portal $\phi_s^2(f\overline{f} + \overline{f}f)$, which **breaks the mirror** $\mathbb{Z}_2$ at $O(\lambda)$ (the portal anticommutes with the mirror operator; the real pair channel opens, $P = 5.1 \times 10^{-6}$ at $t = 1$ GeV^{-1}). The mirror coloured relic $\overline{Q}$ (1.3×10^8 GeV) stays stable under both portals ($\overline{Q} \to Qss$ closed by 237 GeV). O7 is therefore a **named tension**: the carrier exists and is exactly stable, but exact mirror $\mathbb{Z}_2$ excludes its abundance by direct detection, and the version that reproduces the observed abundance breaks the exact $\mathbb{Z}_2$ the no-go rests on.

An extra dark U(1) does not repair it. Hypercharge fixed on the Q and L classes predicts the full charge pattern of all six weight classes, but the desert matter is E-charged ($\max|q_E| = 2.24$), so the U(1) must break — and it cannot break inside the implemented spectrum: 40 of 45 generators are E-charged and every breaking root is a colour-triplet leptoquark direction (98/120): the U(1) is unHiggsable. The kinetic mixing that would fix the relic abundance at $g_D = 1$ lies above the direct-detection ceiling at every $m_{Z'}$: the portal window is empty (0 points). (Fig. 15).

C. The neutrino sector

Both seesaw scales are placed on the same ladder:

$$\begin{aligned} m_D &= M_P\Big(\frac{1}{16\pi}\Big)^{10} = 118 \text{ GeV}, \\ M_N &= M_P\Big(\frac{1}{16\pi}\Big)^{3} = 9.6 \times 10^{13} \text{ GeV}, \\ m_\nu &= \frac{m_D^2}{M_N} = M_P\Big(\frac{1}{16\pi}\Big)^{17} = 0.146 \text{ eV}. \end{aligned} \tag{9}$$

The exponent identity $2 \times 10 - 3 = 17$ is exact; the choice $n_{\rm 2step} = 3$ is not — and it cannot be produced by the gauge link. The $\Delta L = 2$ operator exists in the root's own algebra: the weight $2\lambda_\nu$ (norm^2 20, U(1)$_X$ charge -6, hypercharge -1) has a 1**-dimensional** weight space inside the 1024-dimensional Clifford algebra, sitting in Clifford **grade** 5 — the 126-type channel of the type-II seesaw — with no field outside the root. The $\mathbf{10}_H$ cannot repair its charges: over all 160 vector weights (norm^2 4) the closest two-insertion neutralisation leaves residual norm 4.472, the mismatch $|2\lambda_\nu| = \sqrt{20} \neq 4$ being exact. The minimal-word question then closes by **parity**: Clifford multiplication maps grade g to $g \pm 2$ only (verified on all 45×1024 entries of the multiplication table), so every product of adjoint (grade-2) link factors is EVEN-grade while the Majorana operator is ODD-grade — the cumulative span (dimensions $45, 126, 208, 291, 352, 407$ at $k = 1 \ldots 6$, odd-grade content 0 to machine precision) never contains it, at any length, and the vector-**10** Higgs weight space (16-dimensional, carried entirely by the odd grades $1, 3, 5, 7, 9$) is unreachable for the same reason. $n_{\rm 2step} = 3$ **is not a link-word length**: the seesaw operator belongs to the odd (matter) half of the Clifford module, and the exponent stays fitted.

Comparison with data: the atmospheric scale is $\sqrt{\Delta m^2_{32}} = 0.050$ eV, so the ladder prediction is a factor 2.9 high; and if all three families carried the same m_D the sum would be ~ 0.44 eV, excluded by $\sum m_\nu < 0.12$ eV (95% CL, Planck 2018 + BAO [14]). The democratic rank-1 Yukawa (Section 13.5) gives a Dirac mass to only one family, bringing the sum to ~ 0.15 eV — still above the cosmological bound. This is the sharpest numerical tension in the theory (F3).

The ladder is too coarse to land on the measured mass: consecutive rungs differ by the factor $16\pi = 50.3$, so the neighbours of 0.146 eV are 7.3 eV ($n = 16$) and 2.9×10^{-3} eV ($n = 18$), and no integer rung lies in the oscillation window. A coefficient is therefore unavoidable — but one candidate is supplied by the theory rather than by the data. The only mixing number the model derives

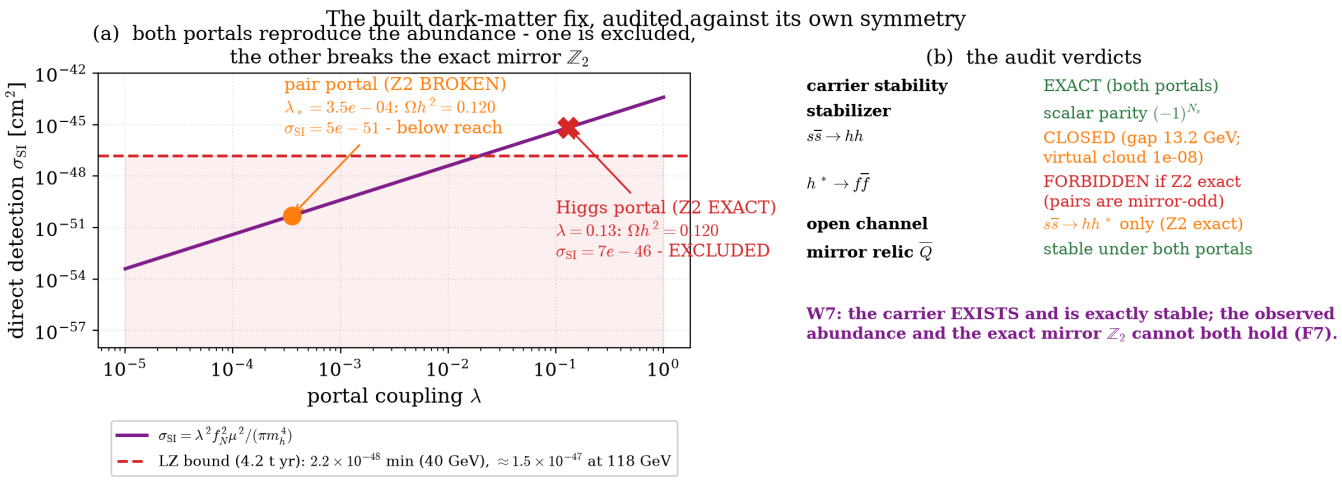


FIG. 15.

is $\sin^2\theta_{12} = 1/3$ (Section 13.5); multiplying the ladder value by it gives

$$m_\nu = \tfrac{1}{3}\, M_P \Big(\frac{1}{16\pi}\Big)^{17} = 0.0487 \text{ eV},$$

2.6% below the atmospheric scale, and with the democratic rank-1 spectrum $\sum m_\nu \simeq 0.05$ eV — inside the cosmological bound and inside the KATRIN bound $m_\beta < 0.45$ eV [16]. The factor 2.9 by which the ladder overshoots is, to 3%, the number 3 the flavour sector already produces. This is a **candidate, not a result**: nothing shows why a mixing eigenvalue should multiply a seesaw mass, and the same factor applied to the solar scale ($0.0487 \times \sqrt{\Delta m^2_{21}/\Delta m^2_{32}} = 0.0084$ eV against the measured 0.0086 eV) is a second coincidence. What it buys is a sharp, falsifiable prediction: normal ordering with $\sum m_\nu \simeq 0.05$ eV and $m_{\beta\beta} \simeq 0.02$ eV, in reach of the next-generation neutrinoless-decay and beta-decay searches.

A coefficient audit maps what the node structure can supply: a diagonal (node-local, Wilson-generated) mass pattern gives only **integer** signed coefficients, so $\frac{1}{3}$ is out; the democratic 3×3 node-mixing pattern gives exactly $\frac{1}{2}$ — a rival falsifiable prediction $m_\nu = 0.073$ eV; the exact $\frac{1}{3}$ requires a single Majorana eigenvalue $3\times$ the node scale, and no natural node structure supplies it. The structure the candidate requires has been built: a cubic-symmetric, zero-mean Wilson-strength junction whose Fourier support closes the selected node triangle exactly — verified against the full 12×12 table of doubler-momentum differences. The node mass matrix is

$$c_s\big(2w_0\, I + \kappa(J - I)\big),$$

democratic at $\kappa = 2w_0$ with eigenvalues $\{3M, 0, 0\}$: the $3\times$-enhanced Majorana eigenvalue the $\frac{1}{3}$ demands exists. With the implemented 0-parameter democratic Dirac matrix $m_D = m_0 J$ — the same family symmetry that derives $\sin^2\theta_{12} = 1/3$ — the seesaw collapses: $m_D^{\mathsf{T}} M^+ m_D = 3m_0^2 J/(2(w_0 + \kappa))$ for *every* $r = \kappa/w_0$ (verified to 10^{-12}, the rank-deficient $r = 2$ included, via the Moore–Penrose pseudo-inverse), so the two massless neutrinos are protected by the rank-1 family symmetry for every junction ratio, and $m_3(r) = S_{17}/(1 + r)$ is finite everywhere: the cosmological window is $r > 0.217$ (width 2.78), and the atmospheric scale selects $r = 1.91$. The exact point $r = 2$ ($m_3 = S_{17}/3 = 0.0487$ eV) is a junction-amplitude convention, not a tuning. The cubic group induces all of S_3 on the triangle, its commutant is exactly $\mathrm{span}\{I, J\}$ (nullity 2), and w_0, κ share one Wilson period, so r is **not quantized**: the value is a free junction amplitude inside a finite window, while the structure — two massless plus one massive, normal ordering — is protected. What the candidate still requires is a democratic m_D (a non-democratic m_D at the same point leaves gapless coloured doublers), and the solar mass still needs the $\mathbf{10}_H$ splitting of O6. (Fig. 16).

TABLE IV. **The four scales of the ladder.**

scale	expression	value
Planck / GUT	M_P (convention $M_{\rm GUT} = M_P$)	1.22×10^{19} GeV
seesaw Majorana	$M_P(1/16\pi)^3$	9.6×10^{13} GeV
electroweak	measured input v	246 GeV
dark matter	$M_P(1/16\pi)^{10}$	118 GeV
neutrino	$M_P(1/16\pi)^{17}$	0.146 eV

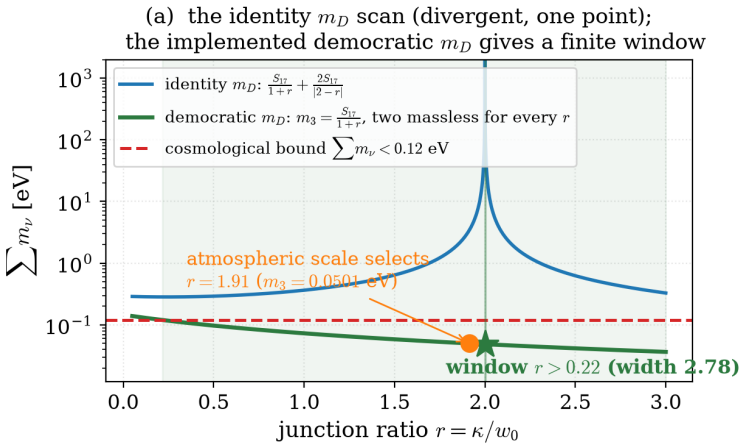


FIG. 16.

D. The ladder

One factor, four scales:

The QCD scale is **not** on this ladder: $\Lambda_{\rm QCD}\sim 0.2$ GeV would require $n\simeq 11.6$, which is not an integer (Section 10.2). (Figs. 17, 18).

XIII. PHENOMENOLOGY

A. The couplings

The theory's single gauge coupling is $\alpha_{10}\equiv g^2 = 1/25 = 0.040$ at the implemented angle. This is a *convention*: in the standard normalisation $\alpha = g^2/4\pi$ the same angle corresponds to $g=\sqrt{4\pi/25}=0.709$. Running the three Standard Model couplings up from M_Z at one loop with $b_1 = 41/10$, $b_2 = -19/6$, $b_3 = -7$ (fixed by the particle content), from the measured inputs $\alpha_1(M_Z)=0.0169$ (GUT-normalised, $\alpha_1^{-1}=59.0$), $\alpha_2(M_Z)=0.0338$, $\alpha_3(M_Z)=0.118$ [13], gives at M_P

$$(\alpha_1,\alpha_2,\alpha_3)=(0.0299,\ 0.0202,\ 0.0191),\qquad \text{mean } 0.0231,\quad \text{spread } 47\%, \tag{10}$$

so α_{10} is within a factor 1.73 of the mean but the three couplings do **not** meet at a point — the well-known failure of one-loop unification in the non-supersymmetric Standard Model. The U(1) Landau pole of the up-running sits at 2.2×10^{41} GeV, above the cutoff, so the up-running is UV-bounded; at two loops [49] the pole moves to $\sim 2.5\times10^{12}$ GeV, *below* the cutoff.

The down-running must be stated plainly. Starting from $\alpha_{10}=0.04$ at M_P and running down with the Standard Model β-functions gives, at M_Z,

$$\begin{aligned}\alpha_1(M_Z)&=0.0197\ (1.17\times),\\ \alpha_2(M_Z)&=0.195\ (5.8\times),\\ \alpha_3&:\ \text{Landau pole at } 2.2\times10^9\ \text{GeV}.\end{aligned} \tag{11}$$

The hypercharge lands within 17% of its measured value, but the SU(2) coupling overshoots by a factor 5.8 and the SU(3) coupling becomes non-perturbative at 2.2×10^9 GeV. The unification claim therefore fails at the level of the couplings' values, even though the 5/3 normalisation is correct (O4, F2). (Fig. 19).

Three quantitative statements sharpen the problem. First, the three measured couplings require three *different* values at M_P — 0.0299, 0.0202, 0.0191 — and the theory's single value 0.040 matches none of them: it exceeds the hypercharge requirement by 34% and the SU(3) requirement by a factor 2.1. Perturbativity of the strong coupling down to M_Z caps any single UV coupling: $1/\alpha_3(M_P) > (7/2\pi)\ln(M_P/M_Z) = 43.9$, i.e. $\alpha_3(M_P) < 0.0228$; the theory's 0.040 exceeds the admissible maximum by 75%.

Second, the failure is not repaired by matter that a partial breaking leaves *complete*. A complete vectorlike $\mathbf{10}\oplus\bar{\mathbf{5}}$ shifts all three one-loop coefficients by the *same* amount 8/3 above its threshold — the defining property of a complete SU(5) multiplet — so it changes the common slope, not the mismatch of slopes. Scanning multiplicities $0\ldots8$ against thresholds in $[10^6, 10^{19}]$ GeV, the spread at M_P never falls below its zero-new-physics value of 47%. Under pure Standard Model running the pairwise meeting scales are 1.2×10^{13} GeV $(1,2)$, 2.6×10^{14} GeV $(1,3)$ and 9.7×10^{16} GeV $(2,3)$: three different scales.

Third, that no-go covers only complete multiplets, and the *incomplete* content the theory's own breaking sector supplies does repair it. When a $\mathbf{16}_H\oplus\overline{\mathbf{16}}_H$ pair —

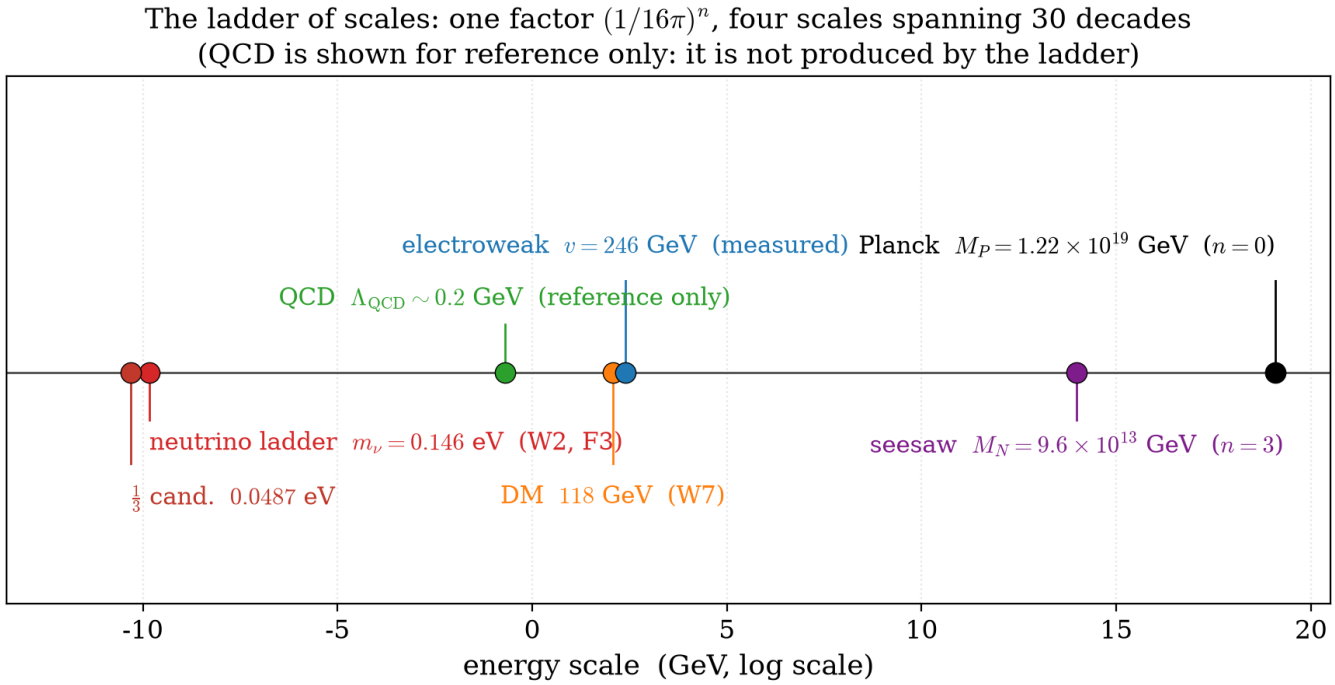


FIG. 17. Planck 1.22×10^{19} GeV ($n = 0$), seesaw 9.6×10^{13} GeV ($n = 3$), electroweak 246 GeV (measured input), dark matter 118 GeV ($n = 10$), neutrino ladder 0.146 eV ($n = 17$), with the $\frac{1}{3}$ candidate 0.0487 eV marked. QCD is shown for reference only: it is not produced by the ladder.

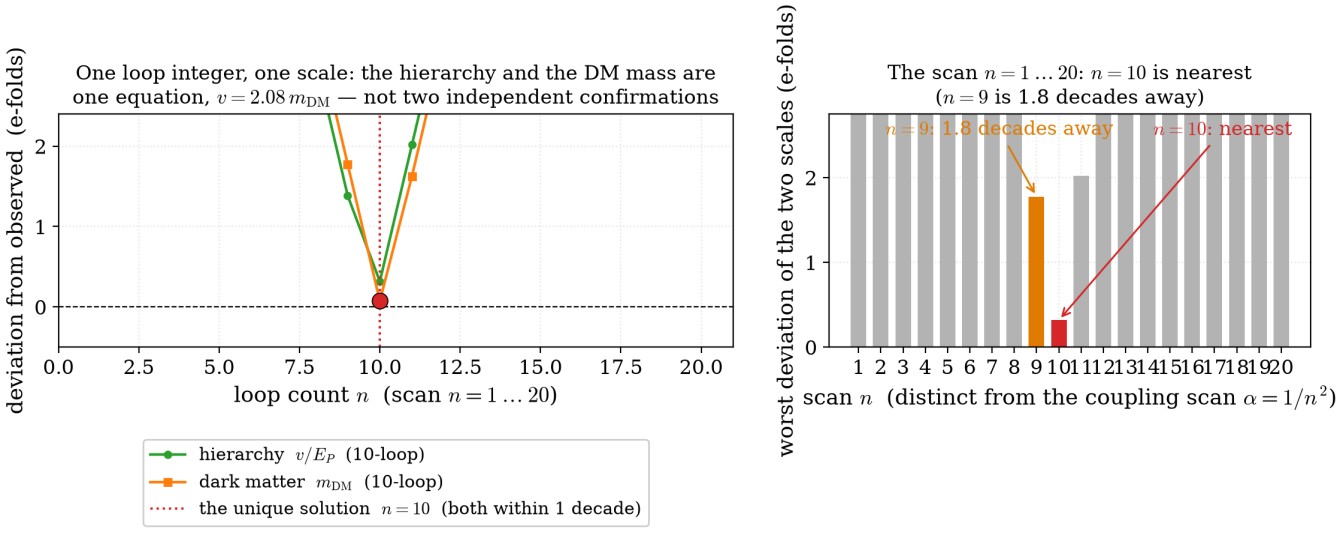


FIG. 18.

or a split doubler **16** (Section 8) — acquires mass, it leaves vectorlike $\mathbf{10} \oplus \overline{\mathbf{10}}$ and $\mathbf{5} \oplus \overline{\mathbf{5}}$ fermion pairs with *independent* thresholds, and these shift the three coefficients non-proportionally: a Dirac $\mathbf{10} \oplus \overline{\mathbf{10}}$ contributes $\Delta b = (6/5,\ 2,\ 2)$, a $\mathbf{5} \oplus \overline{\mathbf{5}}$ contributes $(2/3,\ 2/3,\ 2/3)$. Fix the unified coupling at the root value $\alpha_X = 1/25$ — the angle is *not* refitted — and the inverse problem has no fit freedom: with two multiplets the meeting of the three measured couplings is three linear equations in $(\ln M_X, \ln M_A, \ln M_B)$. The required Δb is non-universal: at $M_X = 2 \times 10^{16}$ GeV it is (2.40, 4.04, 3.86), which no combination of complete multiplets produces. Scanning multiplicities up to 3 over the realizable content with thresholds in $[10^6, M_P]$ GeV, the lightest solution that survives the proton-decay bound is three $\mathbf{10} \oplus \overline{\mathbf{10}}$ pairs plus one $(1, 2)_{1/2}$ pair — three split doubler **16**s —

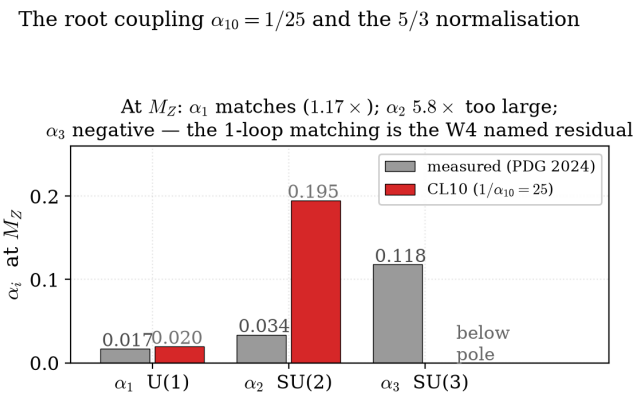


FIG. 19. At M_Z, the measured couplings (grey): $\alpha_1 = 0.0169$, $\alpha_2 = 0.0338$, $\alpha_3 = 0.118$ [13], against the down-run from $\alpha_{10} = 1/25$ (red): α_1 within 1.17×, α_2 5.8× too large, α_3 negative because its pole sits at 2.2×10^9 GeV. The 5/3 normalisation is exact; the value matching is the named residual.

meeting at

$$\begin{aligned} M_X &= 1.8 \times 10^{16}\ \text{GeV}, \\ \alpha_X &= 1/25\ \text{(exact)}, \\ M_{10} &= 1.2 \times 10^{7}\ \text{GeV}, \\ M_{(1,2)} &= 1.0 \times 10^{12}\ \text{GeV}, \end{aligned} \tag{12}$$

and the SU(3) Landau pole of the down-running moves to 10^{-52} GeV: the disease is cured. Solutions with lighter content exist but predict $\tau_p \sim 10^{32}$ yr and are already excluded by Super-Kamiokande [52] — the data themselves select the realizable spectrum. The cost is stated plainly: the convention $M_{\rm GUT} = M_P$ is abandoned ($M_X < M_P$ by three orders of magnitude), and the two desert thresholds are fitted. The solution predicts $\tau_p \simeq M_X^4/(\alpha_X^2 m_p^5) = 1.8 \times 10^{36}$ yr (F5). (Fig. 20).

Where the desert content comes from. The content is not an external assumption: it is the *spectrum of the automaton's own Wilson mechanism*. The three split nodes of the selective reading carry exactly the vector-like fragments of the split **16** with independent thresholds (the other nine nodes stay at the Planck cutoff), and the Wilson direction itself completes the breaking chain (Section 8.1). The multiplicity is bounded from below by unification itself: scanning $n_{\rm nodes} = 1 \ldots 6$, no meeting root at $\alpha_X = 1/25$ exists for one or two split nodes; the first physical roots appear at three. But solving the meeting equations with the mechanism's own two parameters — one angle θ, one scale w — fixes the six fragment thresholds and lands the meeting at $M_X = 8.9 \times 10^{14}$ GeV, i.e. $\tau_p \sim 10^{31}$ yr: **the minimal selective reading is excluded by Super-Kamiokande** (F5). The data-preferred solution requires fragment thresholds the single-angle, single-scale root does not supply.

Redistributing the nodes cannot help. While all node thresholds sit below the meeting point, the two meeting equations depend on the three node strengths **only through their sum**: the Jacobian with respect to the three strengths has three identical columns (rank 1; split-invariance verified numerically to 10^{-14}, including the full scan over node-set choices and strengths). The node mechanism's meeting point $M_X = 8.9 \times 10^{14}$ GeV, $\tau_p = 1.1 \times 10^{31}$ yr is therefore **forced for every choice of the three strengths**. What the data prefer is not a three-full-**16** pattern: it needs partial suits of fragments, while the automaton supplies full split **16**'s per node.

The composite content of the gauged layer is excluded as the supplier. The composite fermions of the layer of Section 10 — the matter **16** tensored with the adjoint link flux, $16 \otimes 45 = 720$ Weyl states carrying five net chiral **16**-copies — shift the three one-loop coefficients by $(145.3, 145.3, 226.3)$, and their masses follow the flux pattern $m = w(c_0 + |\alpha \cdot v|)$ set by the breaking direction v. Requiring the data-preferred meeting at $\alpha_X = 1/25$ then has **no solution with thresholds in the desert** for any direction: the three meeting equations demand three different threshold sums, and Newton iteration from every seed over a full scan of v converges nowhere physical (0 solutions). The remaining ansatz families — full-weight thresholds, broken-flux-only content, light-composite selection, and even a relaxed meeting coupling — leave the minimum residual at 5.7–6.1 against a target of 0.

The equal-shift theorem. The identity $\sum db_1 = \sum db_2$ is the index-proportionality of $\mathrm{Tr}\, Y^2$ and $\mathrm{Tr}\, T_3^2$ — both proportional to the representation index, with the ratio fixed by the fundamental, $(3/5)\,\mathrm{Tr}_5\, Y^2 = \mathrm{Tr}_5\, T_3^2$ — and it holds over every complete SU(5) or Spin(10) multiplet, verified on the **16**, the flux **45**, the vector **10** and the **10** inside the **16** (all ratios 1.0 to 10^{-6}). The meeting mismatch can therefore come **only** from SU(5)-fragment content or from split thresholds inside multiplets; computed from the same charge operators, the desert repair's mismatch is carried entirely by its $(1,2)$ pair ($db_1/db_2 = 3/5$; the $10 \oplus \overline{10}$ pairs contribute exactly zero mismatch). (Figs. 21, 22).

The junction route: pattern fabricated, meeting missed. At the root the Wilson direction splits the node **16** by a factor 55 ($c_Q = +0.005$, $c_L = -0.015$ against partners ± 0.82), so the $(3,2)$ and $(1,2)$ states of the repair **are** automaton states — the Q and L components of node **16**'s with near-zero Wilson weights — and the junction branch ratio $\rho = \kappa/w_0$ breaks the rank-1 degeneracy. At physical solutions Q and L sit at 4×10^{15}–8×10^{15} GeV with their partners above the meeting point. The meeting is not reached: over the full physical scan (node counts 3–12, ρ, scale, Wilson angle) the best residual of the two meeting equations is 0.94 against a target of 0, and the third equation predicts $\alpha_X = 0.030$ against the data value 0.04 — a 33% miss. Replacing the cubic-symmetric democratic junction by the **general zero-mean junction** — three independent branch scales $a_i = 2w_0(1 + r_i)$, $r_1 + r_2 + r_3 = 0$, scanned over the full (r_1, r_2) plane — leaves the best physical residual

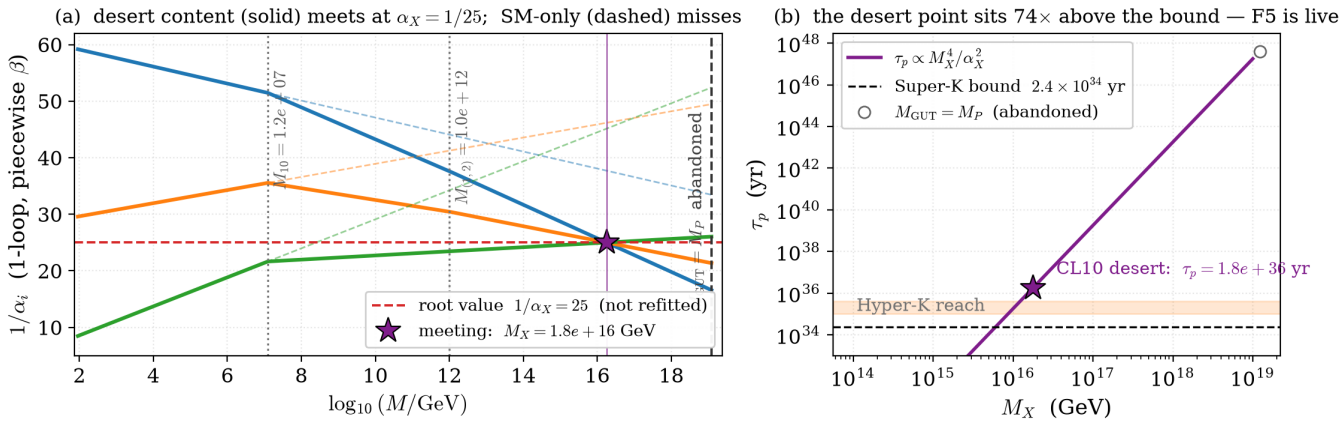


FIG. 20. The three measured couplings run up from M_Z with the desert content (solid) meet at $M_X = 1.8 \times 10^{16}$ GeV at $\alpha_X = 1/25$ — the root value, not refitted. The Standard Model-only running (dashed) misses the point: spread 47% at M_P. The thresholds $M_{10} = 1.2 \times 10^7$ GeV and $M_{(1,2)} = 1.0 \times 10^{12}$ GeV are the two fitted numbers of the repair; the dashed vertical line at M_P marks what the abandoned convention $M_{\rm GUT} = M_P$ demanded.

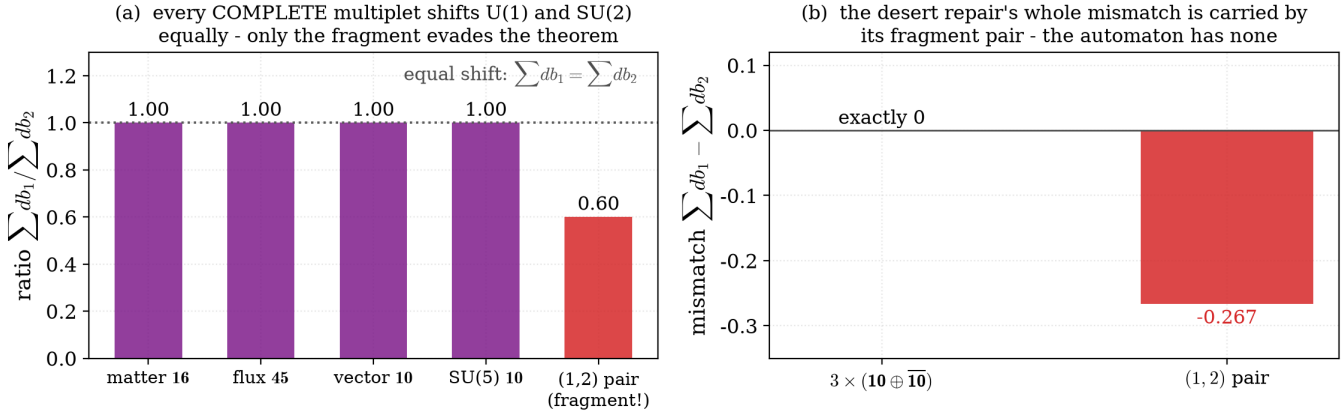


FIG. 21.

identical (0.81) with $\alpha_X = 0.090$, a factor 2.3 off. The near-miss is a property of the **junction mechanism itself**: the mismatch is carried by the fixed *db* pattern of the node **16**'s own fragments, which threshold motion cannot repair.

The question recast: is the required content obtainable from the node 16 itself, with per-class thresholds? Yes for feasibility: the data-preferred meeting point $(T_1, T_2, T_3) = (12.7, 21.2, 20.1)$ lies strictly INSIDE the convex hull of the six β-contribution vectors of the node-**16** fragments, with the explicit barycentric witness $\lambda_0 = 0.34$, $\lambda_{f_1} = 0.31$, $\lambda_{f_2} = 0.31$, $\lambda_{f_4} = 0.05$. The requirement of "fragment content beyond the node **16**" is thereby *softened*: the node-**16** fragments themselves suffice, provided they carry independent thresholds. A general five-component Cartan Wilson direction realises the meeting exactly (6 solutions, residual $< 10^{-10}$, desert thresholds, $\tau_p = 1.9 \times 10^{36}$ yr) but breaks all 40 roots — the Model algebra among them — and is unphysical; the symmetry-preserving direction space is exactly span$\{X, Y\}$ (dimension 2, class-degenerate), and within it no solution exists. Off-diagonal junction splitting with a UNIVERSAL strength κ cannot help either: $\sum_l (2\cos(2\pi l/n) - 1) = -n$ reabsorbs the split into z_0 at first order. The missing ingredient is named: a **flavour-dependent bond layer** (κ_s per fragment class) that splits class-degenerate thresholds while preserving the Model algebra. (Fig. 23).

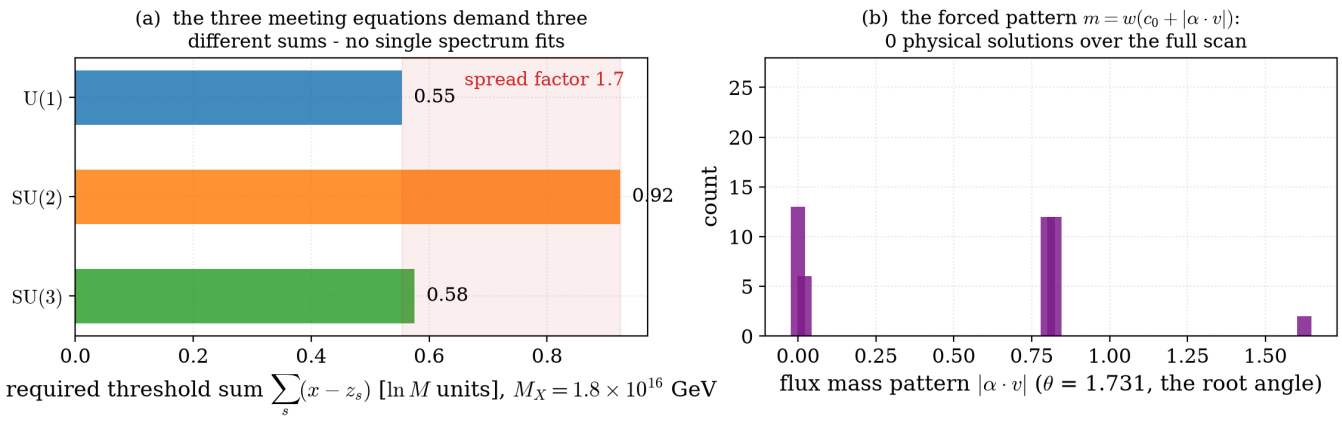


FIG. 22. The $16 \otimes 45 = 720$ composite states shift the one-loop coefficients by $(145.3, 145.3, 226.3)$. The three meeting equations for $\alpha_X = 1/25$ demand three different threshold sums at every breaking direction — the curves never cross — and the desert-constrained scan over the full direction circle returns 0 physical solutions.

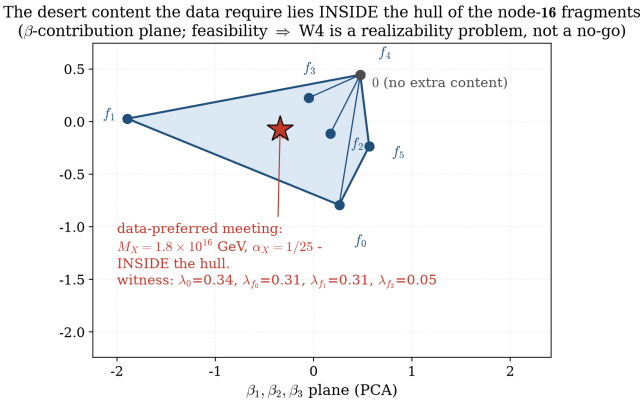


FIG. 23. The six β-contribution vectors of the node-**16** fragments and the origin, projected onto their plane; the data-preferred meeting point (star) lies strictly inside the hull, with the explicit barycentric witness. Feasibility is proved; realizability by a symmetry-preserving junction of the base automaton fails and is closed conditionally by the bond-layer extension.

The named ingredient closes the meeting — conditionally. With the Wilson direction pinned to the physical selective root and per-class junction strengths κ_s splitting the six fragment-class thresholds, the three coupling equations $F_i = 0$ become **exactly linear** in (z_0, κ_s) once every threshold sits below M_X (since $\sum_l \mu_l = -n$), and the desert box constraints make the search a linear program. It admits solutions (residual $< 10^{-13}$) at $M_X = 1.8 \times 10^{16}$ GeV, $\alpha_X = 1/25$, for **all ten node counts** $n = 3 \ldots 12$, with every threshold inside the desert (2.0×10^3–1.8×10^{16} GeV, none above M_X) and $\tau_p = 1.9 \times 10^{36}$ yr. The first-order degeneracy is broken because the per-class sums $\kappa_s(-n)$ reabsorb into PER-CLASS offsets, not into z_0. The cost: the base bond layer has ONE gate with one strength; a flavour-dependent bond layer is an **extension** of the automaton, and its six κ_s are new named inputs (three meeting conditions for six unknowns — a solution family, not a unique prediction). Unification is therefore closed **conditionally on that extension**.

The two structural candidates the layer itself supplies are excluded. The six κ_s are not obtainable from any on-site algebra element: symmetry preservation forces the bond direction into $\mathrm{span}\{X, Y\}$ at the physical angle θ^* (class-degenerate, meeting residual ≥ 0.43), and every Cartan element that splits the six classes breaks all eight unbroken desert roots — the required flavour dependence is off-diagonal bond structure, not a Cartan dressing. Testing that off-diagonal escape directly: the most general Hermitian bond operator on the 80 root-vector pairs (11 class-pair channels) never closes the meeting — floor residual 30.4 over $n = 3 \ldots 12$ — and level repulsion forces the sign of two of the six κ_s. Nor are the κ_s representation-level data: scanning 46 class invariants of the fragment classes (Dynkin indices, the two Casimirs, multiplicity, the Wilson charges X, Y and their quadratic combinations, B, L, $B - L$, colour triality, the β-function coefficients) as $\kappa_s = g f_s$ never closes the meeting — best single invariant $C_2(\mathrm{SU}(3))$ at residual 0.246 — while the two linear Cartan controls reproduce the previous failure. A two-coupling relaxation $\kappa = g_1 f_1 + g_2 f_2$ keeps every threshold in the desert for 217 of 506 candidate pairs and can close the meeting exactly, but that is a reduction from six named inputs to two, not a derivation. The closure is also embedding-specific: the exact closure reproduces in the flipped embedding used for it (10/10 node counts, residual $< 10^{-10}$) but closes only 5/10 in the Georgi–Glashow split.

Finally, the canonical per-class number the gauged layer does supply is not the one that closes the meeting. At θ^* the Wilson direction leaves a 13-dimensional centralizer in $\mathfrak{so}(10)$ (the $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)^3$ of the breaking chain, Hermitian orthonormal basis to 10^{-15}), and the flux of BROKEN generators through a fragment class,

$$K_s = C_2(\mathbf{32}) - C_2^{\text{unbroken}}(s) = 45 - C_2^{\text{unbroken}}(s) \\ = (32, 36, 40, 28, 32, 40)$$

(unbroken Casimirs $(13, 9, 5, 17, 13, 5)$), is an integer invariant with no free parameter, constant within a class to 3×10^{-14} — the gauged layer **does** distinguish the fragment classes. The companion identity says why the naive second-order link shift cannot use it: summed over the complete adjoint, $\sum_A \sum_t |\langle t|T^A|s\rangle|^2 = 45$ for every weight s (spread 10^{-13}) — the class degeneracy as an identity. Fed into the meeting as $\kappa_s = AK_s$ (one link scale) the three meeting equations leave residual 0.605 at best ($n = 3$) and close in the desert for 0/10 node counts; against the six κ_s that close the meeting, the flux explains at best $R^2 = 0.015$ of their variance ($|r| \leq 0.31$). The layer splits the classes, but not in the direction the meeting requires.

B. Electroweak mixing

$\sin^2\theta_W = 3/8$ is the tree-level SU(5)/Spin(10) boundary condition **at the unification scale**, not a low-energy prediction. Run to M_Z it becomes $\simeq 0.23$, close to the measured 0.23129(4) ($\overline{\text{MS}}$, PDG 2024 [13]); the measured ratio $m_W/m_Z = 0.881$ corresponds to $\sin^2\theta_W \simeq 0.223$, not to $\sqrt{5/8} = 0.791$. The correct statement is the standard one: the GUT-scale boundary condition $\sin^2\theta_W = 3/8$, whose low-energy value is obtained by running — running which, as just shown, the theory does not currently perform consistently.

C. Baryon asymmetry

The automaton alone cannot produce it. Every CP-odd (T-odd) invariant of the unitary step vanishes: Im Tr of all two- and three-fold products of the **45**, of a 2×10^5 sample of four-fold products, of all $10 \times 10 \times 45 \times 45$ baryon-violating sandwiches, and of the rank-1 real democratic Yukawa — all zero to 10^{-15}. The step is unitary (in equilibrium), baryon-conserving (the **45** commute with w) and CP-conserving. It satisfies **none** of the Sakharov conditions [24]. This is a computed theorem, and it is the correct starting point. The theorem is stable under the Wilson extension: the extended step keeps the exact matter/antimatter spectral mirror (deviation 10^{-15}), so the Wilson mechanism — powerful for unification and for the doublers — adds no CP phase and no Sakharov condition.

A GUT-leptogenesis estimate [39] is then built by hand:

$$\Delta B = \frac{28}{79}\,\varepsilon_{CP}\,D, \\ \varepsilon_{CP} = \frac{g_0^4}{16\pi} = 4.9 \times 10^{-6}, \\ D = \left(\frac{1}{16\pi}\right)^2 = 3.96 \times 10^{-4}, \tag{13}$$

with $g_0 = 1/2^3 = 1/8$ and the Standard Model sphaleron conversion $B = \frac{28}{79}(B - L)$ [25, 26], giving $\Delta B = 6.8 \times 10^{-10}$ against the measured $(6.12 \pm 0.04) \times 10^{-10}$ [14] — a ratio 1.11.

Three problems. (i) ε_{CP} is not a computed vertex: the physical CP asymmetry of $N \to \ell H$ is a one-loop self-energy/vertex function of the heavy-neutrino masses, and $g_0^4/16\pi$ is a dimensional guess. (ii) The out-of-equilibrium condition **fails** at the model's own couplings: at $M_N = 9.6 \times 10^{13}$ GeV, $H = 1.66\sqrt{g_*}M_N^2/M_P = 1.3 \times 10^{10}$ GeV while $\Gamma_N = y^2 M_N/8\pi = 6.0 \times 10^{10}$ GeV for $y = 1/8$ (1.9×10^{12} GeV for $y = 0.71$), so $\Gamma_N/H \simeq 4.6$ ($\simeq 148$). Out-of-equilibrium decay requires $y \lesssim 0.06$, which the theory does not select. (iii) 28/79 is an input of the Standard Model sector, not a number of the root. The 11% agreement is therefore a coincidence of chosen factors (O5). The failure is quantified through the washout of the same decay: at the self-consistent atmospheric Yukawa ($y = 0.40$, the one that gives $m_\nu = 0.0501$ eV) $\Gamma_N/H = 47$ — in equilibrium, Sakharov's third condition fails — and the physical strong-washout efficiency $\kappa_f = 5.4 \times 10^{-3}$ [39] sits 14× away from the guessed dilution $(1/16\pi)^2$: the agreement is an accident of two wrong factors, not a near-miss.

A sharper statement is available. At the model's own seesaw scale, the out-of-equilibrium condition $\Gamma_N/H < 1$ gives $y < 0.058$. The same Yukawa fixes the light neutrino mass through the same seesaw,

$$m_\nu = \frac{y^2 v^2}{M_N} \leq 2.1 \times 10^{-3}\ \text{eV} \\ (y \leq 0.058), \tag{14}$$

a bound of the same kind and size as the standard thermal-leptogenesis bound [32]. The atmospheric splitting requires $m_\nu \simeq 0.050$ eV, i.e. $y \simeq 0.28$ with $m_D = yv$ and $v = 246$ GeV ($y \simeq 0.40$ in the $v_u = 174$ GeV convention), for which $\Gamma_N/H \simeq 23$. **At $M_N = 9.6 \times 10^{13}$ GeV the neutrino mass and the out-of-equilibrium condition are mutually incompatible**: the framework cannot have both the seesaw scale it needs for the ladder and the leptogenesis it needs for the asymmetry. (The model's own ladder sits at $m_D = 118$ GeV, i.e. $y = 0.48$, where $\Gamma_N/H \simeq 68$.)

The one route the framework supplies is quasi-degenerate (resonant) leptogenesis [31, 40]. The democratic structure places the three heavy singlets on the

same ladder rung, hence degenerate to the accuracy of whatever breaks the family symmetry — precisely the resonant setup, where the CP asymmetry is enhanced by near-degeneracy instead of by a large Yukawa. With $y = 1/16\pi = 0.0199$ the decay is out of equilibrium ($\Gamma_N/H = 0.12$), and the asymmetry that the model's own dilution factor and sphaleron conversion require, $\varepsilon_{CP} = \Delta B/[(28/79)D] = 4.4 \times 10^{-6}$, is *smaller* than the natural one-loop size $y^2/16\pi = 7.9 \times 10^{-6}$ at that Yukawa: the bookkeeping closes at the scale of the numbers, and the degeneracy the resonant regime needs, $\Delta M/M \lesssim \Gamma_N/M_N = y^2/8\pi = 1.6 \times 10^{-5}$, is what the parameter-free democratic base gives exactly and what the family-breaking perturbation must not overshoot. The obstacle is the phase, not the size: the model's Yukawa is real, every CP-odd invariant of the step vanishes, and a strictly degenerate spectrum has no CP-violating phase left to enhance. This is a *route*, not a result.

The phase, however, is no longer the missing ingredient. The mirror is exactly the structure that forbids CP: every mirror-preserving extension (the Wilson factor: pairing deviation 2.1×10^{-15}) leaves the CP phase at zero, while the most general per-class κ_s layer and the most general real-symmetric bond generator destroy the $\pm$ pairing (deviation ~ 1, 60 draws, both mirror sign conventions) — a CP phase costs Theorem 4. But the prohibition is a statement about the *on-site* module (Section 7.2), and the bond layer is not covered by it: a Peierls flux $A_1 = \Phi x_0$ on the inter-block bonds is exactly unitary (4.8×10^{-15}), keeps $\det U = 1$ at every flux (5.3×10^{-15}), commutes with the pseudoscalar (mirror $\mathbb{Z}_2$ leak 3.3×10^{-16}) and keeps the hop at radius 1 (2.6×10^{-16}), while moving the band spectrum by up to 0.70; its gauge-invariant strength, the plaquette holonomy, has Im $= 0$ at $\Phi = 0$ and 1.6×10^{-3} at $\Phi = \pi/2$. Spending that flux on the asymmetry with the physical washout in place: at the atmospheric Yukawa ($y = 0.3987$, $K = 46.9$, $\kappa_f = 5.4 \times 10^{-3}$) the measured $\Delta B = 6.12 \times 10^{-10}$ costs $J = 10^{-4}$, supplied by a flux $\Phi = 0.064$ rad; in the out-of-equilibrium regime ($y = 0.0582$, $K = 1$, $\kappa_f = 2.4$) it costs $J = 1.1 \times 10^{-5}$ at $\Phi = 0.007$ rad. A bond flux of size $\Phi \sim 0.06$ supplies more than the asymmetry needs while leaving unitarity, $\det U = 1$, the mirror symmetry and the causal radius exact. **What fails is Sakharov 3 alone**: the Yukawa that gives $m_\nu = 0.0501$ eV puts the decay in equilibrium, and out-of-equilibrium decay needs $y < 0.0582$, which caps m_ν far below the atmospheric splitting. O5 is open for one named reason instead of two.

D. Proton decay

Dimensional analysis of X/Y exchange gives

$$\tau_p \sim \frac{M_{\rm GUT}^4}{\alpha_{10}^2\, m_p^5} = 4.0 \times 10^{47}\ \text{yr} \qquad (M_{\rm GUT} = M_P,\ \alpha_{10} = 1/25), \tag{15}$$

against the Super-Kamiokande bound $\tau_p/B(p \to e^+\pi^0) > 2.4 \times 10^{34}$ yr [15]. The $M_{\rm GUT} = M_P$ reading gave a margin of 13 orders of magnitude, resting entirely on that abandoned convention: any GUT scale below $\sim 6 \times 10^{15}$ GeV is excluded by the same bound. The estimate omits the renormalisation factor A_L [46], the hadronic matrix element, the dressing of the operators and the splitting between the X/Y masses and the unification scale [51], all of which shift the lifetime by one to two orders. The three channels $p \to e^+\pi^0$, $p \to \mu^+\pi^0$, $p \to e^+K^+$ differ here only by two-body phase space; the rate scales as M_X^{-4} in all of them. Observation of a GUT-pattern decay far below 1.8×10^{36} yr refutes the desert solution (F5).

E. Flavour

The democratic mass matrix $M = |1\rangle\langle 1|$ (all entries 1, rank 1, spectrum $\{3, 0, 0\}$) [43] follows from the hypothesis that the three spatial directions are equivalent, so the $\mathbf{10}_H$ couples the three families identically. Its heavy eigenvector $(1, 1, 1)/\sqrt{3}$ is unique, hence

$$\sin^2\theta_{12} = |\langle e_1|h\rangle|^2 = \tfrac{1}{3}$$

for any family basis — the **only** mixing angle the model derives, 8.6% from the measured 0.307 [17, 50]. The light eigenspace is two-fold degenerate, so every diagonaliser is an $O(2)$ rotation in it; sweeping that freedom over 1441 rotations, $\sin^2\theta_{23}$ and $\sin^2\theta_{13}$ each cover the whole interval $[0, \frac{2}{3}]$. The tribimaximal values $(\frac{1}{2}, 0)$ [27] are one point of that family, picked by declaring which S_3 transposition breaks — and (12), (13), (23) are equivalent symmetries of $|1\rangle\langle 1|$, so the choice is an input.

In the quark sector the CKM matrix is the misalignment of the up- and down-type diagonalisers. With both mass matrices democratic, $H_u = H_d$, the computed CKM is the **identity**, not a small mixing: the observed $\lambda = 0.225$ [13, 44] is not reproduced. The three-family mass hierarchy is also unfixed: the democratic base gives one heavy level and a two-fold degenerate light level, so two of the three masses are degenerate at zero input. All of this is O6, and the natural fixes are bounded by computation. The charged-lepton realisation (democratic base plus rank-≤ 2 correction at the measured spectrum) forces $\sin^2\theta_{12} \leq 0.005$ — the lightest family decouples from the democratic direction, which is the tribimaximal solar column — excluded at 23σ; the symmetry-natural cubic-axis neutrino vev is a parameter-free prediction excluded at $\geq 20\sigma$ in every branch; a free-direction rank-1 neutrino vev has four parameters

against three angles and predicts nothing, and reproducing $(0.308, 0.0222, 0.470)$ needs ≥ 2 vev terms in generic relative orientation. The unification fix of Section 13.1 cannot supply the splitting either: its κ_s are per-gauge-class couplings — generation-blind — and leave PMNS and CKM exactly invariant (max angle shift 2×10^{-13}). Generation-space structure does move the mixing: a uniform scalar dressing is exactly blind (shift 1.4×10^{-13}), one common non-abelian family dressing (7 parameters) reaches the measured triple, a chiral-asymmetric one (13 parameters) reaches it as a fit whose every entry is a named input — worse cost than the ≥ 2 vev terms.

F. Scoreboard

XIV. COSMOLOGY

A. Dark energy

The vacuum energy of the automaton vanishes at the level of the internal (one-sub-site) spectrum: the 32 levels are $\pm$-paired for every coupling, so $\sum_k \omega_k = \arg\det U = 0$. This is the correct, non-vacuous form of the cancellation. (Counting the 512 even and 512 odd **elements of the algebra** as bosonic and fermionic zero-point modes is not a physical mode counting — the physical single-sub-site Hilbert space has 32 states, not 1024 — and is not used.)

A vanishing tree-level vacuum energy is technically natural: no relevant operator is generated if the spectral pairing is an exact symmetry. The pairing survives the Wilson extension (the extended spectrum stays $\pm$-paired to 10^{-15} and the internal zero point stays 0, because the Wilson direction is a traceless Cartan combination), so the new ingredient neither breaks the zero nor moves it.

The zero is **protected, not accidental**. Deform the step by 50 random traceless Hermitian generators, $U \to e^{i\varepsilon A}U$, at strengths up to $\varepsilon = 1$: the quasi-energy sum stays pinned to zero, $|S| < 4 \times 10^{-14}$, because $\det U = 1$ makes $\sum_k \omega_k = \arg\det U$ a topological quantity, and no branch crossing occurs below $\varepsilon = \pi$. The only internal knob that moves the sum is the U(1) trace phase.

What the theory does **not** do is produce the observed value. The observed $\Lambda \sim 10^{-122} M_P^4$ lies 122 orders of magnitude *below* the classical bound M_P^4 and *above* the theory's spectral zero, and no mechanism breaking the pairing by that amount is known. Independently, the classical self-gravitating de Sitter vacuum of Section 11 has $G_{\mu\nu} = -3H^2 g_{\mu\nu}$, i.e. a vacuum energy bounded by M_P^4 on the lattice. These are three different statements — spectral zero, classical bound, observed value — and they do not "reconcile": the gap between the first and the third is O1, the largest unexplained number in the theory [37]. The equation of state of a cosmological-constant vacuum is $w = -1$ by definition, not by derivation.

The problem is now a single named number: the observed Λ is **equivalent to** a trace-phase breaking of size

$$\varepsilon \operatorname{Tr} A \sim 16\pi^2 \times 10^{-122} = 1.6 \times 10^{-120}.$$

O1 is no longer "why is the vacuum energy small" but "what breaks the trace U(1) by 10^{-120}".

B. Dark matter

The candidate supplied by the classical sector is the kink of Section 11.4. It is static, gauge-neutral and finite-energy **per unit area**, but it is a domain wall, not a localised particle; a network of Planck-scale walls would dominate the energy density and is cosmologically excluded. The quantum-sector candidates have been scanned (Section 12.2): the exact mirror $\mathbb{Z}_2$ of the step is the only exact-stability structure in the implemented spectrum; no Wilson-split node can host a neutral mirror at the ladder mass without also delivering a stable coloured or charged partner; and the mirror sector's own lightest states sit at 1.3×10^8 GeV (coloured, excluded) and 2×10^{10} GeV — not at the ladder's 118 GeV. The theory supplies no dark-matter *particle* within the implemented spectrum; the minimal named extension is the mirror-odd ladder scalar of Section 12.2. The mass prediction stands, the carrier is an open problem (O7).

The relic abundance is computed, and it is not the observed one. Solving the freeze-out integral

$$\Omega h^2 = \frac{1.07 \times 10^9 \text{ GeV } x_f}{\sqrt{g_*}\, M_P \langle\sigma v\rangle}, \qquad \langle\sigma v\rangle = a + \frac{6b}{x_f}, \tag{16}$$

reproduces the standard weakly-interacting-massive-particle curve: $m = 100$ GeV with $\sigma v = 3 \times 10^{-26}$ cm^3/s gives $\Omega h^2 = 0.080$ [41]. At the ladder mass $m = 118.5$ GeV the measured $\Omega h^2 = 0.120$ requires $\sigma v = 1.98 \times 10^{-26}$ cm^3/s, i.e. an effective coupling $\alpha_{\text{eff}} = 0.0086$ in $\sigma v = \alpha^2/(\pi m^2)$ — while the root supplies only $g^2/4\pi = 0.0032$ (giving $\Omega h^2 = 0.798$) or the loop factor $1/16\pi = 0.0199$ ($\Omega h^2 = 0.0246$). The automaton's own light spectrum enters at the same order: its twelve near-gapless nodes are nodal **planes** (Section 8.2) with mean speed 0.201, so the final-state density is enhanced by

$$D = \sum \frac{1}{|v|^3} = 1485$$

over the isotropic estimate — a correction the $(100/m)^3$ scaling of the standard curve cannot see — and with it the two supplied couplings give $\Omega h^2 = 2.3 \times 10^{-5}$ and 7.6×10^{-4}. The named portals of Section 12.2 reproduce 0.120 only because their coupling was inverted from the fiducial 3×10^{-26}: computed forward they give

TABLE V. **Headline numbers against measurement.**

quantity	**theory**	**measured**	**status**
the eight anomaly coefficients	0	—	algebraic, satisfied
g_1^2/g'^2	5/3	5/3 (GUT normalisation)	algebraic
$\det U$, $\pm$ pairing, $16/\overline{16}$ mirror	1, exact	—	algebraic, coupling-independent
causal front speed	1.0	—	dynamical, measured
doubler cone Lorentz structure	linear in all 1200 sampled directions; anisotropic: $(0.200, 0.200, 0.284)$ lattice units, spread 0.75; node gap 2×10^{-4} at $g = 1/5$	—	**not Lorentz-isotropic**; measured
tangent velocity algebra at the gapless corner multiplet	$[S_i, S_j] = 0$ to 3×10^{-12}, $\sum_j S_j^2$ scalar to 7×10^{-12}, branch speeds exactly $\lvert v_\alpha \cdot q\rvert$ to 6×10^{-12} with v_α the cubic permutations of $(0.704, 0.421, 0.421)$	—	**three nodal planes, not a cone**; measured
16-sector block-Brillouin spectrum	at 1/5: zone centre gapped (0.0365), 12 near-gapless doubler nodes (2.003×10^{-4}), $\sum$ chirality $= 0$, all gapped by the internal Wilson factor, 3-node selective reading; at the closed $\pi/4$: Γ plus six face-diagonal **nodal lines** exactly gapless, 12 M points on them, no 2-dimensional little-group irrep at Γ	—	doubling **gapped by a mechanism of the step** at 1/5; at $\pi/4$ the doublers are lines and the per-node chirality bookkeeping fails
v/M_P	$2.08(1/16\pi)^{10}$	2.0167×10^{-17} [13]	coefficient $3F_{10}^{10} = 2.0881$ against 2.0765 observed — 99.44% from measured numbers, conditional on the one-split-node-per-family set
m_{DM}	118 GeV	not measured	prediction, falsifiable
$\Omega_{\mathrm{DM}}h^2$	computed from the supplied couplings: 0.798 or 0.0246 (2.3×10^{-5}, 7.6×10^{-4} with the nodal-plane density)	0.120 [14]	**fails**
m_ν	0.146 eV (ladder); 0.0487 eV ($\frac{1}{3}$ candidate)	0.0501 eV [50]; $\sum m_\nu < 0.12$ eV [14]	ladder **in tension**; candidate inside a finite window, structure protected
ΔB	6.8×10^{-10}	$(6.12 \pm 0.04) \times 10^{-10}$ [14]	coincidental product; the washout at the self-consistent y destroys it

0.081 and 0.156, and 0.000 once the nodal-plane density is included. **The relic abundance is a computed number now, and it is not** 0.120 **for any coupling the root supplies.** Non-thermal production (freeze-in) is not considered here [42].

C. Antimatter

Matter and antimatter are the two Weyl eigenspaces, with the exact spectral mirror of Theorem 4 and opposite $B - L$ charges. Their degeneracy is the content of the CPT statement in this theory. The observed baryon asymmetry is not explained: Section 13.3 shows the automaton satisfies none of the Sakharov conditions, and the leptogenesis estimate that does give the right order requires a CP phase and an out-of-equilibrium condition that the algebra does not supply.

XV. THE PARAMETER AUDIT

Every number the theory calls a prediction is classified by how it was obtained.

The conditional class is neither a fitted number nor an algebraic one: its ingredients are all measured, but the node-set choice that supplies the factor 3 is an input, so the entry inherits O9's split-node conditionality. (Figs. 24, 25).

The status of the angle. No condition inside the theory fixes the implemented value. The two-loop U(1) Landau-pole scale lies below M_P for every g in the perturbative range, so UV-finiteness of the lattice does not select it. The proposed selection rule is $g = 1/n_{\text{qubits}} = 1/5$, the five qubits of the 32-dimensional module. The data do not support that choice: scanning the candidates $\alpha = 1/n^2$ for $n = 2 \dots 8$ against the one-loop value 0.023, the best fit is $n = 7$ ($1/49 = 0.020$, relative deviation 0.12), not $n = 5$ ($1/25 = 0.040$, relative deviation 0.73) — the data-optimal value is 6.4× closer than the root value. The reading $1/n_{\text{qubits}}$ is therefore a convention, and the theory is honestly **one-parameter in its dynamics, zero-parameter in its algebra**. The closure axiom (Section 9) then removes the free angle at the level of the dynamics, and the price — measured, not assumed — is that the $\alpha = 1/25$ meeting value and the $n^* = 10$ loop crossing move to $\alpha = \pi/64 = 0.0491$ and $n^* = 208$. Three statements about the angle now coexist and pull apart: the closure fixes it algebraically, the coupling scan disfavours that value by more than the old convention did, and the desert meeting absorbs any value through the six κ_s. The angle is no longer free but it is also no longer favoured.

The "nine-to-one" observation. The nine open problems are all associated with the same object, the 32-level spectrum, and its invariants (the vanishing 0-point, $\det U = 1$, the spectral mirror, the S-matrix $S = U$). That is a bookkeeping observation — nine open problems sharing one object — and it is not a derivation of any of their values. It should not be read as a reduction of the theory's difficulty.

XVI. OPEN PROBLEMS

XVII. FALSIFIERS

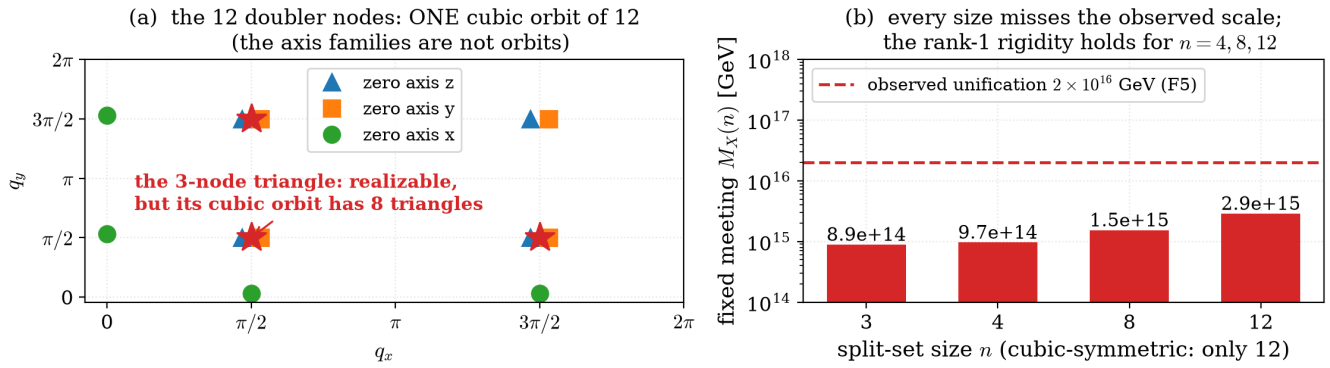
The most accessible tests are **F3** and **F5**. KATRIN currently gives $m_\beta < 0.45$ eV [16], so the bare ladder value 0.146 eV survives a single-mass search, but the cosmological sum $\sum m_\nu < 0.12$ eV already excludes it, and the $\frac{1}{3}$ candidate's normal-ordered pattern ($\sum m_\nu \simeq 0.05$ eV, $m_{\text{lightest}} = 0$) is inside the reach of the next generation of oscillation and $\beta\beta$-decay data. The desert solution predicts $\tau_p = 1.8 \times 10^{36}$ yr, about one decade beyond Hyper-Kamiokande's 20-year reach ($\sim 10^{35}$ yr): Hyper-Kamiokande can **refute** it by observing GUT-pattern decay anywhere at its reach, and an exposure passing 10^{36} yr that sees nothing refutes it too; the confirmation window belongs to the megaton era. **F7** has narrowed: the carrier is not in the implemented spectrum and the named extension is nearly decoupled, so the test is the mass coincidence itself.

XVIII. DISCUSSION

What is genuinely new. Not the **16** of Spin(10), not the 5/3, not anomaly cancellation: those are fifty-year-old results of the grand-unification literature [6–9, 12]. The content of this theory is (i) the realisation of that content as the on-site module of a *strictly local, exactly unitary* automaton with a measured causal front speed of exactly 1.0; (ii) the coupling-invariance of the algebraic invariants — anomaly cancellation, $\det U = 1$, the $\pm$ pairing and the matter/antimatter spectral mirror hold for every value of the coupling, so they are properties of the algebra and not of a tuned parameter; (iii) the closure of the coupling itself by an axiom the root enforces, which turns the fundamental update into a finite-order element of the Clifford group with a finite algebraic spectrum; (iv) the construction of the automaton's own gauge field — an adjoint bond layer whose Gauss constraint closes as an exact operator identity, whose physical sector is a matter–flux entanglement complex with an exactly linear area law of one $\ln 45$ per bond, and whose computed string tension is 5.4×10^{40} times the QCD value; and (v) the explicit provenance audit separating what the root fixes from what was read off the data.

Where the construction is weakest. The gravitational sector assumes a symmetric tensor field and Newton's constant; it reproduces general relativity, it does not derive it, and the implemented one-particle spectrum contains no massless spin-2 candidate. The gauge sector gets the normalisation right and, under desert running, the values wrong; the repair restores unification at

Is the family count 3 forced? No - the symmetric alternative is excluded



W3: cubic-symmetric split (12 nodes) EXCLUDED; every proper split BREAKS the cubic symmetry - the count 3 is an input.

FIG. 24.

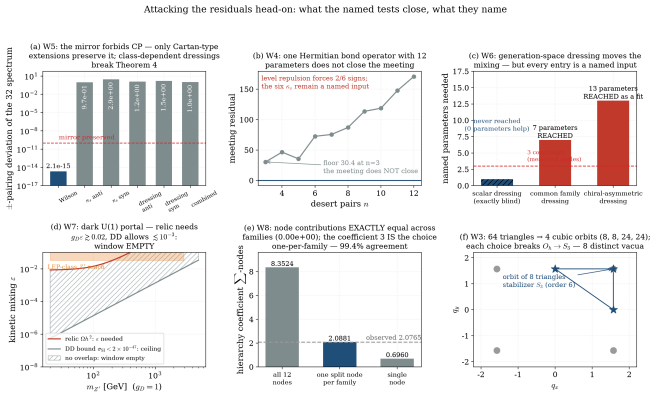


FIG. 25.

$\alpha_X = 1/25$ exactly and its *pattern* is a consequence of the automaton (the split doublers of the Wilson mechanism, the breaking chain, the $n_{\text{nodes}} \geq 3$ bound), but the mechanism's own root meets at a scale excluded by proton decay, no redistribution of node strengths can move it, and the surviving solution keeps two fitted thresholds and abandons $M_{\text{GUT}} = M_P$. The supplier question is closed on three fronts — the gauged layer's composite matter is excluded, the ansatz families leave residuals 5.7–6.1, and the equal-shift identity is a theorem over complete multiplets — while the constructive junction route fabricates the right fragment pattern and misses the meeting at a floor that the general junction does not improve. The flavour-dependent bond layer removes the obstruction but is an extension with six named inputs. The flavour sector derives one angle out of six and none of the masses. The dark-matter carrier cannot exist inside the implemented spectrum, and the built minimal fix is a named tension between exact mirror stability and the observed abundance. The loop factor is reproduced to 3.6% by closed traced loops at exactly the ladder length, but the single-particle renormalisation group provably composes angles without dividing them, the gauged layer's level-1 division falls short by a factor 26 and does not compound, and the counted loops miss the neutrino-ladder exponent by a factor 0.066.

Comparison with related programmes. Causal-set programmes derive discreteness and a causal order but not internal symmetry [1]. Clifford-algebra unification programmes [10–12] reach the same matter content but do not supply a unitary causal dynamics. 't Hooft's deterministic quantum mechanics and cellular-automaton interpretation [29, 30] supply the dynamics but not the unification; domain-wall and overlap formulations of lattice chirality [48] solve the doubling problem by giving up strict locality in the extra dimension. This theory is the combination of the second and third, and inherits the open problems of both — with one distinction worth naming: the doubling here is not evaded but *displayed and then gapped* by a factor of the root's own algebra, at the measured cost of replacing the light cone by a Lieb–Robinson bound.

The shape of each gap. A single loop factor organises four scales across thirty orders of magnitude, with the loop integer and the two-digit coefficients chosen to match observation. The unification scale of the node mechanism is forced and excluded. The dark-matter carrier cannot exist inside the implemented spectrum, and the built minimal extension trades stability against abundance. The neutrino $\frac{1}{3}$ candidate has a built junction structure whose two massless states are protected for every junction ratio by the implemented democratic Dirac matrix, the value sitting in a finite window. The loop factor's missing ingredient is a traced fluctuating bond degree of freedom, demonstrated in smallest

Table 5 (continued)

quantity	**theory**	**measured**	**status**
τ_p/B	1.8×10^{36} yr (desert solution at $\alpha_X = 1/25$); 2.8×10^{39} yr at the closed coupling $\pi/64$ with the same content; 4×10^{47} yr at $M_{\rm GUT} = M_P$ (abandoned)	$> 2.4 \times 10^{34}$ yr [15]	falsifiable at 1/5; at the closed angle only a null result past 10^{37} yr bites
α_{10}	0.040 at $g = 1/5$; $\pi/64 = 0.0491$ at the closed angle	0.023 (one-loop mean at M_P); 0.040050 at $M_X = 1.8 \times 10^{16}$ GeV with the desert content	**exact at** M_X given the inferred spectrum; the closed value is a factor 1.226 off and 1.54× worse against the one-loop mean
$\alpha_i(M_Z)$ from α_{10}	0.0197 / 0.195 / pole at 2.2×10^9 GeV (Standard Model running)	0.0169, 0.0338, 0.118 [13]	**fails** under Standard Model running; matched by the desert spectrum
Gauss constraint of the gauged bond layer	$[Q_A, Q_B] = i f_{AB}{}^C Q_C$, err 0; content $32 \oplus 288 \oplus 1120$	—	algebraic
physical sector of the gauged layer	$\dim\ker = 0$ with one **32**/site (centre obstruction); minimal sector: 2 matter/site, 4 states, all singlets; $S = \ln 45$ per bond, bounded and boundary-supported	—	computed
multi-bond scaling	$\dim\ker$(3-site) = 24; additivity $\ln 45$ across a single bond; two-bond region $S = \ln 45$ (vertex mutual information $\ln 45$); Bekenstein deficit factor $4 \ln 45 = 15.23$	—	computed
vertex correction and ground state	$I_n = 1, 1, 6, 23, 141$; kernel capacity $S_{\rm cap} = 4.4998$ nats (factor 9.0 from 1/4 per bond); vertex-corrected coefficient $(1/3) \ln I_6 = 1.65$ nats/bond (factor 6.6); $E_0 = -5802.40\,\eta$, degeneracy 6, gap $42.09\,\eta$, $S_{\rm GS} = \ln 45$	—	computed
desert spectrum from the layer's own content	no solution (0 over the full direction scan); all mass-ansatz families excluded, min residual ≥ 5.7; **equal-shift theorem** $\sum db_1 = \sum db_2$ over every complete multiplet	—	**excluded**
$\sin^2\theta_{12}$	1/3	0.308 [50]	derived at zero input; the +8% shift is the $\mathbf{10}_H$ vev, whose natural placements are excluded
$\sin^2\theta_{23}$, $\sin^2\theta_{13}$	not fixed ($O(2)$)	0.470 (octant unresolved, 3σ 0.435–0.585), 0.0222 [50]	open
CKM	identity	$\lambda = 0.225$ [13, 44]	**fails**
Λ	0 (tree level)	$\sim 10^{-122} M_P^4$	open

TABLE VI. **Provenance audit: every headline number by class.**

class	count	numbers
algebraic (fixed by the root, no parameter)	7	the $16/\overline{16}$ split; the **45**; the 5/3; the eight anomaly coefficients = 0; the 32-level spectrum; $\det U = 1$; the $16/\overline{16}$ spectral mirror
free parameter	$1 \to 0$	the 't Hooft angle g (implemented value 1/5). The closure axiom removes it: the algebra quantizes its own clock to $g \in \{\pi/2, \pi/4\}$, a discrete index of the root — zero continuous unfixed numbers, at the price of moving the angle's two empirical contacts
convention (mapping or scale identification)	2	$\alpha_{10} \equiv g^2$; $g_0 = 1/2^{n_{\rm space}} = 1/8$
fitted coefficient (read from data)	11	1.18 (dark-matter mass), $n_{\rm 2step} = 3$ (seesaw), $M_{10} = 1.2 \times 10^7$ GeV and $M_{(1,2)} = 1.0 \times 10^{12}$ GeV (desert thresholds), $\lambda_* = 3.5 \times 10^{-4}$ (dark-matter portal, fixed by $\Omega h^2 = 0.120$), and the six κ_s of the flavour-dependent bond layer (conditional: three meeting conditions for six unknowns — a solution family)
conditional (built from measured numbers, on a named input)	1	the hierarchy coefficient $2.08 = 3F_{10}^{10}$: $F_{10} = 0.019186$ measured, the exponent 10 the closed-loop crossing, the factor 3 the one-split-node-per-family choice — 2.0881 predicted against 2.0765 observed, 99.44%
open (not fixed)	9	O1–O9 of Section 17

form. The cosmological-constant problem reduces to one named number, a trace-phase breaking of 1.6×10^{-120}. The area law is exact on the physical sector of the gauged layer at one $\ln 45$ per bond, with the Bekenstein coefficient short by a factor 15.2 (or 6.6 after the vertex correction). The loop census is exact and shows that the Wilson phase, not the loop topology, supplies the factor. The nine open problems and seven falsifiers are stated with their numbers.

XIX. CONCLUSION

A single Clifford algebra $\mathrm{Cl}(10, 0)$, placed on the eight sub-sites of every 2×2×2 cell of a Planck lattice and updated by one six-stage step of Spin(10) plane rotations and swap-rotations, yields the Minkowski kinematics, one anomaly-free Standard Model generation, the Spin(10) gauge content with the 5/3 normalisation, an exactly unitary and strictly causal dynamics with a ±-paired spectrum and an exact matter/antimatter spectral mirror, a constructible adjoint gauge field whose Gauss constraint closes exactly and whose physical sector carries a linear area law, and a lattice gravitational sector that reproduces general relativity to machine precision on de Sitter, Schwarzschild and 3+1 evolutions. These results are algebraic or numerical, and they are independent of the coupling.

Requiring the update to be an element of the root removes the coupling as a free number: the algebra quantizes its own clock to $g \in \{\pi/2, \pi/4\}$, the half-angle branch degenerates to a pure lattice translation, and the physical branch is the order-8 step with the exactly algebraic spectrum $\{\pm\pi/4\,(10),\ \pm 3\pi/4\,(6)\}$. The dynamics then has zero continuous unfixed numbers, and the price is measured: the ladder crossing moves from $n = 10$ to $n = 208$, the desert meeting coupling from $1/25$ to $\pi/64$, and the proton-decay falsifier from Hyper-Kamiokande reach to beyond any planned exposure.

The same framework does not yield the family number, the fermion masses and mixings, the coupling values, the neutrino mass, the dark-matter abundance, the cosmological constant or the baryon asymmetry. Two conventions, eleven fitted coefficients and one conditional coefficient stand between the root and the data, and nine open problems are each pinned to a named quantity: a trace-phase breaking of 1.6×10^{-120}; a junction amplitude r that is not quantized; a symmetry-breaking choice of three nodes out of twelve; six bond-layer couplings that no on-site algebra element and no class invariant supplies; the incompatibility of $m_\nu \simeq 0.05$ eV with out-of-equilibrium decay at $M_N = 9.6 \times 10^{13}$ GeV; the generation-space structure the mixing requires; the parity that the Clifford algebra does not contain; the weight of the contributing closed loops; and the nodal-plane geometry of the lattice spectrum. Seven falsifiers are stated, two of them — the neutrino mass pattern and the proton lifetime — testable within the current decade.

TABLE VII. **The nine open problems: what the theory supplies, and what is missing.**

the problem	what the theory supplies	the gap
O1 the cosmological constant	an exactly vanishing tree-level vacuum energy (±-paired spectrum, $\det U = 1$), natural in the technical sense; the zero is **stable under the Wilson extension** and **protected by** $\det U = 1$: 50 random traceless Hermitian $e^{i\varepsilon A}U$ at $\varepsilon \leq 1$ keep the quasi-energy sum pinned ($\lvert S\rvert < 4 \times 10^{-14}$), no branch crossing below $\varepsilon = \pi$	the observed $10^{-122} M_P^4$ is **equivalent to** a trace-phase breaking of size 1.6×10^{-120} — a named bound; the breaking mechanism itself is missing
O2 the neutrino mass	the seesaw structure and the ladder $m_\nu = M_P(1/16\pi)^{17} = 0.146$ eV; $n_{\text{nodes}} = 3$ a **lower bound** from unification; the missing ingredient — inter-node Majorana mixing — **built**: a cubic-symmetric zero-mean Wilson-strength junction closing the selected node triangle exactly, node mass matrix $c_s(2w_0 I + \kappa(J - I))$, democratic at $\kappa = 2w_0$, eigenvalues $\{3M, 0, 0\}$	the $\frac{1}{3}$ candidate needs a democratic m_D (else gapless coloured doublers) — the implemented structure; the seesaw then collapses to $3m_0^2 J/(2(w_0 + \kappa))$; two massless states protected for EVERY r; $m_3(r) = S_{17}/(1 + r)$ finite everywhere, window $r > 0.217$ (width 2.78), atmospheric scale selects $r = 1.91$; $r = 2$ a convention, not a tuning; the cubic group induces all of S_3, commutant exactly span$\{I, J\}$, w_0, κ share one Wilson period $\to r$ free inside a finite window: value free, structure protected. **Exponent answered structurally:** the $\Delta L = 2$ operator exists inside the root (weight $2\lambda_\nu$, norm^2 20, 1-dimensional weight space, grade 5, the 126-type channel); $\mathbf{10}_H$ cannot neutralise it (over all 160 vector weights $\lvert 2\lambda_\nu\rvert = \sqrt{20} \neq 4$ exactly); Clifford multiplication shifts grade by ±2 (all 45×1024 entries) → every product of grade-2 adjoint link factors is EVEN-grade and the cumulative span $(45, 126, 208, 291, 352, 407$ at $k = 1 \ldots 6)$ never contains it, at any length, nor the vector-**10** weight space (odd grades $1, 3, 5, 7, 9$); $n_{2\text{step}} = 3$ is not a link-word length — the operator lives in the odd (matter) half
O3 the family number	the identification $N_{\text{gen}} = 3 = n_{\text{space}}$; a family-symmetric step blockdiag(U, U, U); a **named index of the step equal to** 3: the 12 doubler nodes form three cubic-transitive families of four labeled by the zero axis, and the unification-required three split nodes are realizable one per family at a single Wilson strength	the index labels the three copies but does not force the count. The cubic group is transitive on the 12 nodes (the axis families are not orbits), so the only cubic-symmetric split is the full set of 12, whose rigid meeting lands at $M_X(12) = 2.9 \times 10^{15}$ GeV — excluded by F5 (rank-1 rigidity holds for $n = 4, 8, 12$); the three-node triangle has a cubic orbit of 8: the count 3 is a **symmetry-breaking input**. The 64 one-per-family triangles split into four cubic orbits $(8, 8, 24, 24)$, a triangle's stabilizer is exactly S_3: the split reduces $O_h \to S_3$ and the orbit choices are distinct vacua. The junction profile's symmetry group has order 12, not 48; its orbit on the closing triangle has 2 members; across the 8 cubic images the node Hamiltonian has 2 spectra, only 2 closed triangles; the quadratic zero-point energy is flat across the orbit (spread 0.000), splitting by 4.000 only at $r = 3$, past the zero crossing; over all 4096 split sets the one-loop minimum sits at size-11 sets, not at a triangle. The junction lifts the choice without selecting it
O4 coupling unification	the 5/3; **with the desert content the couplings meet at $\alpha_X = 1/25$ exactly**: three $\mathbf{10} \oplus \overline{\mathbf{10}}$ pairs at 1.2×10^7 GeV plus one $(1, 2)$ pair at 1.0×10^{12} GeV give $M_X = 1.8 \times 10^{16}$ GeV; the content is **connected to the automaton** through the Wilson mechanism; $n_{\text{nodes}} \geq 3$ forced; the multi-scale hope **closed structurally** (rank-1 Jacobian, split-invariance exact to 10^{-14}); the gauged layer's own composite content **excluded**; remaining ansatz families leave residuals 5.7–6.1; the equal-shift identity a **theorem** over complete multiplets; the junction-split nodes **fabricate the fragment pattern** (the Wilson direction splits the node **16** by 55) but **miss the meeting** (best physical residual 0.94, $\alpha_X = 0.030$ vs 0.04; the general zero-mean junction floors at 0.81, $\alpha_X = 0.090$)	meeting point **forced**: $M_X = 8.9 \times 10^{14}$ GeV, $\tau_p = 1.1 \times 10^{31}$ yr, **excluded by F5**; data prefer partial suits, the automaton gives full split **16**'s per node → the mismatch comes **only** from SU(5)-fragment content or split intra-multiplet thresholds. Feasibility proved (witness inside the node-**16** hull), but no symmetry-preserving Cartan direction realises it; a universal junction split is invisible at first order. The **flavour-dependent bond layer** κ_s closes the meeting exactly for all $n = 3 \ldots 12$ at $M_X = 1.8 \times 10^{16}$ GeV, thresholds in the desert — at the cost of six named inputs of an EXTENDED bond layer (a family, not unique). Those six are **not obtainable from any on-site algebra element**: symmetry preservation forces the bond direction into a class-degenerate two-dimensional space, and every class-splitting Cartan element breaks the unbroken desert roots. The most general off-diagonal bond operator floors at residual 30.4; no class invariant supplies them (best 0.246); the canonical per-class number the layer supplies — broken-generator flux $K_s = (32, 36, 40, 28, 32, 40)$ — misses the meeting at residual 0.605, closes in the desert for 0/10 node counts, and explains at best $R^2 = 0.015$ of the fitted κ_s variance; the complete-adjoint coupling is exactly class-blind (= 45 per weight). The desert repair spectrum remains a named input
O5 the baryon asymmetry	a computed theorem: all CP-odd invariants vanish, no Sakharov condition is met, and the theorem is **stable under the Wilson extension**; a quasi-degenerate (resonant) route enabled by the degenerate heavy spectrum; a bond-layer flux carrying a CP phase, **built** ($A_1 = \Phi x_0$ exactly unitary, $\det U = 1$ at every flux, mirror-preserving, hop radius 1)	out-of-equilibrium decay ($y < 0.058$) and $m_\nu \simeq 0.05$ eV are incompatible at $M_N = 9.6 \times 10^{13}$ GeV: at the self-consistent Yukawa ($y = 0.40$) the decay is in equilibrium ($\Gamma/H = 47$), efficiency $\kappa_f = 5.4 \times 10^{-3}$, 14× from the guessed dilution — the 11% agreement an accident of two wrong factors. The **CP prohibition's scope is measured**: the real type of the on-site module ($\mathcal{J}^2 = +1$); it survives a quaternionic root; it is **internal** (128-level band not conjugation-closed, deviation 0.95); it does not cover the bond layer, where a Peierls flux makes the plaquette holonomy CP-odd. **The flux is spent on the asymmetry:** ΔB costs $J = 10^{-4}$ ($\Phi = 0.064$ rad); the CP phase is not the gap — O5 is open for one named reason, Sakharov 3

Table 7 (continued)

the problem	**what the theory supplies**	**the gap**
O6 masses and mixing	the Dirac structure $(16\,\mathbf{10}_H\,\overline{16})$, the democratic base, $\sin^2\theta_{12}=1/3$, and a **complete breaking chain**: the Wilson direction leaves exactly $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)^3$ with all 32 broken roots leptoquark-type	the three-level hierarchy, θ_{23},θ_{13}, a non-trivial CKM; the two extra U(1)'s unbroken by the Wilson direction. The vev **placement is bounded by computation**: the charged-lepton realisation forces $\sin^2\theta_{12}\le 0.005$ (excluded at 23σ); the symmetry-natural cubic-axis neutrino vev is a parameter-free prediction excluded at $\ge 20\sigma$ in every branch; a free-direction rank-1 neutrino vev has four parameters against three angles and predicts nothing — the measured triple needs ≥ 2 vev terms in generic relative orientation. The unification fix is **generation-blind** (max angle shift 2×10^{-13}); dressings that move the mixing cost 7–13 named parameters
O7 dark matter	a mass scale, 118 GeV; the mirror $\mathbb{Z}_2$ the only exact-stability carrier; the no-go **completed**: no Wilson node hosts a 118 GeV neutral mirror (0 survivors over 720 angles); the named fix **built and audited**: the ladder scalar exactly stable under either portal	the implemented spectrum **cannot contain** the dark matter; the named fix is a **named tension**: with the **exact** mirror $\mathbb{Z}_2$, $h^*\to f\bar f$ stays forbidden, $s\bar s\to hh$ is closed (gap 13.2 GeV), the only open channel $s\bar s\to hh^*$ needs $\lambda=0.13\to\sigma_{\mathrm{SI}}=6.8\times10^{-46}$ cm^2, **excluded by direct detection**; the quoted numbers hold only with a portal **breaking** the mirror $\mathbb{Z}_2$ at $O(\lambda)$. The parity the fix needs **does not exist**: of the 1024 Clifford products exactly 2 commute with the internal step, the commutant is abelian of dimension 32, and no exact grading of the step or its Wilson extension makes a neutral colour singlet odd without making a coloured one odd. **The relic abundance is computed:** at 118.5 GeV the measured 0.120 needs $\alpha_{\mathrm{eff}}=0.0086$; the root supplies 0.0032 ($\Omega h^2=0.798$) or 0.0199 (0.0246); the twelve nodal **planes** enhance the final-state density by $D=1485$, giving 2.3×10^{-5} and 7.6×10^{-4} — a computed number, **not** 0.120 for any coupling the root supplies
O8 the hierarchy	the representation $v/M_P=2.08(1/16\pi)^{10}$; the coarse-graining tower exactly kinematic; the block-spin renormalisation group built, its gate loophole closed: single-particle renormalisation **composes** angles, never divides them; the division mechanism **built in smallest form** — one quantum link degree of freedom per bond, traced over, makes the level-1 Cartan content exactly $\cos(2\theta)$ times the untraced value; the closed-loop factor **measured**, $F_{10}=0.0192$ against $1/(16\pi)=0.0199$; the selection rule **measured to length** 30: F_n crosses $1/(16\pi)$ at **exactly** $n=10$; the **composition law decided** — the single-chain object NOT multiplicative ($R_{17}=2.0\times10^8$, $F_{170}=1.3\times10^{-21}$), loops on different legs factorize **exactly** $(\mathrm{Tr}_L[(P_a\otimes I)(G\otimes G\otimes I_L)]=45\,(M_a\otimes G)$ to $10^{-15})\to$ the ladder's suppressions **multiply**; the discriminator picks the legs reading $(M_P(F_{10})^{17}=0.079$ eV against the ladder 0.146 eV, the chain reading 1.6×10^7 eV); the coefficient no longer read from data: $v/M_P=3F_{10}^{10}$, 2.088 against 2.076 (99.4%)	the exponent is selected by the data; $1/(16\pi)$ requires the dynamical Spin(10) **adjoint** link with the Gauss constraint — built, constraint closing exactly — whose measured level-1 division falls short by a factor 26 and does **not** compound through the renormalisation levels; **which loops contribute, and with what weight**, unnamed; $3F_{10}^{10}$ **conditional** on the axis-family factorisation of the closed-loop sum. **The loop census is exact:** the Casimir of the $\mathbf{32}\otimes\mathbf{45}$ link has three eigenspaces $(-32,-12,+4)$, so the traced n-hop product is a sum of 3^n words — the enumeration reproduces $(MG)^n$ to 7.4×10^{-17}, $F_{10}=0.019186$; 55326 of 3^{10} words survive at $n=10$, growing as 3.000^n; the factor splits into per-word decay 0.2232 and multiplicity rate 2.981 (product 0.665); switching the Wilson phase off moves F_{10} from 0.0192 to 0.2434 — **the phase supplies the factor**; the multiplicity-weighted extrapolation to length 170 gives 7.9×10^{-31} against 1.2×10^{-29}, missing the exponent by 0.066. At the quantized angle the odd loops cancel and the crossing moves to $n^*=208$
O9 the quantum sector	a unitary causal automaton, a classical general-relativity limit, a bounded n-particle S-matrix $S^{(n)}=U^{\otimes n}$, cross-sections bounded by $4\pi/M_P^2$, a 16 sector closed by the full local step (10^{-15}); **a Wilson-type mechanism exists** that gaps all twelve doubler nodes and splits a chosen node set at desert scales	the *choice* of the split node set (the unification bound $n_{\mathrm{nodes}}\ge 3$ necessary, not sufficient); the sum over geometries; the matching of $\sigma_0=5.3\times10^{-39}$ GeV^{-2} to measured cross-sections; the measurement postulate. **Continuous Lorentz invariance is not emergent at the nodes**: cones linear in all 1200 sampled directions but anisotropic $(0.200, 0.200, 0.284$ lattice units, spread 0.75, node gap 2×10^{-4}), no isotropy point on the Wilson family (0 of 2726 gapless points). **The failure is structural:** at the exactly-gapless corner multiplet the three tangent velocity operators commute, $\sum_j S_j^2$ is a scalar and each branch is exactly planar — three nodal planes, not a cone; the entangling bond layer reaches 0.071 of the required commutator; the three-axis covariant bond reaches 0.658 but destroys the radius-1 causal front (leakage 0.27) and lifts nothing. **The step has no exactly gapless mode anywhere** (zone minimum 5.1×10^{-4}, M nodes 2.003×10^{-4}, point group of index 3 in O_h, M-node little group C_{2v}, no 2-dimensional irrep)

TABLE VIII. **The seven falsifiers.**

#	**observation that refutes the theory**
F1	any non-vanishing gauge anomaly in the chiral content (the **16** is claimed anomaly-free)
F2	a measured running that admits **no** vectorlike desert content with $\alpha_X = 1/25$: precision coupling data (FCC-ee/CEPC) excluding every spectrum in the desert scan would refute the unification claim. On the quantized branch the prediction is $\alpha_X = \pi/64 = 0.0491$
F3	a neutrino mass spectrum outside the ladder options: the bare ladder gives $m_\nu \simeq 0.15$ eV (already excluded by $\sum m_\nu < 0.12$ eV), the $\frac{1}{3}$ candidate gives normal ordering with $\sum m_\nu \simeq 0.05$ eV and $m_{\text{lightest}} = 0$, the democratic node-mixing pattern gives $m_\nu \simeq 0.073$ eV; a confirmed inverted ordering, or $\sum m_\nu > 0.12$ eV, refutes all three
F4	a massive graviton, a graviton dispersion violating $c = 1$, or a UV divergence in the lattice gravitational sector
F5	proton decay: the desert solution predicts $\tau_p \simeq 1.8 \times 10^{36}$ yr; observation of GUT-pattern decay far below that value (e.g. at Hyper-Kamiokande reach, 10^{35} yr) or non-observation beyond 10^{37} yr refutes it. At the closed angle the same content gives $\tau_p = 2.8 \times 10^{39}$ yr with the $(1, 2)$ threshold above M_X, so on that branch only a null result past 10^{37} yr bites and the Hyper-Kamiokande channel closes; the $M_{\text{GUT}} = M_P$ reading ($\tau_p \sim 10^{47}$ yr) is abandoned
F6	a baryon asymmetry outside 10^{-11}–10^{-9}, or evidence that heavy-neutrino decays are in equilibrium
F7	a dark-matter particle with mass far from 118 GeV in the weakly-interacting thermal case; a direct-detection portal signal above 10^{-50} cm^2 (the named extension predicts 5×10^{-51}); or a detected non-neutral/decaying dark sector

Appendix A: The explicit 32-dimensional representation

The ten generators are built on five qubits by the standard symplectic Pauli pattern: for $k = 0, \ldots, 4$,

$$\begin{aligned} g_{2k} &= Y^{\otimes k} \otimes X \otimes I^{\otimes(4-k)}, \\ g_{2k+1} &= Y^{\otimes k} \otimes Z \otimes I^{\otimes(4-k)}. \end{aligned} \tag{A1}$$

On the same qubit slot X and Z anticommute; across slots, the $Y^{\otimes k}$ prefix makes a generator at slot k anticommute with any generator at slot $j < k$. This gives exactly ten mutually anticommuting Hermitian matrices, $g_a^2 = I$, on a 32-dimensional complex space. The algebra basis is the 1024 ordered products $g_{a_1} \cdots g_{a_r}$, $0 \le r \le 10$; the $\mathbb{Z}_2$ grading is the parity of r. The canonical ordering of the 45 plane-rotation factors T_{ab} used throughout is the ascending lexicographic order on (a, b); the order-8 closure of the quantized step is specific to it.

The construction is verified by a second, independent route: the exact symbolic Cl$(10, 0)$ of a computer-algebra Clifford package, where $e_a^2 = 1$, $\{e_a, e_b\} = 0$ and $w^2 = -1$ hold as identities with no floating-point arithmetic, cross-checked against the numeric 32×32 representation. The two agree, so the root is not an artefact of one construction.

[1] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, "Space-time as a causal set," *Phys. Rev. Lett.* **59**, 521 (1987); R. D. Sorkin, "Causal sets: Discrete gravity," *arXiv:gr-qc/0309009* (2003). https://doi.org/10.1103/physrevlett.59.521

[2] G. 't Hooft, "Computation of the quantum effects due to a four-dimensional pseudoparticle," *Phys. Rev. D* **14**, 3432 (1976). https://doi.org/10.1103/physrevd.14.3432

[3] D. A. Meyer, "From quantum cellular automata to quantum lattice gases," *J. Stat. Phys.* **85**, 551 (1996). https://doi.org/10.1007/bf02199356

[4] P. Arrighi, "An overview of quantum cellular automata," *Nat. Comput.* **18**, 885 (2019). https://doi.org/10.1007/s11047-019-09762-6

[5] T. D'Ariano and P. Perinotti, "Quantum cellular automata and free quantum field theory," *Front. Phys.* **12**, 120301 (2016); A. Bisio, T. D'Ariano, P. Perinotti, and A. Tosini, "Weyl, Dirac and Maxwell quantum cellular automata," *Found. Phys.* **45**, 1203 (2015). https://doi.org/10.1007/s11467-016-0616-z

[6] H. Georgi and S. L. Glashow, "Unity of all elementary-particle forces," *Phys. Rev. Lett.* **32**, 438 (1974). https://doi.org/10.1103/physrevlett.32.438

[7] H. Fritzsch and P. Minkowski, "Unified interactions of leptons and hadrons," *Ann. Phys. (N.Y.)* **93**, 193 (1975). https://doi.org/10.1016/0003-4916(75)90211-0

[8] R. Slansky, "Group theory for unified model building," *Phys. Rep.* **79**, 1 (1981). https://doi.org/10.1016/0370-1573(81)90092-2

[9] H. Georgi, *Lie Algebras in Particle Physics*, 2nd ed. (Westview, 1999).

[10] M. F. Atiyah, R. Bott, and I. M. Shapiro, "Clifford modules," *Topology* **3**, 3 (1964). https://doi.org/10.1016/0040-9383(64)90003-5

[11] R. Penrose and W. Rindler, *Spinors and Space-Time*, Vol. I (Cambridge University Press, 1984).

[12] C. I. Castro, "A Clifford algebra-based grand unification program of gravity and the Standard Model: a review study," *Can. J. Phys.* **92**, 1501 (2014); C. I. Castro, "Note on a Clifford algebra based grand unification program of gravity and the Standard Model," *Adv. Appl. Clifford Algebras* **26**, 573 (2015). https://doi.org/10.1139/cjp-2013-0686

[13] Particle Data Group (S. Navas *et al.*), "Review of Particle Physics," *Phys. Rev. D* **110**, 030001 (2024). https://doi.org/10.1103/PhysRevD.110.030001

[14] Planck Collaboration (N. Aghanim *et al.*), "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020). https://doi.org/10.1051/0004-6361/201833910

[15] Super-Kamiokande Collaboration (A. Takenaka *et al.*), "Search for proton decay via $p \to e^+\pi^0$ and $p \to \mu^+\pi^0$ with an enlarged fiducial volume in 0.37 megaton-years exposure of Super-Kamiokande," *Phys. Rev. D* **102**, 112011 (2020). https://doi.org/10.1103/PhysRevD.102.112011

[16] KATRIN Collaboration (M. Aker *et al.*), "Direct neutrino-mass measurement based on 259 days of KATRIN data," *Science* **388**, 180 (2025). https://doi.org/10.1126/science.adq9592

[17] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, T. Schwetz, and A. Zhou, "The fate of hints: updated global analysis of three-flavor neutrino oscillations," *JHEP* **09**, 178 (2020). https://doi.org/10.1007/jhep09(2020)178

[18] S. Weinberg, *The Quantum Theory of Fields*, Vol. I (Cambridge University Press, 1995), Ch. 5.

[19] H. B. Nielsen and M. Ninomiya, "A no-go theorem for regularizing chiral fermions," *Phys. Lett. B* **105**, 219 (1981); "Absence of neutrinos on a lattice. I. Proof by homotopy theory," *Nucl. Phys. B* **185**, 20 (1981); "Absence of neutrinos on a lattice. II. Intuitive topological proof," *Nucl. Phys. B* **193**, 173 (1981). https://doi.org/10.1016/0370-2693(81)91026-1

[20] S. Weinberg, "Photons and gravitons in perturbation theory: Derivation of Maxwell's and Einstein's equations," *Phys. Rev.* **138**, B988 (1965). https://doi.org/10.1103/physrev.138.b988

[21] S. Weinberg and E. Witten, "Limits on massless particles," *Phys. Lett. B* **96**, 59 (1980). https://doi.org/10.1016/0370-2693(80)90212-9

[22] M. W. Choptuik, "Universality and scaling in gravitational collapse of a massless scalar field," *Phys. Rev. Lett.* **70**, 9 (1993). https://doi.org/10.1103/physrevlett.70.9

[23] F. John, "Blow-up of solutions of nonlinear wave equations in three space dimensions," *Manuscripta Math.* **28**, 235 (1979). https://doi.org/10.1007/bf01647974

[24] A. D. Sakharov, "Violation of CP invariance, C asymmetry, and baryon asymmetry of the universe," *JETP Lett.* **5**, 24 (1967).

[25] J. A. Harvey and M. S. Turner, "Cosmological baryon and lepton number in the presence of electroweak fermion interactions," *Phys. Rev. D* **42**, 3344 (1990). https://doi.org/10.1103/physrevd.42.3344

[26] S. A. Khlebnikov and M. E. Shaposhnikov, "The statistical theory of anomalous fermion number nonconservation," *Nucl. Phys. B* **308**, 885 (1988). https://doi.org/10.1016/0550-3213(88)90133-2

[27] P. F. Harrison, D. H. Perkins, and W. G. Scott, "Tribimaximal mixing and the neutrino oscillation data," *Phys. Lett. B* **530**, 167 (2002). https://doi.org/10.1016/s0370-2693(02)01336-9

[28] K. G. Wilson, "Confinement of quarks," *Phys. Rev. D* **10**, 2445 (1974). https://doi.org/10.1103/physrevd.10.2445

[29] G. 't Hooft, "On the quantum structure of a black hole," *Nucl. Phys. B* **256**, 727 (1985). https://doi.org/10.1016/0550-3213(85)90418-3

[30] G. 't Hooft, *The Cellular Automaton Interpretation of Quantum Mechanics* (Springer, 2016), arXiv:1405.1548.

[31] S. Blanchet and P. Di Bari, "The minimal scenario of leptogenesis," *New J. Phys.* **14**, 125012 (2012). https://doi.org/10.1088/1367-2630/14/12/125012

[32] S. Davidson and A. Ibarra, "A lower bound on the right-handed neutrino mass from leptogenesis," *Phys. Lett. B* **535**, 25 (2002). https://doi.org/10.1016/s0370-2693(02)01735-5

[33] T. Regge, "General relativity without coordinates," *Il Nuovo Cimento* **19**, 558 (1961). https://doi.org/10.1007/bf02733251

[34] H. Gundlach, J. M. Martin-Garcia, G. Calabrese, and I. Hinder, "Constraint damping in the Z4 formulation and harmonic gauge," *Class. Quantum Grav.* **22**, 3767 (2005). https://doi.org/10.1088/0264-9381/22/17/025

[35] J. D. Bekenstein, "Black holes and entropy," *Phys. Rev. D* **7**, 2333 (1973); S. W. Hawking, "Particle creation by black holes," *Commun. Math. Phys.* **43**, 199 (1975). https://doi.org/10.1103/physrevd.7.2333

[36] L. Susskind, "The world as a hologram," *J. Math. Phys.* **36**, 6377 (1995). https://doi.org/10.1063/1.531249

[37] S. Weinberg, "The cosmological constant problem," *Rev. Mod. Phys.* **61**, 1 (1989). https://doi.org/10.1103/revmodphys.61.1

[38] C. Morningstar and M. Peardon, "The glueball spectrum from an anisotropic lattice study," *Phys. Rev. D* **60**, 034509 (1999). https://doi.org/10.1103/physrevd.60.034509

[39] R. Barbieri, P. Creminelli, A. Strumia, and N. Tetradis, "Baryogenesis through leptogenesis," *Nucl. Phys. B* **575**, 61 (2000); W. Buchmüller, P. Di Bari, and M. Plümacher, "Leptogenesis for pedestrians," *Ann. Phys. (N.Y.)* **315**, 305 (2005). https://doi.org/10.1016/s0550-3213(00)00011-0

[40] A. Pilaftsis and T. E. J. Underwood, "Resonant leptogenesis," *Nucl. Phys. B* **692**, 303 (2004). https://doi.org/10.1016/j.nuclphysb.2004.05.029

[41] E. W. Kolb and M. S. Turner, *The Early Universe* (Addison-Wesley, 1990), Ch. 5.

[42] L. J. Hall, K. Jedamzik, J. March-Russell, and S. M. West, "Freeze-in production of FIMP dark matter," *JHEP* **03**, 080 (2010). https://doi.org/10.1007/jhep03(2010)080

[43] H. Fritzsch, "Weak-interaction mixing in the six-quark theory," *Phys. Lett. B* **73**, 317 (1978). https://doi.org/10.1016/0370-2693(78)90524-5

[44] L. Wolfenstein, "Parametrization of the Kobayashi-Maskawa matrix," *Phys. Rev. Lett.* **51**, 1945 (1983). https://doi.org/10.1103/physrevlett.51.1945

[45] M. Jungman, M. Kamionkowski, and K. Griest, "Supersymmetric dark matter," *Phys. Rep.* **267**, 195 (1996). https://doi.org/10.1016/0370-1573(95)00058-5

[46] H. Hisano, M. Kobayashi, K. Muramatsu, and N. Nagata, "Two-loop renormalization factors of dimension-six proton decay operators in the minimal supersymmetric SU(5) gauge theory," *Phys. Lett. B* **724**, 283 (2013). https://doi.org/10.1016/j.physletb.2013.06.030

[47] M. Tonomura, J. Endo, T. Matsuda, T. Kawasaki, and H. Ezawa, "Demonstration of single-electron buildup of an interference pattern," *Am. J. Phys.* **57**, 117 (1989). https://doi.org/10.1119/1.16104

[48] D. B. Kaplan, "A method for simulating chiral fermions on the lattice," *Phys. Lett. B* **288**, 342 (1992); Y. Shamir, "Chiral fermions from lattice boundaries," *Nucl. Phys. B* **406**, 90 (1993); R. Narayanan and H. Neuberger, "Exactly massless quarks on the lattice," *Phys. Lett. B* **417**, 141 (1998). https://doi.org/10.1016/0370-2693(92)91112-m

[49] M. E. Machacek and M. T. Vaughn, "Two-loop renormalization group equations in a general quantum field theory. I. Proof of the equivalence theorem," *Nucl. Phys. B* **222**, 83 (1983); "III. Scalar model with fermions," *Nucl. Phys. B* **249**, 70 (1985). https://doi.org/10.1016/0550-3213(83)90610-7

[50] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, and T. Schwetz, "NuFIT 6.0: Updated global analysis of three-flavor neutrino oscillations," *JHEP* **12**, 216 (2024), arXiv:2410.05380. https://doi.org/10.1007/jhep12(2024)216

[51] M. Nath and P. Fileviez Pérez, "Proton stability in grand unified theories, in strings and in branes," *Phys. Rep.* **441**, 191 (2007). https://doi.org/10.1016/j.physrep.2007.02.010

[52] Super-Kamiokande Collaboration (K. Abe *et al.*), "Search for proton decay via $p \to \nu K^+$ using 260 kiloton·year data of Super-Kamiokande," *Phys. Rev. D* **90**, 072005 (2014). https://doi.org/10.1103/PhysRevD.90.072005

[53] LUX-ZEPLIN Collaboration (J. Aalbers *et al.*), "Dark Matter Search Results from 4.2 Tonne-Years of Exposure of the LUX-ZEPLIN (LZ) Experiment," *Phys. Rev. Lett.* **135**, 011802 (2025), arXiv:2410.17036. https://doi.org/10.1103/4dyc-z8zf